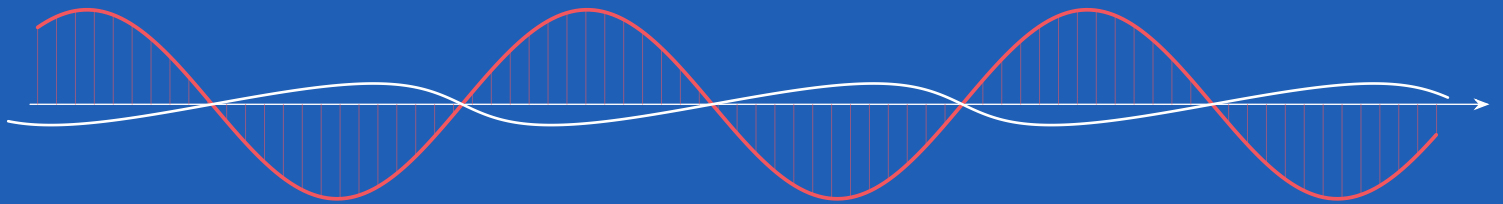


Electromagnetics

الکترومغناطیس



Ferdowsi University of Mashhad

دانشگاه فردوسی مشهد

Instructor: سید محمد سعید ماجدی

Compiled and edited by: Aidin Shekari

Electromagnetics

الکترومغناطیس

Lecture Notes

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دانشگاه فردوسی مشهد

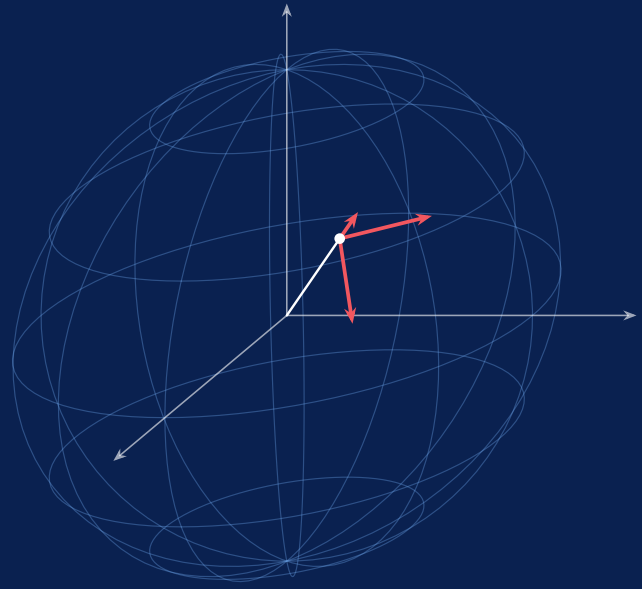
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CHAPTER

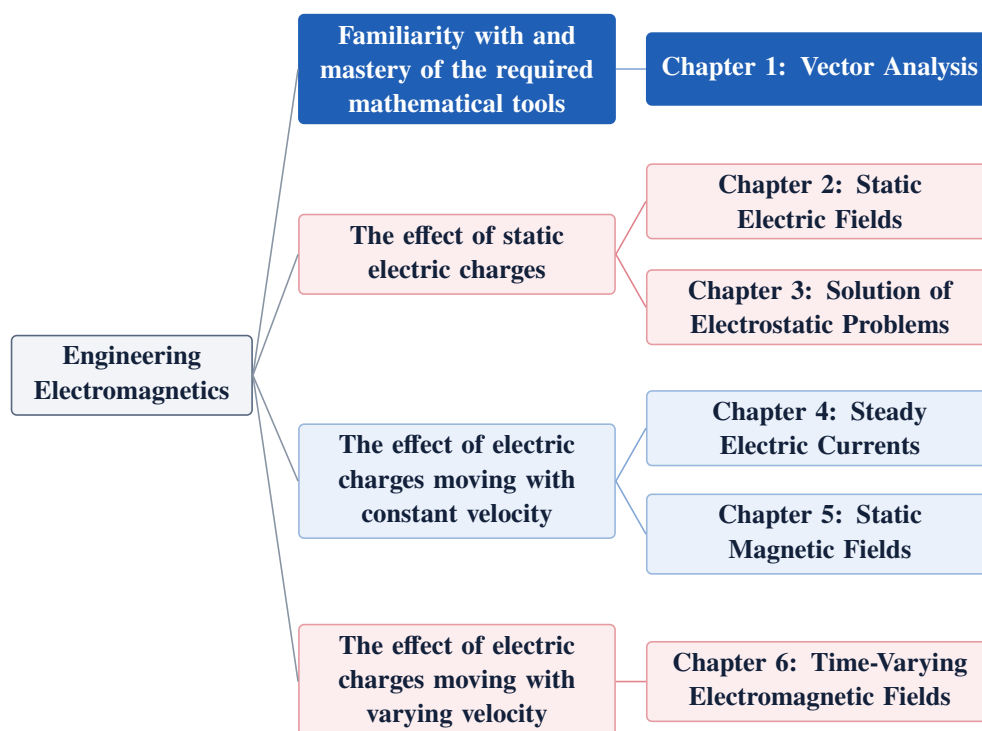
Vector Analysis

آنالیز برداری

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1.1 Course Outline

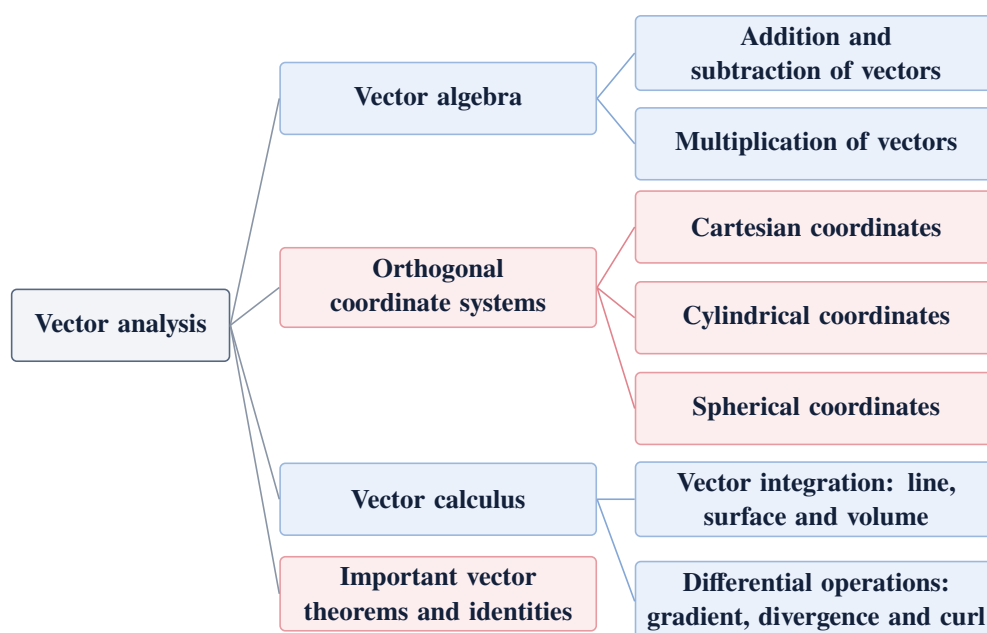


1.2 Vector Analysis

- **Vector analysis** is a mathematical tool with which the expression and understanding of electromagnetic concepts become simple.
- Quantities in electromagnetics are divided into the following two groups.
 - **Scalar:** a quantity that is completely specified by its magnitude (positive or negative). For example: charge, electric current, energy, ...
 - **Vector:** a quantity that is described by a magnitude and a direction. For example the electric and magnetic field intensity, ...
- Both of the above quantities can be functions of position and time.

Example 1-1

- In your opinion, of what type are the following quantities?
 - Temperature
 - Power
 - Force
 - Potential difference
 - Velocity
- Apart from the above, which other scalar and vector quantities in physics can you name?



- How can two vector quantities be added or subtracted graphically?
- How can vectors be multiplied? What is the meaning of this product? Where is it used?

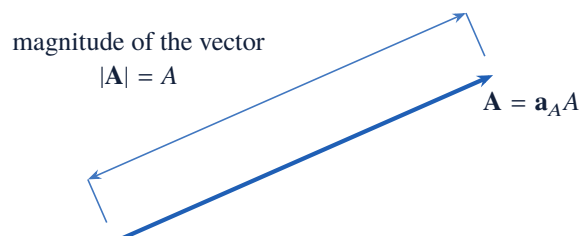
$$\mathbf{A} = 2\mathbf{a}_x + 3\mathbf{a}_y + \mathbf{a}_z \quad \mathbf{B} = \mathbf{a}_x - 2\mathbf{a}_y + 3\mathbf{a}_z \quad \mathbf{C} = -\mathbf{a}_x + 4\mathbf{a}_y - \mathbf{a}_z$$

- Find the magnitude of the projection of vector $\mathbf{A}$ on vector $\mathbf{B}$.
- What is the angle between vectors $\mathbf{A}$ and $\mathbf{B}$?
- What is the area of the triangle formed by the two vectors $\mathbf{A}$ and $\mathbf{B}$?
- Find the unit vector normal to the plane containing the two vectors $\mathbf{A}$ and $\mathbf{B}$.
- What is the volume of the parallelepiped formed by the 3 vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$?

1.3 Vector Algebra

- Every vector has a magnitude and a direction.

$$\mathbf{A} = \mathbf{a}_A A, \quad \mathbf{a}_A = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\mathbf{A}}{A} \quad (\text{unit vector})$$



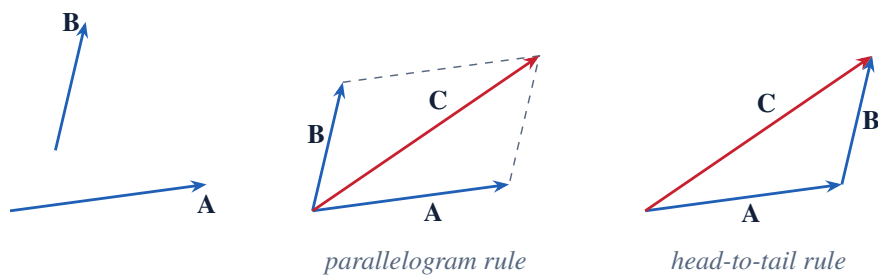
① Remark

For every vector quantity both the magnitude and the direction must be known. Leaving the direction of a vector quantity unspecified means ignoring part of the information of that quantity.

1.4 Addition of Vectors

► Adding two vectors graphically

- The **parallelogram rule**
- The **head-to-tail rule**



► Laws governing the addition of vectors

- **Commutative law**

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

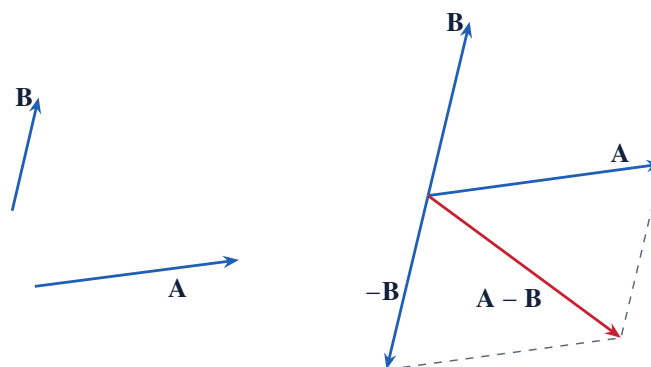
- **Associative law**

$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$$

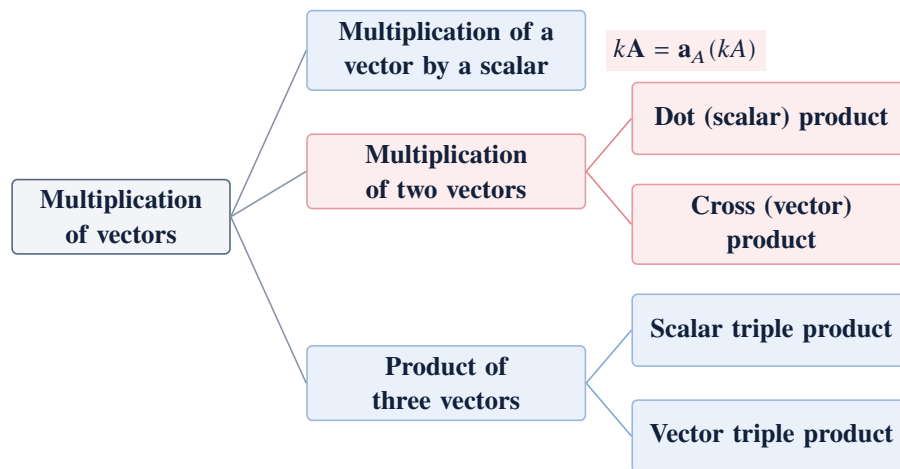
1.5 Subtraction of Vectors

► Vector subtraction

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$$



1.6 Multiplication of Vectors



- Multiplication of a vector by a scalar:

$$k\mathbf{A} = \mathbf{a}_A(kA)$$

1.7 Dot or Scalar Product

≡ Definition | Mathematical definition

$$\mathbf{A} \cdot \mathbf{B} \triangleq AB \cos \theta_{AB}$$

- It equals the product of the magnitude of one vector and the projection of the other vector on the first.

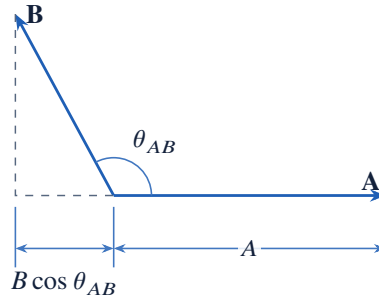


Figure 1.1 | θ_{AB} : the smaller angle between the two vectors when their tails are joined

► Properties of the dot product

- The dot product of two vectors is a scalar quantity.
- It is less than or equal to the product of their magnitudes.
- Depending on the angle between them it can be positive or negative.
- If the vectors are perpendicular to each other, it is zero.
- It is commutative and distributive.

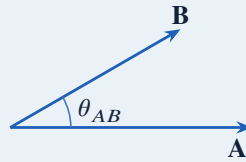
$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}, \quad \mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$$

- The dot product of a vector with itself

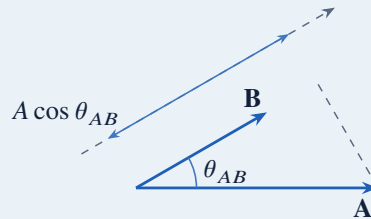
$$\mathbf{A} \cdot \mathbf{A} = A^2, \quad A = +\sqrt{\mathbf{A} \cdot \mathbf{A}}$$

Example 1-2

Using the concept of the dot product, find the projection of vector $\mathbf{A}$ on $\mathbf{B}$.



► **Solution.**



Magnitude of the projection of vector $\mathbf{A}$ on $\mathbf{B}$:

$$A \cos \theta_{AB} = \frac{\mathbf{A} \cdot \mathbf{B}}{B}$$

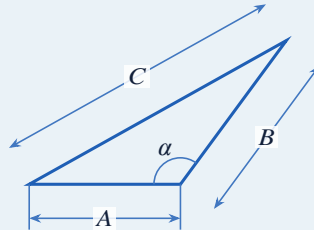
Projection of vector $\mathbf{A}$ on $\mathbf{B}$:

$$\frac{\mathbf{A} \cdot \mathbf{B}}{B} \mathbf{a}_B = \left(\frac{\mathbf{A} \cdot \mathbf{B}}{B} \right) \frac{\mathbf{B}}{B} = \frac{\mathbf{A} \cdot \mathbf{B}}{B^2} \mathbf{B}$$

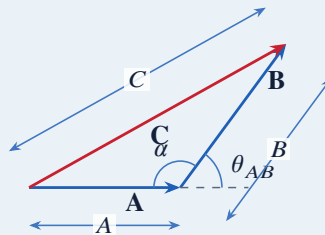
Example 1-3

Prove the law of cosines for a triangle using the concept of the dot product.

$$C = \sqrt{A^2 + B^2 - 2AB \cos \alpha}$$



► **Solution.**



$$\mathbf{C} = \mathbf{A} + \mathbf{B}$$

$$\begin{aligned} C^2 &= \mathbf{C} \cdot \mathbf{C} = (\mathbf{A} + \mathbf{B}) \cdot (\mathbf{A} + \mathbf{B}) \\ &= \mathbf{A} \cdot \mathbf{A} + \mathbf{B} \cdot \mathbf{B} + 2\mathbf{A} \cdot \mathbf{B} \\ &= A^2 + B^2 + 2AB \cos \theta_{AB} \end{aligned}$$

$$\cos \theta_{AB} = \cos(180^\circ - \alpha) = -\cos \alpha$$

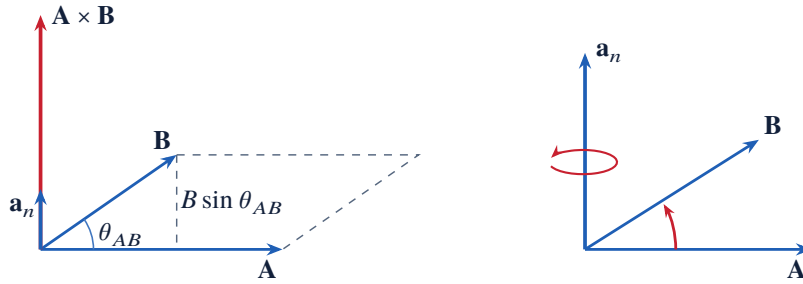
$$C^2 = A^2 + B^2 - 2AB \cos \alpha$$

1.8 Cross or Vector Product

Definition

$$\mathbf{A} \times \mathbf{B} \triangleq \mathbf{a}_n |AB \sin \theta_{AB}|$$

- The magnitude of this product equals the area of the parallelogram formed by the vectors **A** and **B**, and its direction is perpendicular to both vectors.



- Properties of the cross product

- The cross product of two vectors is a vector quantity.
- It is not commutative

$$\mathbf{B} \times \mathbf{A} = -\mathbf{A} \times \mathbf{B}$$

- It is distributive

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$$

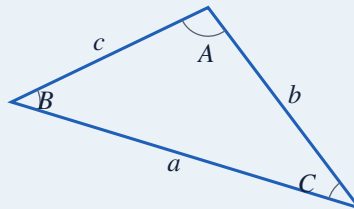
- It is not associative

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) \neq (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$$

Example 1-4

Prove the law of sines for a triangle using the concept of the cross product.

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$



► Solution.

Area of the triangle:

$$\left| \frac{1}{2} \mathbf{a} \times \mathbf{b} \right| = \left| \frac{1}{2} \mathbf{b} \times \mathbf{c} \right| = \left| \frac{1}{2} \mathbf{c} \times \mathbf{a} \right|$$

$$ab \sin C = bc \sin A = ca \sin B$$

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

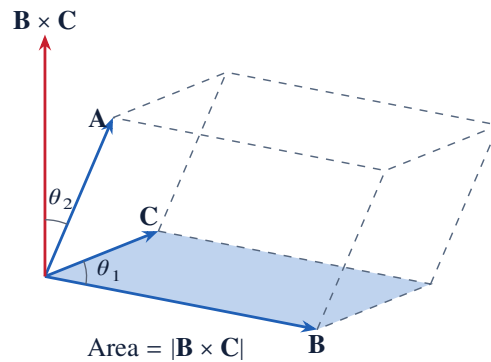
1.9 Product of Three Vectors

➤ Scalar triple product

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

➤ Its magnitude equals the volume of the parallelepiped formed by the three vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$.

- Area of the base: $|\mathbf{B} \times \mathbf{C}| = |BC \sin \theta_1|$
- Height: $|A \cos \theta_2|$
- Volume of the parallelepiped: $|ABC \sin \theta_1 \cos \theta_2|$



➤ Vector triple product

- Known as the back-cab rule

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

🔍 Example 1-5

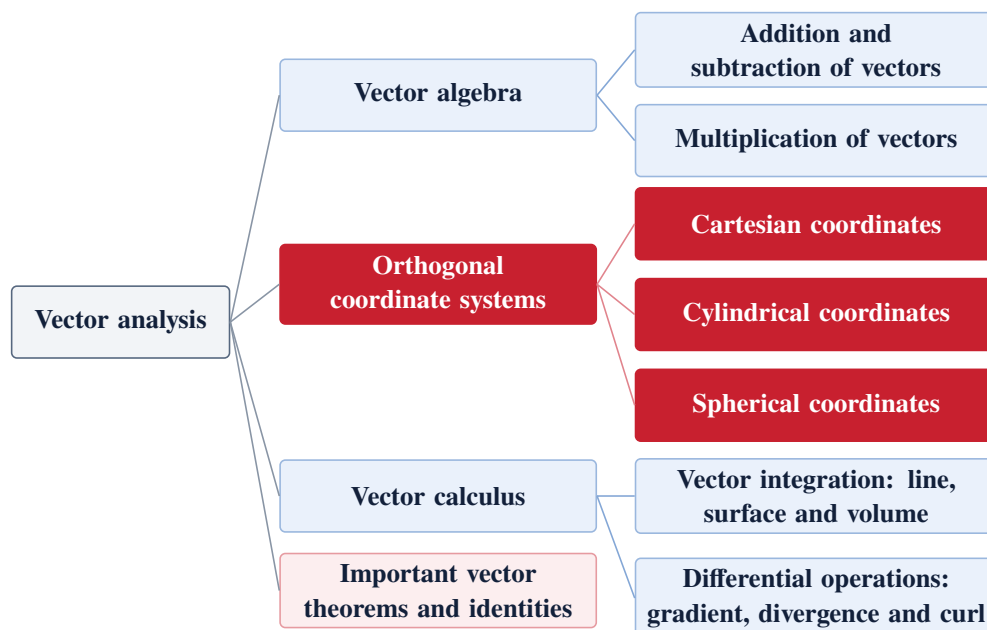
The expression $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$ is equal to which of the following expressions?

- $\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$
- $-\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) + \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$
- $-\mathbf{A}(\mathbf{C} \cdot \mathbf{B}) + \mathbf{B}(\mathbf{C} \cdot \mathbf{A})$
- $\mathbf{A}(\mathbf{C} \cdot \mathbf{B}) - \mathbf{B}(\mathbf{C} \cdot \mathbf{A})$

➤ Solution.

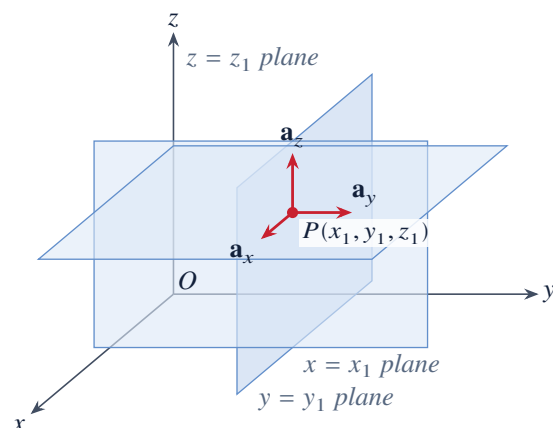
$$\begin{aligned} (\mathbf{A} \times \mathbf{B}) \times \mathbf{C} &= -\mathbf{C} \times (\mathbf{A} \times \mathbf{B}) \\ &= -\mathbf{A}(\mathbf{C} \cdot \mathbf{B}) + \mathbf{B}(\mathbf{C} \cdot \mathbf{A}) \end{aligned}$$

1.10 Orthogonal Coordinate Systems



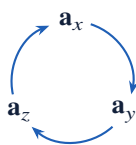
- What are the principal surfaces in the different coordinate systems?
- How are the unit vectors defined in the different coordinate systems?
- How are the position vector of a point and the distance vector between two points computed in the different coordinate systems?
- Determine the differential length, surface and volume in the different coordinate systems, for use in integration.
- Find the representation of the vector $\mathbf{A} = \mathbf{a}_r(3 \cos \phi) - \mathbf{a}_\phi 2r + \mathbf{a}_z 5$ in Cartesian coordinates.
- The position of point P in spherical coordinates is $(8, 120^\circ, 330^\circ)$. Express the position of this point in cylindrical and Cartesian coordinates.
- To describe the spatial variations of physical quantities, we must be able to describe all points of space uniquely and appropriately.
 - This requires the use of an appropriate coordinate system.
- The laws of electromagnetics do not depend on the coordinate system.
- For simplicity of the calculations, a coordinate system that suits the geometry of the problem is used.
- In three-dimensional space a point can be located at the intersection of three surfaces.
 - If these three surfaces are mutually perpendicular, we have an orthogonal coordinate system.
- The three orthogonal systems used in this course
 - Cartesian (rectangular) coordinates
 - Cylindrical coordinates
 - Spherical coordinates

1.11 Cartesian Coordinate System



- Every point $P(x_1, y_1, z_1)$ is the intersection of the three planes $x = x_1$, $y = y_1$ and $z = z_1$.
- Unit vector $\mathbf{a}_x \rightarrow$ the unit vector normal to the x -constant plane, in the positive direction
- Unit vector $\mathbf{a}_y \rightarrow$ the unit vector normal to the y -constant plane, in the positive direction
- Unit vector $\mathbf{a}_z \rightarrow$ the unit vector normal to the z -constant plane, in the positive direction
- Cross and dot products of the unit vectors

$$\mathbf{a}_x \times \mathbf{a}_y = \mathbf{a}_z, \quad \mathbf{a}_y \times \mathbf{a}_z = \mathbf{a}_x, \quad \mathbf{a}_z \times \mathbf{a}_x = \mathbf{a}_y$$



$$\mathbf{a}_x \cdot \mathbf{a}_y = \mathbf{a}_y \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_x = 0$$

$$\mathbf{a}_x \cdot \mathbf{a}_x = \mathbf{a}_y \cdot \mathbf{a}_y = \mathbf{a}_z \cdot \mathbf{a}_z = 1$$

- Representation of an arbitrary vector $\mathbf{A}$ in Cartesian coordinates

⚡ Important result

$$\mathbf{A} = \mathbf{a}_x A_x + \mathbf{a}_y A_y + \mathbf{a}_z A_z$$

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

► **Position vector**

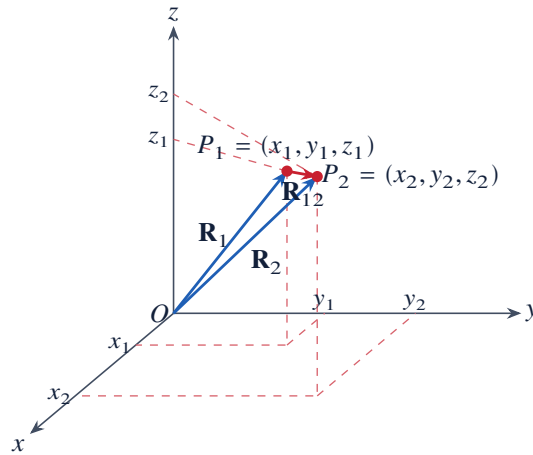
- The position vector of point P is the directed distance from the origin to P .

$$\mathbf{R}_1 = x_1 \mathbf{a}_x + y_1 \mathbf{a}_y + z_1 \mathbf{a}_z, \quad \mathbf{R}_2 = x_2 \mathbf{a}_x + y_2 \mathbf{a}_y + z_2 \mathbf{a}_z$$

► **Distance vector**

- The displacement vector from one point to another.

$$\mathbf{R}_{12} = \mathbf{R}_2 - \mathbf{R}_1 = (x_2 - x_1) \mathbf{a}_x + (y_2 - y_1) \mathbf{a}_y + (z_2 - z_1) \mathbf{a}_z$$



$$\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z, \quad \mathbf{B} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z$$

► **Dot product of the two vectors $\mathbf{A}$ and $\mathbf{B}$**

⚡ **Important result**

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$$

► **Cross product of the two vectors $\mathbf{A}$ and $\mathbf{B}$**

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= A_x B_x \mathbf{a}_x \times \mathbf{a}_x + A_x B_y \mathbf{a}_x \times \mathbf{a}_y + A_x B_z \mathbf{a}_x \times \mathbf{a}_z \\ &\quad + A_y B_x \mathbf{a}_y \times \mathbf{a}_x + A_y B_y \mathbf{a}_y \times \mathbf{a}_y + A_y B_z \mathbf{a}_y \times \mathbf{a}_z \\ &\quad + A_z B_x \mathbf{a}_z \times \mathbf{a}_x + A_z B_y \mathbf{a}_z \times \mathbf{a}_y + A_z B_z \mathbf{a}_z \times \mathbf{a}_z \\ &= (A_y B_z - A_z B_y) \mathbf{a}_x + (A_z B_x - A_x B_z) \mathbf{a}_y + (A_x B_y - A_y B_x) \mathbf{a}_z \end{aligned}$$

⚡ **Important result**

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

- Differential length → **vector**

$$d\ell = \mathbf{a}_x dx + \mathbf{a}_y dy + \mathbf{a}_z dz$$

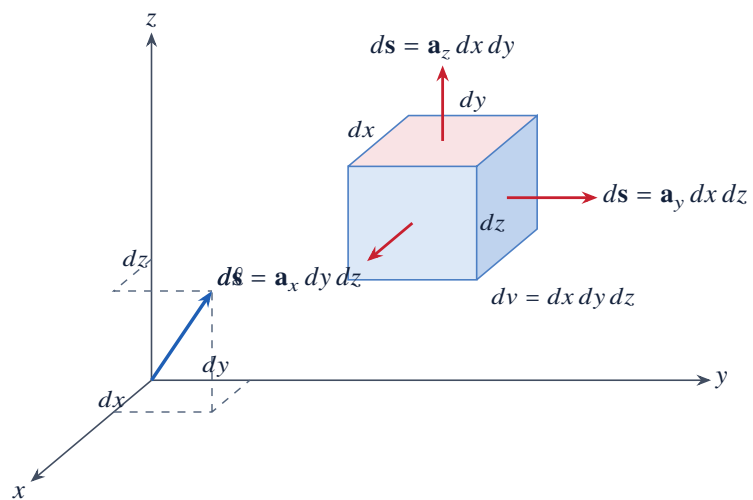
- Differential surface → **vector**

$$d\mathbf{S} = dS \mathbf{a}_n$$

$$d\mathbf{S} = dy dz \mathbf{a}_x, \quad d\mathbf{S} = dx dz \mathbf{a}_y, \quad d\mathbf{S} = dx dy \mathbf{a}_z$$

- Differential volume → **scalar**

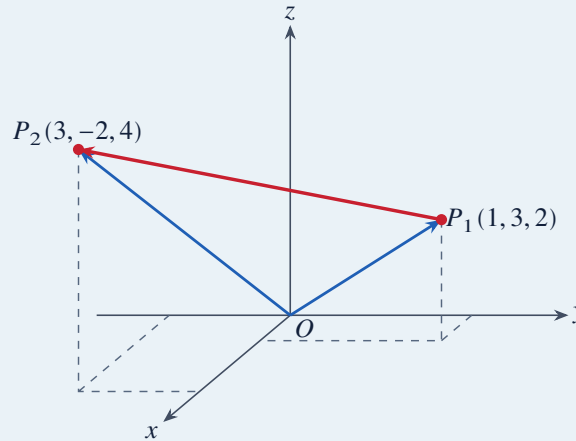
$$dv = dx dy dz$$



Example 1-6

- (a) Find the vector whose initial point is $P_1(1, 3, 2)$ and whose terminal point is $P_2(3, -2, 4)$.
 (b) What is the length of this vector?

► Solution.



$$\begin{aligned}
 \overrightarrow{P_1P_2} &= \overrightarrow{OP_2} - \overrightarrow{OP_1} \\
 (a) \quad &= (\mathbf{a}_x 3 - \mathbf{a}_y 2 + \mathbf{a}_z 4) - (\mathbf{a}_x + \mathbf{a}_y 3 + \mathbf{a}_z 2) \\
 &= \mathbf{a}_x 2 - \mathbf{a}_y 5 + \mathbf{a}_z 2 \\
 (b) \quad P_1P_2 &= |\overrightarrow{P_1P_2}| = \sqrt{2^2 + (-5)^2 + 2^2} = \sqrt{33}
 \end{aligned}$$

Example 1-7

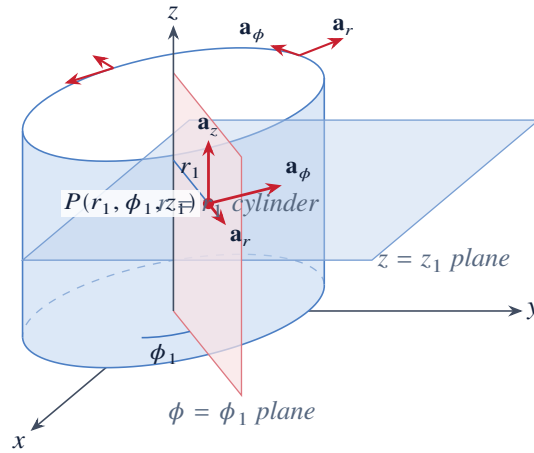
Given $\mathbf{A} = \mathbf{a}_x 5 - \mathbf{a}_y 2 + \mathbf{a}_z$, find the unit vector $\mathbf{B}$ such that

- (a) $\mathbf{B} \parallel \mathbf{A}$
- (b) $\mathbf{B}$ lies in the xy -plane and is perpendicular to $\mathbf{A}$

► Solution.

$$\begin{aligned}
 (a) \quad \mathbf{B} &= \frac{\mathbf{A}}{A} = \frac{\mathbf{a}_x 5 - \mathbf{a}_y 2 + \mathbf{a}_z}{\sqrt{5^2 + 2^2 + 1^2}} = \frac{\mathbf{a}_x 5 - \mathbf{a}_y 2 + \mathbf{a}_z}{\sqrt{30}} \\
 (b) \quad &\begin{cases} B = \sqrt{B_x^2 + B_y^2 + B_z^2} = 1 \\ B_z = 0 \\ \mathbf{B} \cdot \mathbf{A} = 5B_x - 2B_y = 0 \end{cases} \Rightarrow \begin{cases} B_x = \frac{2}{\sqrt{29}} \\ B_y = \frac{5}{\sqrt{29}} \end{cases}
 \end{aligned}$$

1.12 Cylindrical Coordinate System



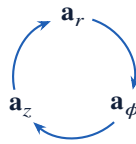
- Every point $P(r_1, \phi_1, z_1)$ is the intersection of the cylinder $r = r_1$, the half-plane $\phi = \phi_1$ and the plane $z = z_1$.
- Unit vector $\mathbf{a}_r \rightarrow$ the unit vector normal to the r -constant cylinder, in the positive direction
- Unit vector $\mathbf{a}_\phi \rightarrow$ the unit vector normal to the ϕ -constant half-plane, in the positive direction
- Unit vector $\mathbf{a}_z \rightarrow$ the unit vector normal to the z -constant plane, in the positive direction

① Remark

The directions of the vectors $\mathbf{a}_r$ and $\mathbf{a}_\phi$ change from point to point in space. This must be taken into account in integration.

- Cross and dot products of the unit vectors

$$\mathbf{a}_r \times \mathbf{a}_\phi = \mathbf{a}_z, \quad \mathbf{a}_\phi \times \mathbf{a}_z = \mathbf{a}_r, \quad \mathbf{a}_z \times \mathbf{a}_r = \mathbf{a}_\phi$$



$$\mathbf{a}_r \cdot \mathbf{a}_r = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = \mathbf{a}_z \cdot \mathbf{a}_z = 1$$

$$\mathbf{a}_r \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_r = 0$$

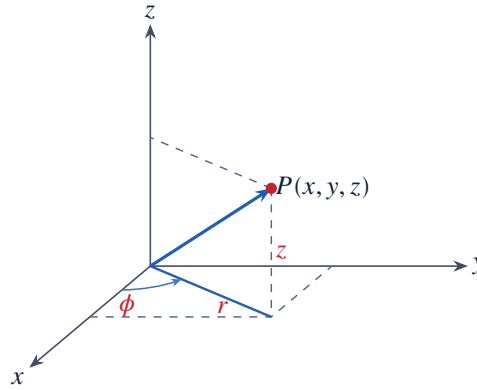
- Representation of an arbitrary vector $\mathbf{A}$ in cylindrical coordinates

⚡ Important result

$$\mathbf{A} = \mathbf{a}_r A_r + \mathbf{a}_\phi A_\phi + \mathbf{a}_z A_z$$

$$A = \sqrt{A_r^2 + A_\phi^2 + A_z^2}$$

► Position vector

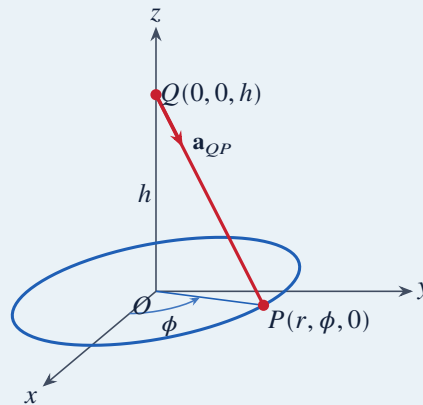


$$\mathbf{R} = r\mathbf{a}_r + z\mathbf{a}_z$$

Example 1-8

The cylindrical coordinates of an arbitrary point P are $(r, \phi, 0)$. Find the unit vector from the point $z = h$ on the z -axis toward the point P .

► Solution.



$$\overrightarrow{QP} = \overrightarrow{OP} - \overrightarrow{OQ} = (\mathbf{a}_r r) - (\mathbf{a}_z h)$$

$$\mathbf{a}_{QP} = \frac{\overrightarrow{QP}}{|\overrightarrow{QP}|} = \frac{1}{\sqrt{r^2 + h^2}} (\mathbf{a}_r r - \mathbf{a}_z h)$$

► Differential length → vector

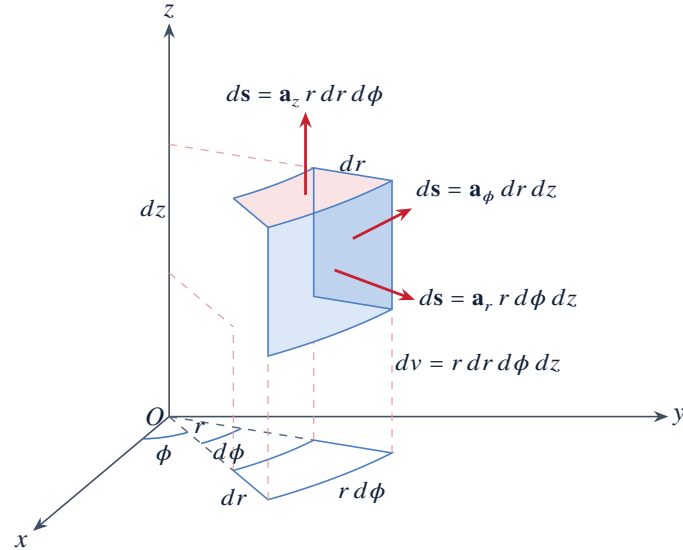
$$d\ell = \mathbf{a}_r dr + \mathbf{a}_\phi r d\phi + \mathbf{a}_z dz$$

► Differential surface → vector

$$d\mathbf{S} = r d\phi dz \mathbf{a}_r, \quad d\mathbf{S} = dr dz \mathbf{a}_\phi, \quad d\mathbf{S} = r dr d\phi \mathbf{a}_z$$

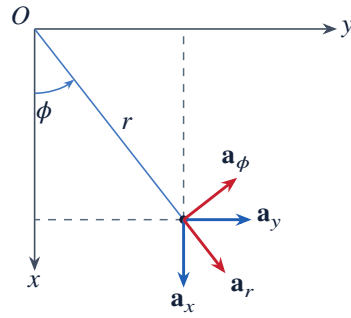
► Differential volume → scalar

$$dv = r dr d\phi dz$$



- Transformation of the representation of a vector from cylindrical to Cartesian coordinates

$$\mathbf{A} = \mathbf{a}_r A_r + \mathbf{a}_\phi A_\phi + \mathbf{a}_z A_z \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_x A_x + \mathbf{a}_y A_y + \mathbf{a}_z A_z$$



$$\mathbf{a}_r = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y, \quad \mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y, \quad \mathbf{a}_z = \mathbf{a}_z$$

$$\begin{aligned} \mathbf{A} &= (\cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y) A_r + (-\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y) A_\phi + \mathbf{a}_z A_z \\ &\Rightarrow \mathbf{A} = \mathbf{a}_x (\cos \phi A_r - \sin \phi A_\phi) + \mathbf{a}_y (\sin \phi A_r + \cos \phi A_\phi) + \mathbf{a}_z A_z \end{aligned}$$

- Transformation of the representation of a vector from Cartesian to cylindrical coordinates

$$\mathbf{A} = \mathbf{a}_x A_x + \mathbf{a}_y A_y + \mathbf{a}_z A_z \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_r A_r + \mathbf{a}_\phi A_\phi + \mathbf{a}_z A_z$$

$$\mathbf{a}_x = \cos \phi \mathbf{a}_r - \sin \phi \mathbf{a}_\phi, \quad \mathbf{a}_y = \sin \phi \mathbf{a}_r + \cos \phi \mathbf{a}_\phi, \quad \mathbf{a}_z = \mathbf{a}_z$$

$$\begin{aligned} \mathbf{A} &= (\cos \phi \mathbf{a}_r - \sin \phi \mathbf{a}_\phi) A_x + (\sin \phi \mathbf{a}_r + \cos \phi \mathbf{a}_\phi) A_y + \mathbf{a}_z A_z \\ &\Rightarrow \mathbf{A} = \mathbf{a}_r (\cos \phi A_x + \sin \phi A_y) + \mathbf{a}_\phi (-\sin \phi A_x + \cos \phi A_y) + \mathbf{a}_z A_z \end{aligned}$$

- Transformation of the representation of a vector from cylindrical to Cartesian coordinates

⚡ Important result

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix}$$

- Transformation of the representation of a vector from Cartesian to cylindrical coordinates

⚡ Important result

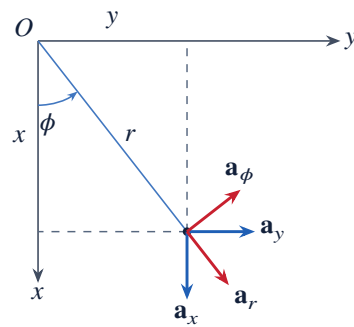
$$\begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

- Transformation from cylindrical to Cartesian coordinates

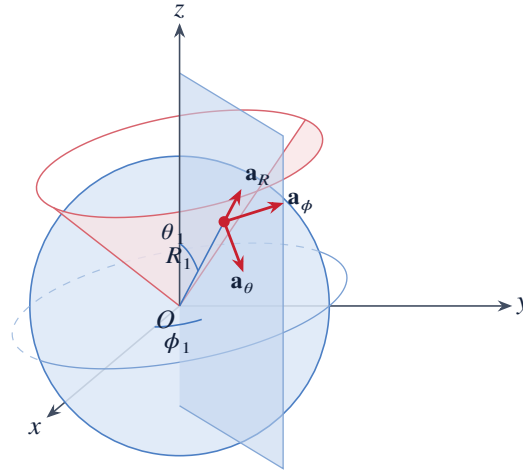
$$x = r \cos \phi, \quad y = r \sin \phi, \quad z = z$$

- Transformation from Cartesian to cylindrical coordinates

$$r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}, \quad z = z$$



1.13 Spherical Coordinate System



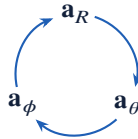
- Every point $P(R_1, \theta_1, \phi_1)$ is the intersection of the sphere $R = R_1$, the cone $\theta = \theta_1$ and the half-plane $\phi = \phi_1$
- Unit vector $\mathbf{a}_R \rightarrow$ the unit vector normal to the R -constant sphere, in the positive direction
- Unit vector $\mathbf{a}_\theta \rightarrow$ the unit vector normal to the θ -constant cone, in the positive direction
- Unit vector $\mathbf{a}_\phi \rightarrow$ the unit vector normal to the ϕ -constant half-plane, in the positive direction

Remark

The directions of the vectors $\mathbf{a}_R$, $\mathbf{a}_\theta$ and $\mathbf{a}_\phi$ change from point to point in space. This must be taken into account in integration.

- Cross and dot products of the unit vectors

$$\mathbf{a}_R \times \mathbf{a}_\theta = \mathbf{a}_\phi, \quad \mathbf{a}_\theta \times \mathbf{a}_\phi = \mathbf{a}_R, \quad \mathbf{a}_\phi \times \mathbf{a}_R = \mathbf{a}_\theta$$



$$\mathbf{a}_R \cdot \mathbf{a}_R = \mathbf{a}_\theta \cdot \mathbf{a}_\theta = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = 1$$

$$\mathbf{a}_R \cdot \mathbf{a}_\theta = \mathbf{a}_\theta \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_R = 0$$

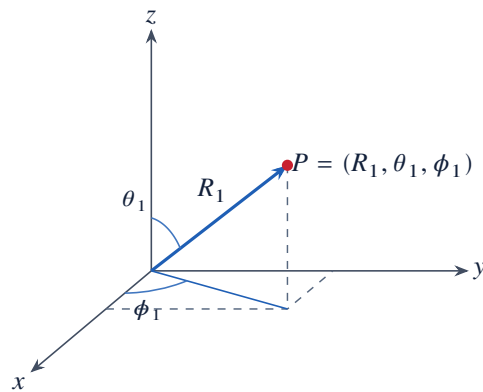
- Representation of an arbitrary vector $\mathbf{A}$ in spherical coordinates

⚡ Important result

$$\mathbf{A} = \mathbf{a}_R A_R + \mathbf{a}_\theta A_\theta + \mathbf{a}_\phi A_\phi$$

$$A = \sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$$

- Position vector



$$\mathbf{R} = R_1 \mathbf{a}_R$$

- Differential length → **vector**

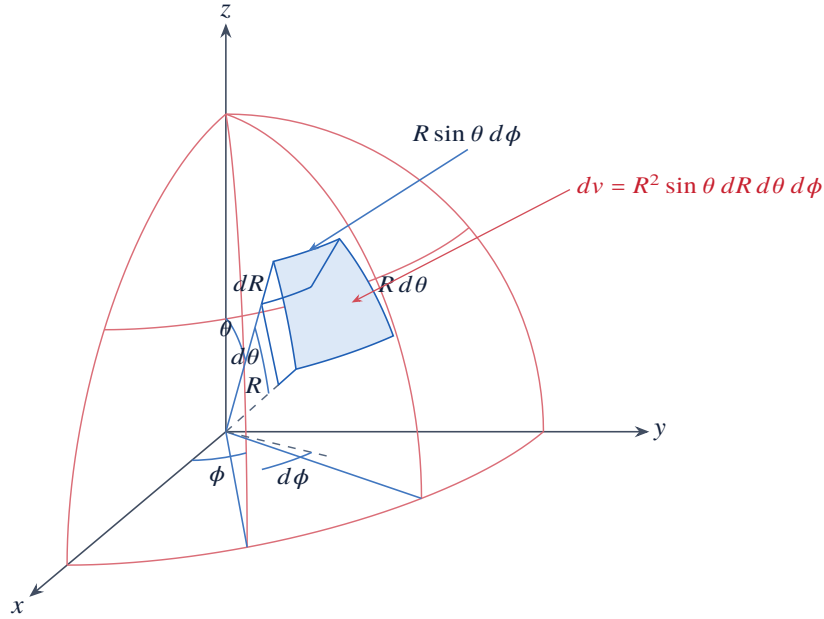
$$d\ell = \mathbf{a}_R dR + \mathbf{a}_\theta R d\theta + \mathbf{a}_\phi R \sin \theta d\phi$$

- Differential surface → **vector**

$$d\mathbf{S} = R^2 \sin \theta d\theta d\phi \mathbf{a}_R, \quad d\mathbf{S} = R \sin \theta dR d\phi \mathbf{a}_\theta, \quad d\mathbf{S} = R dR d\theta \mathbf{a}_\phi$$

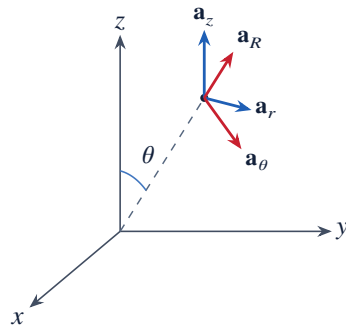
- Differential volume → **scalar**

$$dv = R^2 \sin \theta dR d\theta d\phi$$



- Transformation of the representation of a vector from spherical to cylindrical coordinates

$$\mathbf{A} = \mathbf{a}_R A_R + \mathbf{a}_\theta A_\theta + \mathbf{a}_\phi A_\phi \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_r A_r + \mathbf{a}_\phi A_\phi + \mathbf{a}_z A_z$$



$$\mathbf{a}_R = \sin \theta \mathbf{a}_r + \cos \theta \mathbf{a}_z, \quad \mathbf{a}_\theta = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_z, \quad \mathbf{a}_\phi = \mathbf{a}_\phi$$

$$\begin{aligned} \mathbf{A} &= (\sin \theta \mathbf{a}_r + \cos \theta \mathbf{a}_z) A_R \\ &\quad + (\cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_z) A_\theta \\ &\quad + \mathbf{a}_\phi A_\phi \end{aligned} \quad \Rightarrow \quad \begin{aligned} \mathbf{A} &= \mathbf{a}_r (\sin \theta A_R + \cos \theta A_\theta) \\ &\quad + \mathbf{a}_\phi A_\phi \\ &\quad + \mathbf{a}_z (\cos \theta A_R - \sin \theta A_\theta) \end{aligned}$$

- Transformation of the representation of a vector from cylindrical to spherical coordinates

$$\mathbf{A} = \mathbf{a}_r A_r + \mathbf{a}_\phi A_\phi + \mathbf{a}_z A_z \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_R A_R + \mathbf{a}_\theta A_\theta + \mathbf{a}_\phi A_\phi$$

$$\mathbf{a}_r = \sin \theta \mathbf{a}_R + \cos \theta \mathbf{a}_\theta, \quad \mathbf{a}_\phi = \mathbf{a}_\phi, \quad \mathbf{a}_z = \cos \theta \mathbf{a}_R - \sin \theta \mathbf{a}_\theta$$

$$\begin{aligned} \mathbf{A} &= (\sin \theta \mathbf{a}_R + \cos \theta \mathbf{a}_\theta) A_r \\ &\quad + \mathbf{a}_\phi A_\phi \\ &\quad + (\cos \theta \mathbf{a}_R - \sin \theta \mathbf{a}_\theta) A_z \end{aligned} \quad \Rightarrow \quad \begin{aligned} \mathbf{A} &= \mathbf{a}_R (\sin \theta A_r + \cos \theta A_z) \\ &\quad + \mathbf{a}_\theta (\cos \theta A_r - \sin \theta A_z) \\ &\quad + \mathbf{a}_\phi A_\phi \end{aligned}$$

- Transformation of the representation of a vector from spherical to cylindrical coordinates

⚡ Important result

$$\begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_R \\ A_\theta \\ A_\phi \end{bmatrix}$$

- Transformation of the representation of a vector from cylindrical to spherical coordinates

⚡ Important result

$$\begin{bmatrix} A_R \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta & 0 & \cos \theta \\ \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix}$$

- Transformation of the representation of a vector from Cartesian to spherical coordinates

$$\mathbf{A} = \mathbf{a}_x A_x + \mathbf{a}_y A_y + \mathbf{a}_z A_z \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_R A_R + \mathbf{a}_\theta A_\theta + \mathbf{a}_\phi A_\phi$$

$$\begin{cases} \mathbf{a}_x = \cos \phi \mathbf{a}_r - \sin \phi \mathbf{a}_\phi \\ \mathbf{a}_y = \sin \phi \mathbf{a}_r + \cos \phi \mathbf{a}_\phi \\ \mathbf{a}_z = \mathbf{a}_z \end{cases} \quad \begin{cases} \mathbf{a}_r = \sin \theta \mathbf{a}_R + \cos \theta \mathbf{a}_\theta \\ \mathbf{a}_\phi = \mathbf{a}_\phi \\ \mathbf{a}_z = \cos \theta \mathbf{a}_R - \sin \theta \mathbf{a}_\theta \end{cases}$$

$$\Rightarrow \begin{cases} \mathbf{a}_x = \sin \theta \cos \phi \mathbf{a}_R + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi \\ \mathbf{a}_y = \sin \theta \sin \phi \mathbf{a}_R + \cos \theta \sin \phi \mathbf{a}_\theta + \cos \phi \mathbf{a}_\phi \\ \mathbf{a}_z = \cos \theta \mathbf{a}_R - \sin \theta \mathbf{a}_\theta \end{cases}$$

⚡ Important result

$$\begin{bmatrix} A_R \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

- Transformation of the representation of a vector from spherical to Cartesian coordinates

$$\mathbf{A} = \mathbf{a}_R A_R + \mathbf{a}_\theta A_\theta + \mathbf{a}_\phi A_\phi \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_x A_x + \mathbf{a}_y A_y + \mathbf{a}_z A_z$$

$$\begin{cases} \mathbf{a}_R = \sin \theta \mathbf{a}_r + \cos \theta \mathbf{a}_z \\ \mathbf{a}_\theta = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_z \\ \mathbf{a}_\phi = \mathbf{a}_\phi \end{cases} \quad \begin{cases} \mathbf{a}_r = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y \\ \mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y \\ \mathbf{a}_z = \mathbf{a}_z \end{cases}$$

$$\Rightarrow \begin{cases} \mathbf{a}_R = \sin \theta \cos \phi \mathbf{a}_x + \sin \theta \sin \phi \mathbf{a}_y + \cos \theta \mathbf{a}_z \\ \mathbf{a}_\theta = \cos \theta \cos \phi \mathbf{a}_x + \cos \theta \sin \phi \mathbf{a}_y - \sin \theta \mathbf{a}_z \\ \mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y \end{cases}$$

⚡ Important result

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_R \\ A_\theta \\ A_\phi \end{bmatrix}$$

- Transformation from cylindrical to spherical coordinates

$$R = \sqrt{r^2 + z^2}, \quad \theta = \tan^{-1} \frac{r}{z}, \quad \phi = \phi$$

- Transformation from spherical to cylindrical coordinates

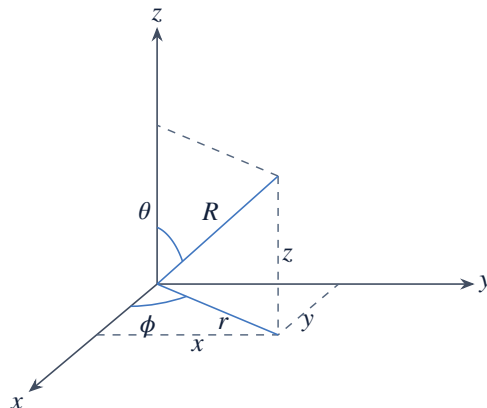
$$r = R \sin \theta, \quad \phi = \phi, \quad z = R \cos \theta$$

- Transformation from spherical to Cartesian coordinates

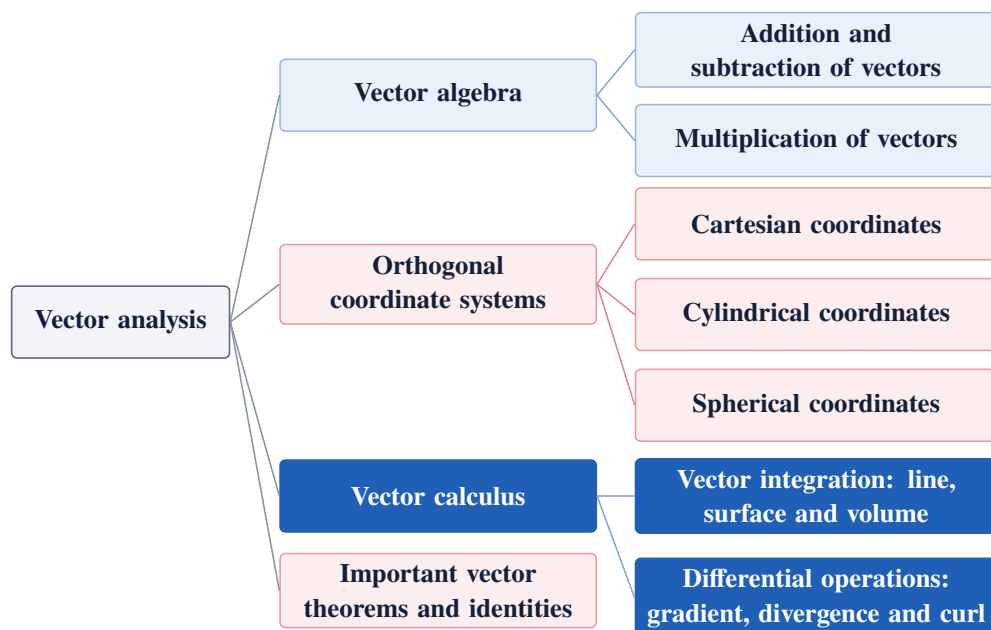
$$x = R \sin \theta \cos \phi, \quad y = R \sin \theta \sin \phi, \quad z = R \cos \theta$$

- Transformation from Cartesian to spherical coordinates

$$R = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z}, \quad \phi = \tan^{-1} \frac{y}{x}$$



1.14 Integrals Containing Vector Functions



- The kinds of integrals containing vector functions that we deal with

$$\begin{aligned}
 &\int_C \mathbf{F} d\ell \quad \iint_S \mathbf{F} ds \quad \iiint_V \mathbf{F} dv \\
 &\int_C V d\ell \\
 &\int_C \mathbf{F} \cdot d\ell \quad \iint_S \mathbf{A} \cdot ds \\
 &\int_C \mathbf{F} \times d\ell
 \end{aligned}$$

Remark

Important points about integration involving vector functions

- Using an appropriate coordinate system
- Correct determination of the differential length, surface and volume
- Attention to the variation of the unit vectors $\mathbf{a}_r$, $\mathbf{a}_R$, $\mathbf{a}_\theta$ and $\mathbf{a}_\phi$ over the range of integration

- The integral of a vector function over a specified curve, surface and volume

$$\int_C \mathbf{F} d\ell \quad \iint_S \mathbf{F} ds \quad \iiint_V \mathbf{F} dv$$

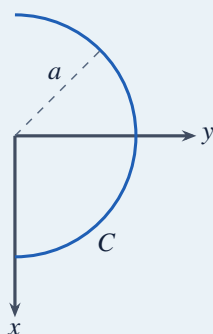
- The result is a vector.
- Important applications in electromagnetics: computing the electric field intensity due to a line, surface and volume charge distribution

$$\begin{cases} \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int_C \frac{\rho_\ell d\ell' (\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3} \\ \mathbf{E} = \frac{1}{4\pi\epsilon_0} \iint_S \frac{\rho_s ds' (\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3} \\ \mathbf{E} = \frac{1}{4\pi\epsilon_0} \iiint_V \frac{\rho_v dv' (\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3} \end{cases}$$

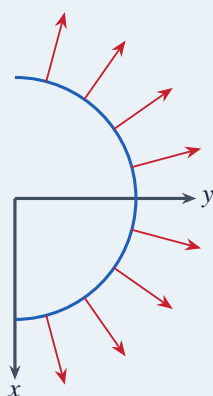
Example 1-9

Evaluate the following integral over the curve C .

$$\int_C \mathbf{F} d\ell, \quad \mathbf{F} = 3a\mathbf{a}_r$$



► **Solution.**



$$d\ell = r d\phi = a d\phi$$

$$\int_C \mathbf{F} d\ell = \int_0^\pi 3a\mathbf{a}_r a d\phi = 3a \int_0^\pi (\cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y) d\phi = 6a\mathbf{a}_y$$

- ▶ The integral of a scalar function along a specified path

$$\int_C V \, d\ell$$

- ▶ The result is a vector.
- ▶ Important applications in electromagnetics: computing the magnetic potential due to a line current

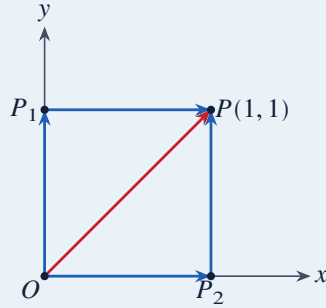
$$\mathbf{A} = \frac{\mu_0 I}{4\pi} \oint_C \frac{d\ell'}{|\mathbf{R} - \mathbf{R}'|}$$

Example 1-10

Evaluate the following integral along the following paths

$$\int_O^P r^2 d\ell, \quad r^2 = x^2 + y^2$$

- Path OP
- Path OP_1P
- Path OP_2P



➤ Solution.

- Along the path OP

$$\begin{aligned} \int_O^P r^2 d\ell &= \mathbf{a}_r \int_0^{\sqrt{2}} r^2 dr = \mathbf{a}_r \frac{2\sqrt{2}}{3} \\ &= \frac{2\sqrt{2}}{3} (\mathbf{a}_x \cos 45^\circ + \mathbf{a}_y \sin 45^\circ) \\ &= \mathbf{a}_x \frac{2}{3} + \mathbf{a}_y \frac{2}{3} \end{aligned}$$

- Along the path OP_1P

$$\begin{aligned} \int_O^P (x^2 + y^2) d\ell &= \mathbf{a}_y \int_O^{P_1} y^2 dy + \mathbf{a}_x \int_{P_1}^P (x^2 + 1) dx \\ &= \mathbf{a}_y \frac{1}{3} y^3 \Big|_0^1 + \mathbf{a}_x \left(\frac{1}{3} x^3 + x \right) \Big|_0^1 \\ &= \mathbf{a}_x \frac{4}{3} + \mathbf{a}_y \frac{1}{3} \end{aligned}$$

- Along the path OP_2P

$$\begin{aligned} \int_O^P (x^2 + y^2) d\ell &= \mathbf{a}_x \int_O^{P_2} x^2 dx + \mathbf{a}_y \int_{P_2}^P (1 + y^2) dy \\ &= \mathbf{a}_x \frac{1}{3} x^3 \Big|_0^1 + \mathbf{a}_y \left(y + \frac{1}{3} y^3 \right) \Big|_0^1 \\ &= \mathbf{a}_x \frac{1}{3} + \mathbf{a}_y \frac{4}{3} \end{aligned}$$

► Scalar line integral

$$\int_C \mathbf{F} \cdot d\ell$$

► The result is a scalar.

► If $\mathbf{F}$ is a force

- the work done by the force in moving a body from point P_1 to point P_2 along the specified path C

► Important applications in electromagnetics

- Computing the electric potential difference between the points P_1 and P_2 :

$$V = - \int_{P_1}^{P_2} \mathbf{E} \cdot d\ell$$

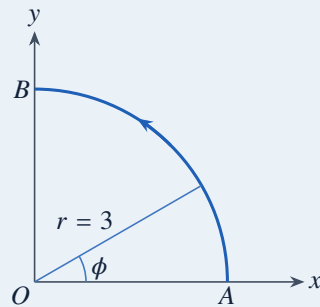
- Ampère's circuital law:

$$\mu_0 I = - \oint_C \mathbf{B} \cdot d\ell$$

Example 1-11

Evaluate the following integral along the quarter-circle path of the figure.

$$\int_A^B \mathbf{F} \cdot d\ell, \quad \mathbf{F} = \mathbf{a}_x xy - \mathbf{a}_y 2x$$



► Solution.

$$\begin{bmatrix} F_r \\ F_\phi \\ F_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} xy \\ -2x \\ 0 \end{bmatrix}$$

$$\mathbf{F} = \mathbf{a}_r(xy \cos \phi - 2x \sin \phi) - \mathbf{a}_\phi(xy \sin \phi + 2x \cos \phi)$$

$$d\ell = \mathbf{a}_\phi 3 d\phi \quad \mathbf{F} \cdot d\ell = -3(xy \sin \phi + 2x \cos \phi) d\phi$$

$$x = 3 \cos \phi \quad y = 3 \sin \phi$$

$$\int_A^B \mathbf{F} \cdot d\ell = \int_0^{\pi/2} -3(9 \sin^2 \phi \cos \phi + 6 \cos^2 \phi) d\phi = -9 \left(1 + \frac{\pi}{2} \right)$$

- Scalar surface integral

$$\iint_S \mathbf{A} \cdot d\mathbf{s}$$

- The result is a scalar.
- Result of the integral → the flux of the vector field $\mathbf{A}$ passing through the surface S
- Vector differential surface

$$d\mathbf{s} = \mathbf{a}_n ds$$

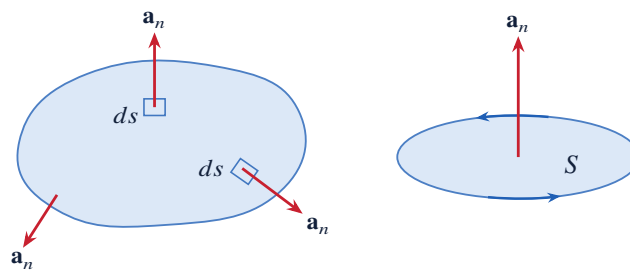


Figure 1.2 | Direction of $\mathbf{a}_n$ for a closed surface: normal to the surface, pointing out of the volume; for an open surface: depends on the direction of travel along the rim of the open surface, by the right-hand rule

- Important applications in electromagnetics

- Gauss's law in free space:

$$\oiint_S \mathbf{E} \cdot d\mathbf{s} = \frac{Q}{\epsilon_0}$$

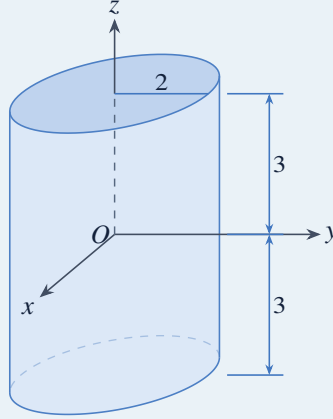
- Computing the electric current:

$$\iint_S \mathbf{J} \cdot d\mathbf{s} = I$$

Example 1-12

Evaluate the following integral over the closed surface about the z -axis specified by $z = \pm 3$ and $r = 2$

$$\oint_S \mathbf{F} \cdot d\mathbf{s}, \quad \mathbf{F} = \mathbf{a}_r \frac{k_1}{r} + \mathbf{a}_z k_2 z$$



► Solution.

$$\oint_S \mathbf{F} \cdot d\mathbf{s} = \oint_S \mathbf{F} \cdot \mathbf{a}_n ds = \int_{\text{top face}} \mathbf{F} \cdot \mathbf{a}_n ds + \int_{\text{bottom face}} \mathbf{F} \cdot \mathbf{a}_n ds + \int_{\text{side wall}} \mathbf{F} \cdot \mathbf{a}_n ds$$

► Top face

$$z = 3, \quad \mathbf{a}_n = \mathbf{a}_z, \quad \mathbf{F} \cdot \mathbf{a}_n = k_2 z = 3k_2, \quad ds = r dr d\phi$$

$$\int_{\text{top face}} \mathbf{F} \cdot \mathbf{a}_n ds = \int_0^{2\pi} \int_0^2 3k_2 r dr d\phi = 12\pi k_2$$

► Bottom face

$$z = -3, \quad \mathbf{a}_n = -\mathbf{a}_z, \quad \mathbf{F} \cdot \mathbf{a}_n = -k_2 z = 3k_2, \quad ds = r dr d\phi$$

$$\int_{\text{bottom face}} \mathbf{F} \cdot \mathbf{a}_n ds = 12\pi k_2$$

► Side wall

$$r = 2, \quad \mathbf{a}_n = \mathbf{a}_r, \quad \mathbf{F} \cdot \mathbf{a}_n = \frac{k_1}{r} = \frac{k_1}{2}, \quad ds = r d\phi dz = 2 d\phi dz$$

$$\int_{\text{side wall}} \mathbf{F} \cdot \mathbf{a}_n ds = \int_{-3}^3 \int_0^{2\pi} k_1 d\phi dz = 12\pi k_1$$

$$\oint_S \mathbf{F} \cdot d\mathbf{s} = 12\pi k_2 + 12\pi k_2 + 12\pi k_1 = 12\pi(k_1 + 2k_2)$$

► Vector line integral

$$\int_C \mathbf{F} \times d\ell$$

- The result is a vector.
- Important applications in electromagnetics

– Biot–Savart law:

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \oint_C \frac{d\ell \times (\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3}$$

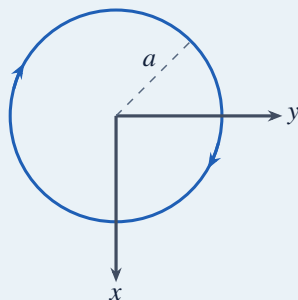
– Computing the magnetic force on a current-carrying wire in a magnetic field:

$$\mathbf{F} = I \oint_C d\ell \times \mathbf{B}$$

Example 1-13

Evaluate the following integral along the path C .

$$\int_C \mathbf{F} \times d\ell, \quad \mathbf{F} = 2\mathbf{a}_z$$



► Solution.

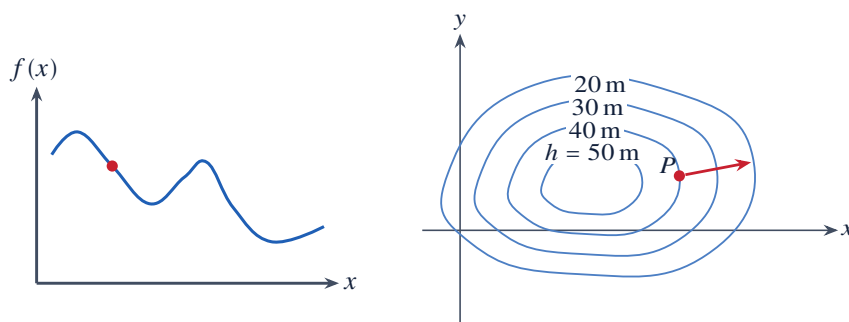
$$d\ell = r d\phi \mathbf{a}_\phi = a d\phi \mathbf{a}_\phi$$

$$\begin{aligned} \int_C \mathbf{F} \times d\ell &= \int_{2\pi}^0 2\mathbf{a}_z \times a d\phi \mathbf{a}_\phi = -2a \int_{2\pi}^0 \mathbf{a}_r d\phi \\ &= -2a \int_{2\pi}^0 (\cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y) d\phi = 0 \end{aligned}$$

1.15 Differential Operations - Gradient

► Prerequisite concepts for the discussion of the gradient

- The concept of the derivative of a function of one variable
 - The rate of change of the function at the point of interest
- The concept of the directional derivative of a function of several variables
 - The rate of change of the function at the point of interest and in a specified direction



△ Definition | Gradient

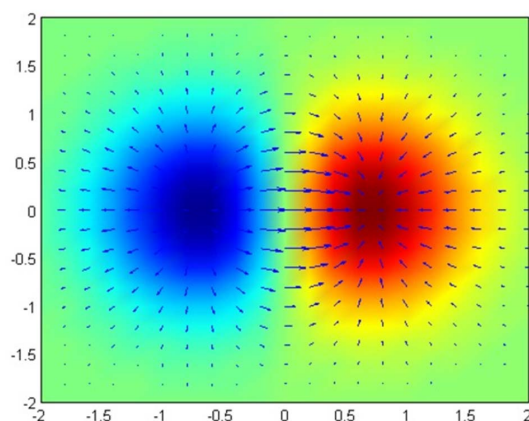
The gradient of a **scalar quantity** is a **vector quantity** that lies in the direction in which the scalar quantity increases most, and whose magnitude equals the maximum rate of increase of the scalar quantity.

$$\text{grad } v = \nabla v = \mathbf{a}_n \frac{dv}{dn}$$

- The directional derivative of v in the direction of $d\ell$:

$$\frac{dv}{d\ell} = \nabla v \cdot \mathbf{a}_\ell$$

- The gradient of v is normal to the surfaces of constant v .



- The gradient in Cartesian coordinates is as follows

⚡ Important result

$$\nabla v = \frac{\partial v}{\partial x} \mathbf{a}_x + \frac{\partial v}{\partial y} \mathbf{a}_y + \frac{\partial v}{\partial z} \mathbf{a}_z$$

- The general expression of the gradient

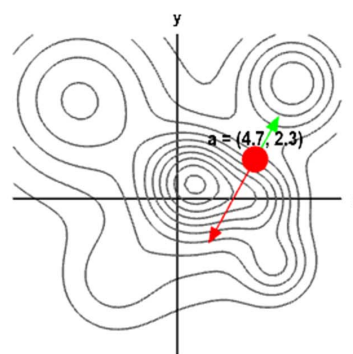
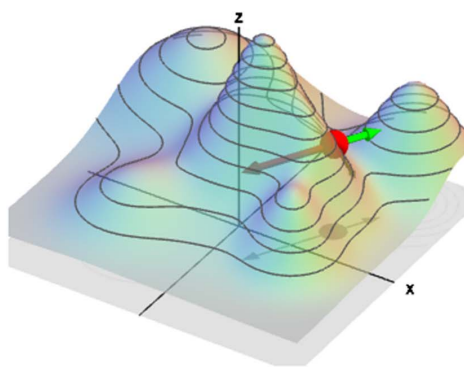
⚡ Important result

$$\nabla V = \mathbf{a}_{u_1} \frac{\partial V}{h_1 \partial u_1} + \mathbf{a}_{u_2} \frac{\partial V}{h_2 \partial u_2} + \mathbf{a}_{u_3} \frac{\partial V}{h_3 \partial u_3}$$

Metric coefficients	Cartesian	Cylindrical	Spherical
h_1	1	1	1
h_2	1	r	R
h_3	1	1	$R \sin \theta$

- **The concept of the gradient**

- Representing a hill by contours of height $\rightarrow h(x, y)$
- If a ball is placed at point P , in which direction does it move and what is the magnitude of the force acting on it?
 - The answer is proportional to $-\nabla h$.



Example 1-14

The electric field intensity $\mathbf{E}$ can be obtained as the negative of the gradient of the scalar electric potential V . Find $\mathbf{E}$ at the point $(1, 1, 0)$ if

$$V = E_0 R \cos \theta$$

► **Solution.**

$$\mathbf{E} = -\nabla V$$

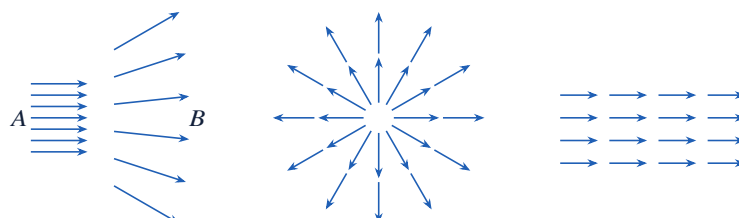
$$\begin{aligned} \mathbf{E} &= - \left[\mathbf{a}_R \frac{\partial}{\partial R} + \mathbf{a}_\theta \frac{1}{R} \frac{\partial}{\partial \theta} + \mathbf{a}_\phi \frac{1}{R \sin \theta} \frac{\partial}{\partial \phi} \right] E_0 R \cos \theta \\ &= -(\mathbf{a}_R \cos \theta - \mathbf{a}_\theta \sin \theta) E_0 \end{aligned}$$

$$\mathbf{E} = -\mathbf{a}_z E_0$$

1.16 Differential Operations - Divergence

► Prerequisite concepts for the discussion of the divergence

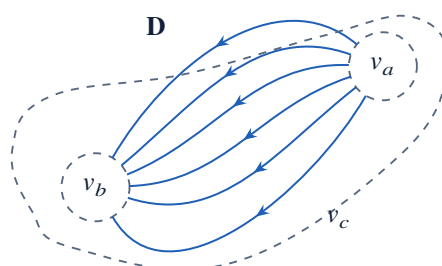
- In studying vector fields it is easier to represent the variations of the field graphically by flux lines. These are directed lines or curves that indicate the direction of the vector field at each point.



- The flux of a vector field is analogous to the flow of a fluid such as water.
- If $\mathbf{A}$ is a flux density vector, the total flux leaving the surface S equals

$$\oint_S \mathbf{A} \cdot d\mathbf{s}$$

- In a volume with a closed surface, there will be an excess of inward or outward flux only when the volume contains a sink or a source, respectively.
 - The net outward flux from v_a is positive: there is a source in the volume v_a
 - The net outward flux from v_b is negative: there is a sink in the volume v_b
 - The net outward flux from v_c is zero.



≡ Definition | Divergence

The divergence of a **vector field** $\mathbf{A}$ at a point is the net outward flux of $\mathbf{A}$ per unit volume as the volume about the point tends to zero

$$\text{div } \mathbf{A} \triangleq \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{s}}{\Delta v}$$

- In Cartesian coordinates the divergence is as follows

⚡ Important result

$$\operatorname{div} \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \quad \nabla \cdot \mathbf{A} \equiv \operatorname{div} \mathbf{A}$$

- The general expression of the divergence

⚡ Important result

$$\nabla \cdot \mathbf{A} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (h_2 h_3 A_1) + \frac{\partial}{\partial u_2} (h_1 h_3 A_2) + \frac{\partial}{\partial u_3} (h_1 h_2 A_3) \right]$$

🔗 Example 1-15

The magnetic flux density $\mathbf{B}$ outside a long current-carrying wire is circular and inversely proportional to the distance from the axis of the wire. Find the divergence of $\mathbf{B}$.

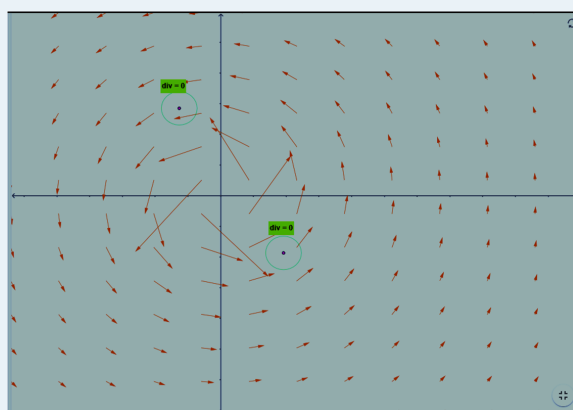
► Solution.

- Assuming the wire coincides with the z -axis

$$\mathbf{B} = \mathbf{a}_\phi \frac{k}{r}$$

$$\nabla \cdot \mathbf{B} = \frac{1}{r} \frac{\partial}{\partial r} (r B_r) + \frac{1}{r} \frac{\partial B_\phi}{\partial \phi} + \frac{\partial B_z}{\partial z}$$

$$\nabla \cdot \mathbf{B} = 0$$



1.17 Differential Operations - Curl

► Prerequisite concepts for the discussion of the curl

- The circulation of a vector field around a closed path

$$\text{Circulation of } \mathbf{A} \text{ around contour } C \triangleq \oint_C \mathbf{A} \cdot d\boldsymbol{\ell}$$

- The physical meaning of the circulation depends on the kind of field that the vector $\mathbf{A}$ represents.
 - If $\mathbf{A}$ is the force acting on a body, the circulation of $\mathbf{A}$ is the work done by the force in moving the body around the path.

$\triangleq$ Definition | Curl

The curl of a **vector field** $\mathbf{A}$ at a point is a **vector** whose magnitude is the maximum net circulation of $\mathbf{A}$ per unit area as the area about the point tends to zero, and whose direction is the direction of the normal to the area when the area is oriented so as to make the net circulation maximum.

$$\text{curl } \mathbf{A} \equiv \nabla \times \mathbf{A} \triangleq \lim_{\Delta s \rightarrow 0} \frac{1}{\Delta s} \left[\mathbf{a}_n \oint_C \mathbf{A} \cdot d\boldsymbol{\ell} \right]_{\max}$$

- In Cartesian coordinates the curl is as follows

⚡ Important result

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

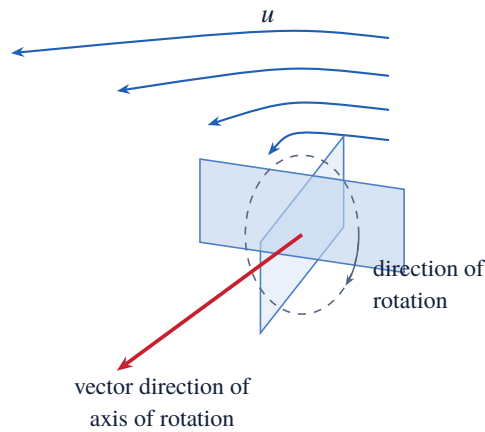
- The general expression of the curl

⚡ Important result

$$\nabla \times \mathbf{A} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} \mathbf{a}_{u_1} h_1 & \mathbf{a}_{u_2} h_2 & \mathbf{a}_{u_3} h_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 A_1 & h_2 A_2 & h_3 A_3 \end{vmatrix}$$

► **The concept of the curl**

- The rotation of a paddle wheel in a stream of water with velocity vector u
 - The maximum angular velocity of the paddle wheel is proportional to the magnitude of the curl of the velocity vector.
 - The axis of the wheel, by the right-hand rule, lies in the direction of the curl.



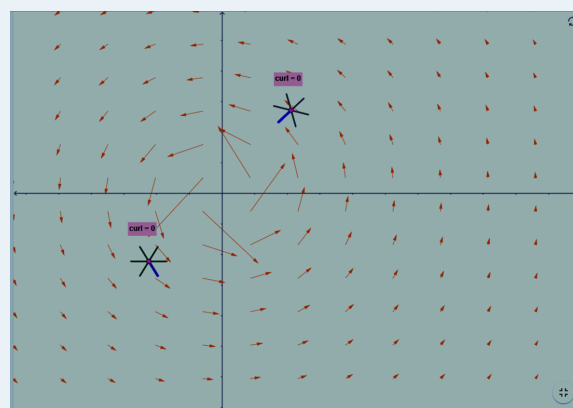
Example 1-16

Find the curl of the following vector

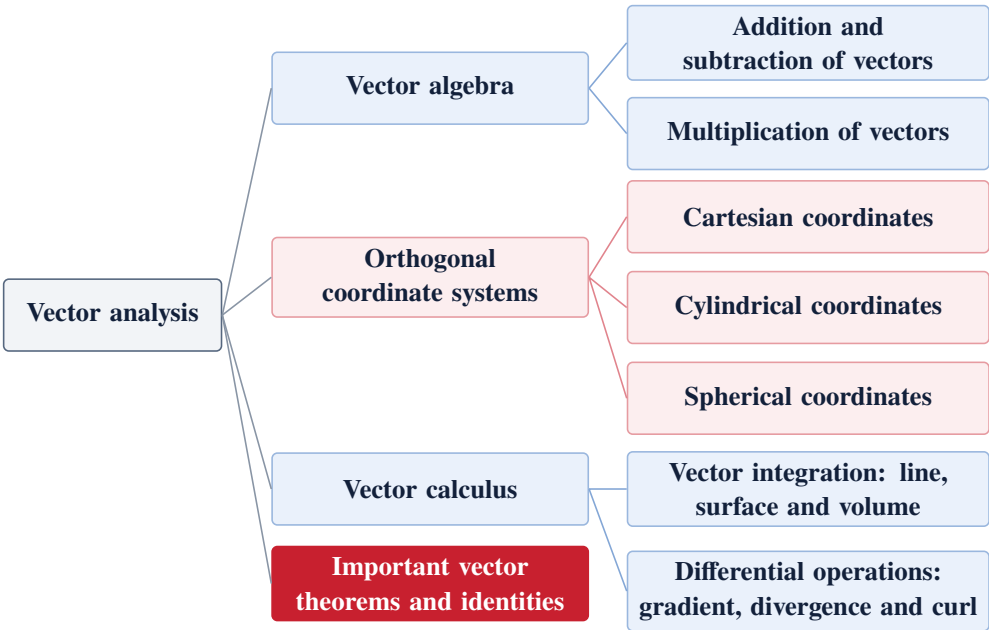
$$\mathbf{A} = \mathbf{a}_\phi \left(\frac{k}{r} \right)$$

► **Solution.**

$$\nabla \times \mathbf{A} = \frac{1}{r} \begin{vmatrix} \mathbf{a}_r & \mathbf{a}_\phi r & \mathbf{a}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ 0 & k & 0 \end{vmatrix} = 0$$



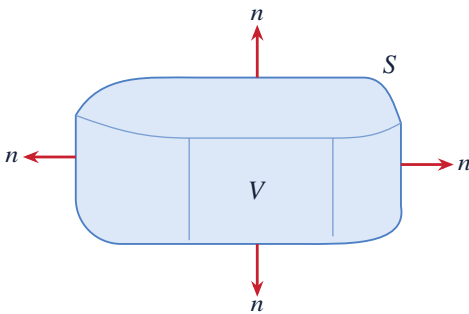
1.18 Important Vector Theorems



▽ Theorem | Divergence theorem

The volume integral of the divergence of a vector field equals the total outward flux of the vector through the surface enclosing the volume.

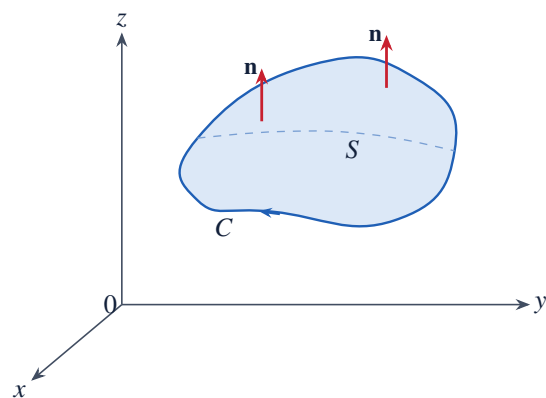
$$\int_V \nabla \cdot \mathbf{A} \, dv = \oint_S \mathbf{A} \cdot d\mathbf{s}$$



▽ Theorem | Stokes's theorem

The surface integral of the curl of a vector field over an open surface equals the closed line integral of the vector along the contour bounding the surface.

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell}$$

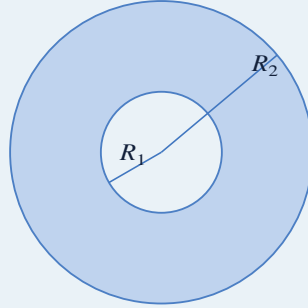
**▽ Theorem** | Helmholtz's theorem

A vector field is determined to within an additive constant if both its divergence and its curl are specified.

Example 1-17

Verify the divergence theorem for the shell region enclosed by the spherical surfaces $R = R_1$ and $R = R_2$ ($R_2 > R_1$) centred at the origin, for the following vector field.

$$\mathbf{F} = \mathbf{a}_R kR$$



► **Solution.**

$$\int_V \nabla \cdot \mathbf{A} dv = \oint_S \mathbf{A} \cdot d\mathbf{s}$$

► For the outer surface

$$R = R_2, \quad d\mathbf{s} = \mathbf{a}_R R_2^2 \sin \theta d\theta d\phi$$

$$\int_{\text{outer surface}} \mathbf{F} \cdot d\mathbf{s} = \int_0^{2\pi} \int_0^\pi (kR_2) R_2^2 \sin \theta d\theta d\phi = 4\pi k R_2^3$$

► For the inner surface

$$R = R_1, \quad d\mathbf{s} = -\mathbf{a}_R R_1^2 \sin \theta d\theta d\phi$$

$$\int_{\text{inner surface}} \mathbf{F} \cdot d\mathbf{s} = - \int_0^{2\pi} \int_0^\pi (kR_1) R_1^2 \sin \theta d\theta d\phi = -4\pi k R_1^3$$

$$\oint_S \mathbf{F} \cdot d\mathbf{s} = 4\pi k (R_2^3 - R_1^3)$$

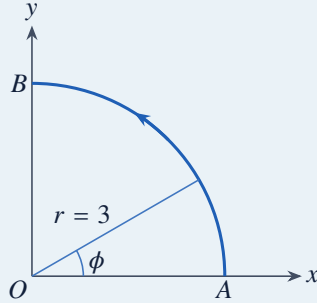
$$\nabla \cdot \mathbf{F} = \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 F_R) = \frac{1}{R^2} \frac{\partial}{\partial R} (kR^3) = 3k$$

$$\int_V \nabla \cdot \mathbf{F} dv = (\nabla \cdot \mathbf{F}) V = 4\pi k (R_2^3 - R_1^3)$$

Example 1-18

Verify Stokes's theorem over a quarter of a circular disk of radius 3 in the first quadrant for the following vector field.

$$\mathbf{F} = \mathbf{a}_x xy - \mathbf{a}_y 2x$$



► **Solution.**

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell}$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & -2x & 0 \end{vmatrix} = -\mathbf{a}_z (2 + x)$$

$$\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{s} = \int_0^{\pi/2} \int_0^3 -\mathbf{a}_z (2 + r \cos \phi) \cdot r dr d\phi \mathbf{a}_z = -9 \left(1 + \frac{\pi}{2} \right)$$

► From B to O

$$x = 0, \text{ and } \mathbf{F} \cdot d\boldsymbol{\ell} = \mathbf{F} \cdot (\mathbf{a}_y dy) = 2x dy = 0$$

► From O to A

$$y = 0, \text{ and } \mathbf{F} \cdot d\boldsymbol{\ell} = \mathbf{F} \cdot (\mathbf{a}_x dx) = xy dx = 0$$

$$\oint_{ABOA} \mathbf{F} \cdot d\boldsymbol{\ell} = \int_A^B \mathbf{F} \cdot d\boldsymbol{\ell} = -9 \left(1 + \frac{\pi}{2} \right)$$

(examined in Example 1-11)

1.19 Important Vector Identities

⚡ Important result | Identity I

The curl of the gradient of any scalar quantity is zero

$$\nabla \times (\nabla V) \equiv 0$$

- A curl-free vector field is called an **irrotational** or **conservative** field.
- If a vector quantity is curl-free, it can be expressed as the gradient of a scalar quantity.

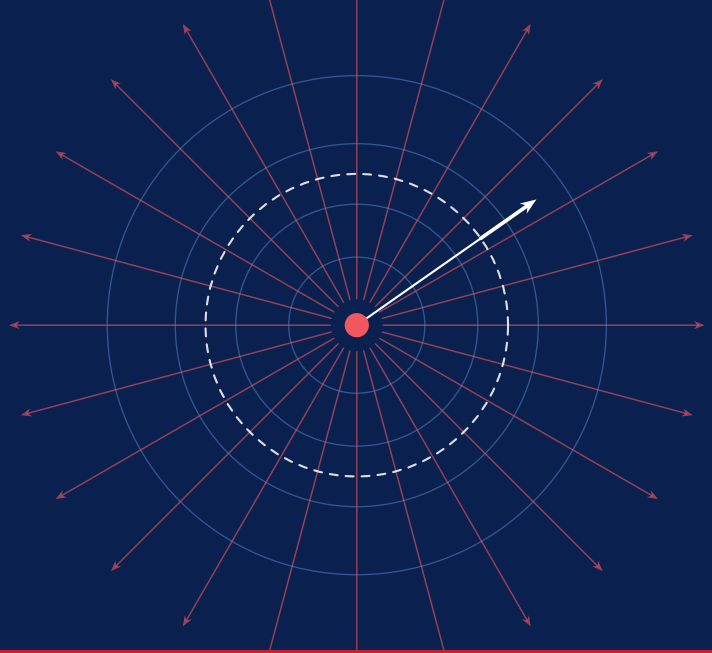
⚡ **Important result** | Identity II

The divergence of the curl of any vector field is zero.

$$\nabla \cdot (\nabla \times \mathbf{A}) \equiv 0$$

- A divergence-free vector field is called a **solenoidal** field.
- If a vector quantity is divergence-free, it can be expressed as the curl of a vector quantity.

2



CHAPTER

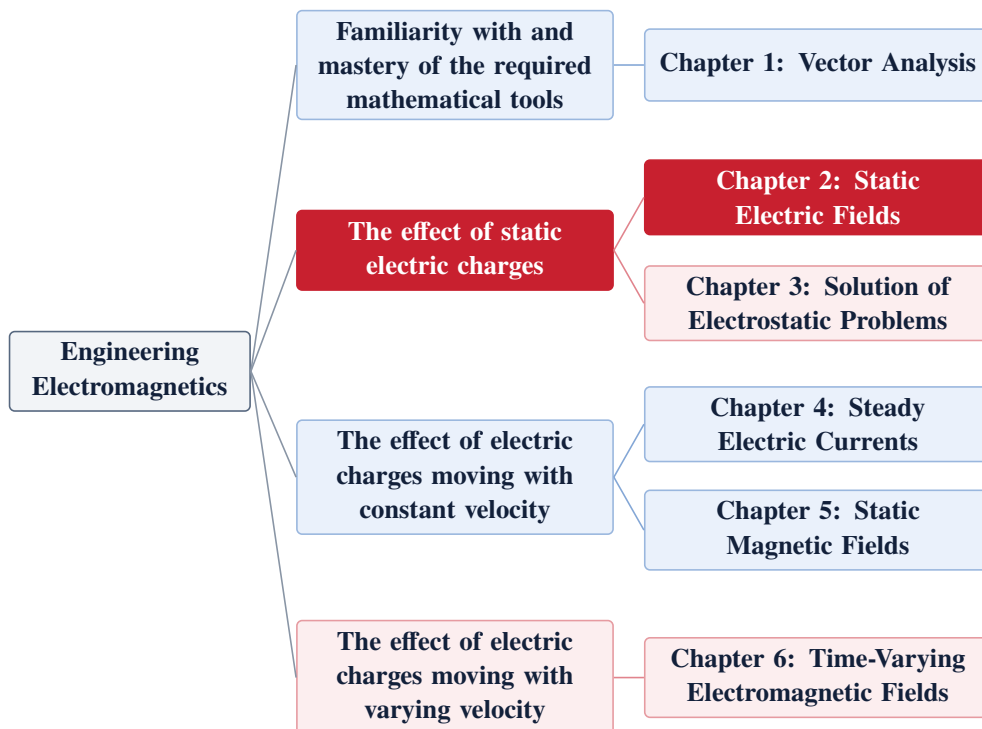
Static Electric Fields

میدان‌های الکتریکی ساکن

In this chapter

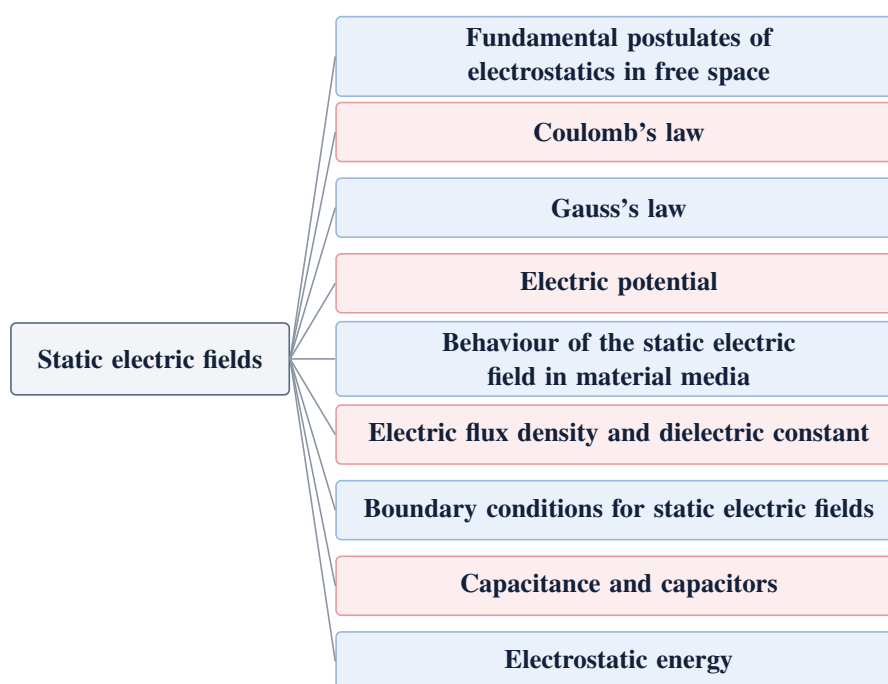
2.1	Course Outline	45	2.8	Conductors in Static Electric Field	69
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2.1 Course Outline



2.2 Static Electric Fields

- In electrostatics the electric charges are at rest and the electric fields do not change with time.
 - In this case there are no magnetic fields.
- In introductory physics the development of electrostatics usually begins with Coulomb's experimental law for the force between two charges.
- In this course, instead of following the historical development of electrostatics, we introduce the subject by postulating the divergence and the curl of the electric field intensity in free space.



- What are the fundamental postulates of electrostatics from which the other relations and laws of the subject can be derived?
- How are Coulomb's and Gauss's laws derived from these postulates?
- How is the electric field of a point charge, of a number of discrete point charges and of a continuous charge distribution computed?

2.3 Fundamental Postulates of Electrostatics in Free Space

► Electric field intensity

- The force on a small test charge placed in the field, divided by the charge. $\rightarrow$ N/C (newtons per coulomb), equivalent to V/m (volts per metre)

$\triangleq$ Definition

$$\mathbf{E} = \lim_{q \rightarrow 0} \frac{\mathbf{F}}{q} \quad (\text{V/m})$$

- The electric field intensity is proportional to the force and in the same direction.
- The two fundamental postulates of electrostatics in free space

⚡ Important result

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \nabla \times \mathbf{E} = 0$$

- ρ : volume charge density (C/m^3); ϵ_0 : permittivity of free space (F/m)
- $\mathbf{E}$ is not solenoidal unless $\rho = 0$
- $\mathbf{E}$ is irrotational
- Integral form of the first equation $\rightarrow$ integration over an arbitrary volume V

$$\int_V \nabla \cdot \mathbf{E} \, dv = \frac{1}{\epsilon_0} \int_V \rho \, dv$$

- $\int_V \rho \, dv$: the total charge contained in the volume V enclosed by the surface S
- Using the divergence theorem:

⚡ Important result

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \frac{Q}{\epsilon_0}$$

- This relation is a form of **Gauss's law**
 - The total outward flux of the electric field intensity through any closed surface equals the total charge enclosed in the surface divided by ϵ_0 .
- Integral form of the second equation $\rightarrow$ integration over an arbitrary open surface S

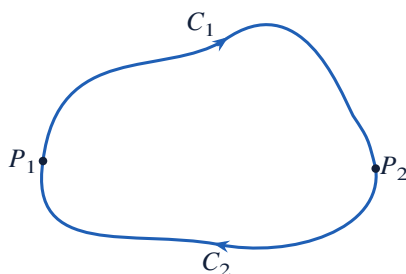
$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = 0$$

- Using Stokes's theorem:

⚡ Important result

$$\oint_C \mathbf{E} \cdot d\ell = 0$$

- The scalar line integral of the static electric field intensity around any closed path is zero.
- The scalar product $\mathbf{E} \cdot d\ell$ integrated over any path is the voltage along that path.
- This equation is an expression of **Kirchhoff's voltage law**
 - The algebraic sum of the voltage drops around any closed path is zero.
- The line integrals of the irrotational field $\mathbf{E}$ are independent of the path and depend only on the initial and final points.



$$\oint_C \mathbf{E} \cdot d\ell = 0$$

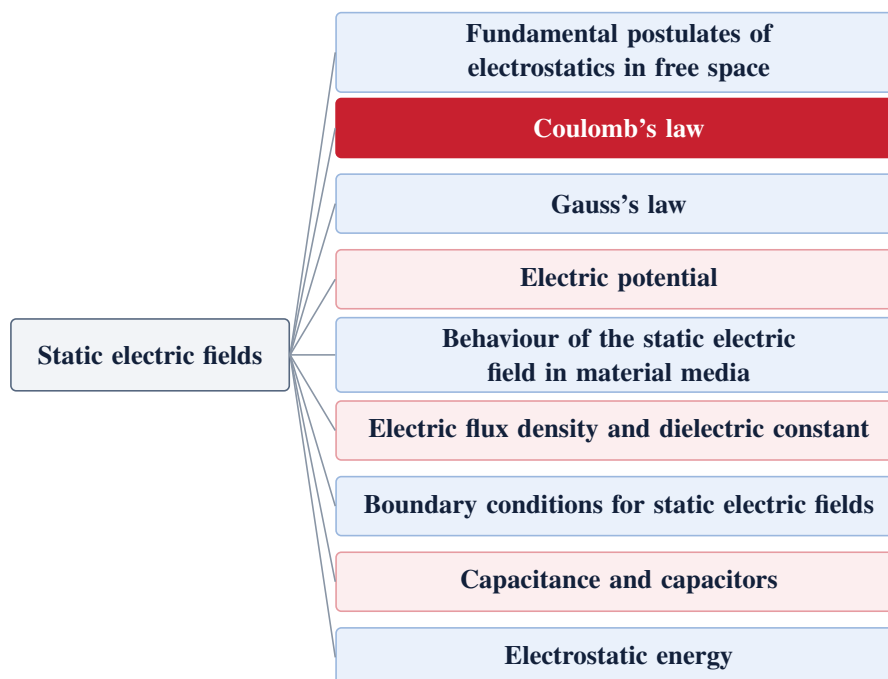
$$\int_{C_1} \mathbf{E} \cdot d\ell + \int_{C_2} \mathbf{E} \cdot d\ell = 0$$

$$\int_{P_1}^{P_2} \mathbf{E} \cdot d\ell + \int_{P_2}^{P_1} \mathbf{E} \cdot d\ell = 0$$

$$\int_{\text{Along } C_1}^{P_2}_{P_1} \mathbf{E} \cdot d\ell = - \int_{\text{Along } C_2}^{P_1}_{P_2} \mathbf{E} \cdot d\ell$$

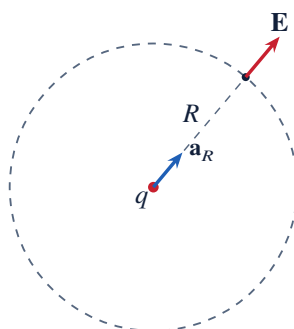
$$\int_{\text{Along } C_1}^{P_2}_{P_1} \mathbf{E} \cdot d\ell = \int_{\text{Along } C_2}^{P_2}_{P_1} \mathbf{E} \cdot d\ell$$

2.4 Coulomb's Law



► The electric field intensity of a point charge q in free space

- We consider a hypothetical spherical surface of radius R centred at q .
- Since a point charge has no preferred direction, we expect the field to be radial everywhere and its intensity to be constant over the spherical surface.



► The electric field intensity on the spherical surface:

$$\mathbf{E} = E_R \mathbf{a}_R$$

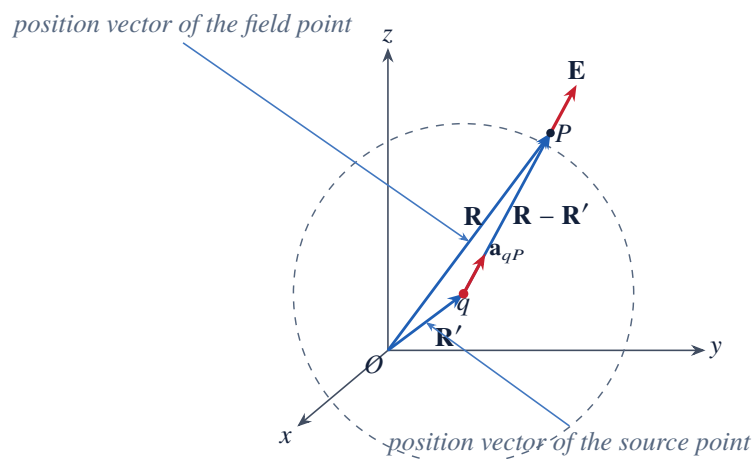
$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \oint_S (\mathbf{a}_R E_R) \cdot \mathbf{a}_R ds = \frac{q}{\epsilon_0}$$

$$E_R \oint_S ds = E_R (4\pi R^2) = \frac{q}{\epsilon_0}$$

⚡ Important result

$$\mathbf{E} = \mathbf{a}_R E_R = \mathbf{a}_R \frac{q}{4\pi\epsilon_0 R^2} \quad (\text{V/m})$$

- If the charge is not at the origin of the coordinate system



$$\mathbf{E}_P = \mathbf{a}_{qP} \frac{q}{4\pi\epsilon_0 |\mathbf{R} - \mathbf{R}'|^2}, \quad \mathbf{a}_{qP} = \frac{\mathbf{R} - \mathbf{R}'}{|\mathbf{R} - \mathbf{R}'|}$$

⚡ Important result

$$\mathbf{E}_P = \frac{q(\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0 |\mathbf{R} - \mathbf{R}'|^3} \quad (\text{V/m})$$

- When a point charge q_2 is placed in the electric field of another point charge q_1 , the force acting on it is

⚡ Important result | Mathematical statement of Coulomb's law

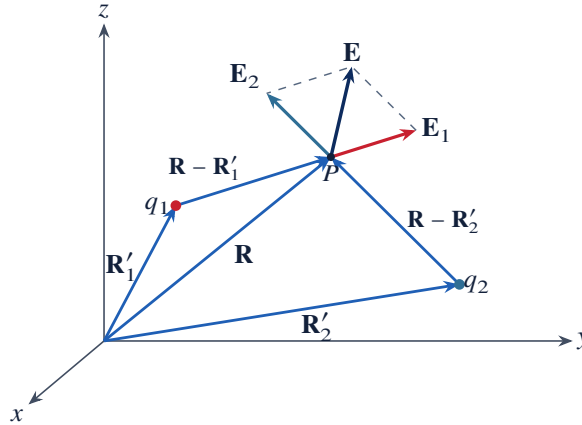
$$\mathbf{F}_{12} = q_2 \mathbf{E}_{12} = \mathbf{a}_R \frac{q_1 q_2}{4\pi\epsilon_0 R^2} \quad (\text{N})$$



► The electric field intensity of a set of discrete charges

⚡ Important result

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^n \frac{q_k(\mathbf{R} - \mathbf{R}'_k)}{|\mathbf{R} - \mathbf{R}'_k|^3} \quad (\text{V/m})$$



► **The electric field intensity of a continuous charge distribution**

- The field of a differential charge dq is

$$d\mathbf{E} = \frac{dq(\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0|\mathbf{R} - \mathbf{R}'|^3}$$

► **Line charge distribution**

$$dq = \rho_\ell d\ell' \quad \mathbf{E} = \int_{L'} \frac{\rho_\ell d\ell' (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0|\mathbf{R} - \mathbf{R}'|^3}$$

► **Surface charge distribution**

$$dq = \rho_s ds' \quad \mathbf{E} = \int_{S'} \frac{\rho_s ds' (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0|\mathbf{R} - \mathbf{R}'|^3}$$

► **Volume charge distribution**

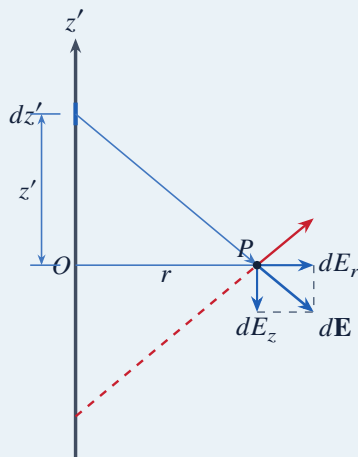
$$dq = \rho_v dv' \quad \mathbf{E} = \int_{V'} \frac{\rho_v dv' (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0|\mathbf{R} - \mathbf{R}'|^3}$$

► **Steps in computing the electric field intensity of a continuous charge distribution**

- Choosing an appropriate coordinate system
- Correctly determining the differential element $d\ell'$, ds' and dv' (note that in these relations the magnitudes of the differential length and surface vectors are used)
- Correctly determining the position vector of the field point ($\mathbf{R}$)
- Correctly determining the position vector of the source point ($\mathbf{R}'$)
- Evaluating the integral
 - Note that we integrate with respect to the primed coordinates.

Example 2-1

Determine the electric field intensity of an infinitely long straight line charge of uniform density ρ_ℓ .



► Solution.

$$\mathbf{E} = \int_{L'} \frac{\rho_\ell d\ell' (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0 |\mathbf{R} - \mathbf{R}'|^3}$$

► Choosing cylindrical coordinates

$$d\ell' = dz' \quad \mathbf{R} = r\mathbf{a}_r \quad \mathbf{R}' = z'\mathbf{a}_z$$

$$\mathbf{E} = \frac{\rho_\ell}{4\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{r\mathbf{a}_r - z'\mathbf{a}_z}{(r^2 + z'^2)^{3/2}} dz'$$

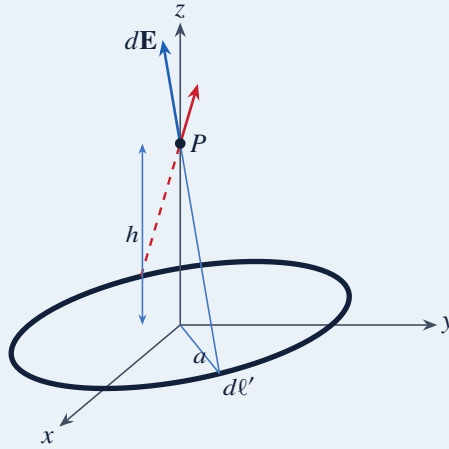
► It is intuitively clear that the field has no component along z .

$$\mathbf{E} = \mathbf{a}_r E_r = \mathbf{a}_r \frac{\rho_\ell r}{4\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{dz'}{(r^2 + z'^2)^{3/2}}$$

$$\mathbf{E} = \mathbf{a}_r \frac{\rho_\ell}{2\pi\epsilon_0 r} \quad (\text{V/m})$$

Example 2-2

A ring of radius a carries a line charge density ρ_ℓ . Find the electric field intensity on the axis of the ring at a distance h from it.



► Solution.

$$\mathbf{E} = \int_{L'} \frac{\rho_\ell d\ell' (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0 |\mathbf{R} - \mathbf{R}'|^3}$$

- Choosing cylindrical coordinates

$$d\ell' = a d\phi' \quad \mathbf{R} = h\mathbf{a}_z \quad \mathbf{R}' = a\mathbf{a}_{r'}$$

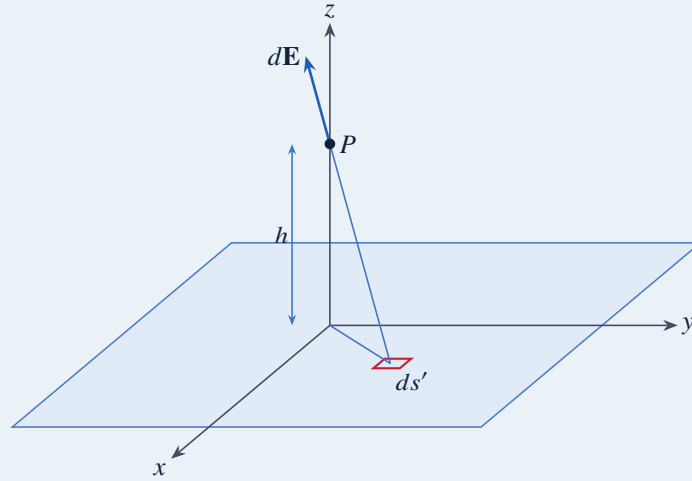
$$\mathbf{E} = \frac{\rho_\ell}{4\pi\epsilon_0} \int_0^{2\pi} \frac{h\mathbf{a}_z - a\mathbf{a}_{r'}}{(a^2 + h^2)^{3/2}} a d\phi'$$

- It is intuitively clear that the field has no component along r .

$$\mathbf{E} = \frac{\rho_\ell a h}{4\pi\epsilon_0 (a^2 + h^2)^{3/2}} \mathbf{a}_z \int_0^{2\pi} d\phi' = \frac{\rho_\ell a h}{2\epsilon_0 (a^2 + h^2)^{3/2}} \mathbf{a}_z$$

Example 2-3

Determine the electric field intensity of an infinite planar charge with a uniform surface charge density ρ_s .



► **Solution.**

$$\mathbf{E} = \int_{S'} \frac{\rho_s ds' (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0 |\mathbf{R} - \mathbf{R}'|^3}$$

► Choosing cylindrical coordinates

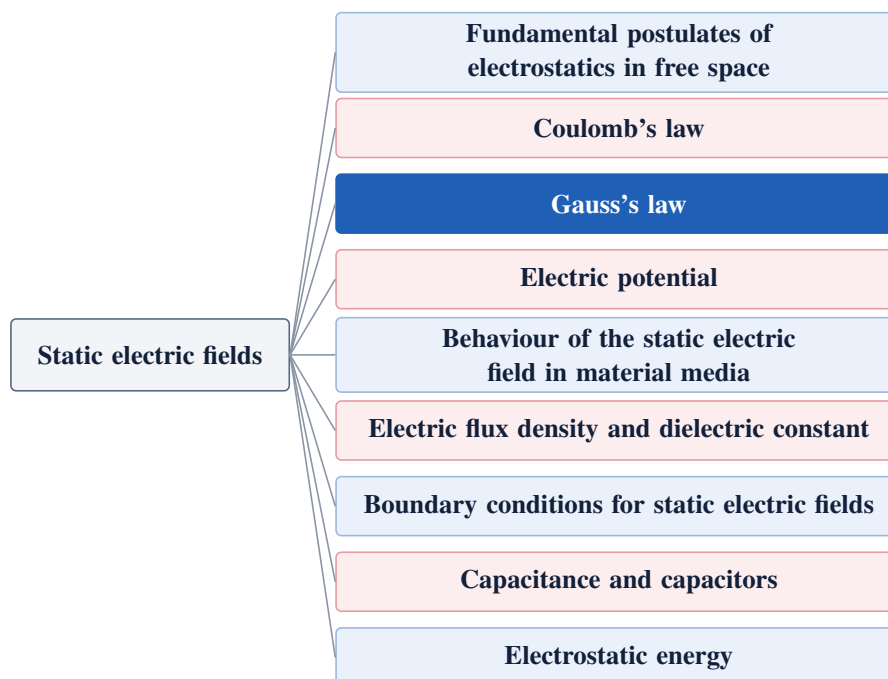
$$ds' = r' dr' d\phi' \quad \mathbf{R} = h\mathbf{a}_z \quad \mathbf{R}' = r'\mathbf{a}_{r'}$$

$$\mathbf{E} = \frac{\rho_s}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\infty \frac{h\mathbf{a}_z - r'\mathbf{a}_{r'}}{(h^2 + r'^2)^{3/2}} r' dr' d\phi'$$

► It is intuitively clear that the field has no component along r .

$$\mathbf{E} = \frac{\rho_s h \mathbf{a}_z}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\infty \frac{r' dr' d\phi'}{(h^2 + r'^2)^{3/2}} = \frac{\rho_s}{2\epsilon_0} \mathbf{a}_z$$

2.5 Gauss's Law



⚡ Important result

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \frac{Q}{\epsilon_0}$$

- Q : the total charge contained in the volume V enclosed by the surface S
- The total outward flux of the electric field intensity through any closed surface equals the total charge enclosed in the surface divided by ϵ_0 .
- The surface S can be any closed surface, chosen for convenience.

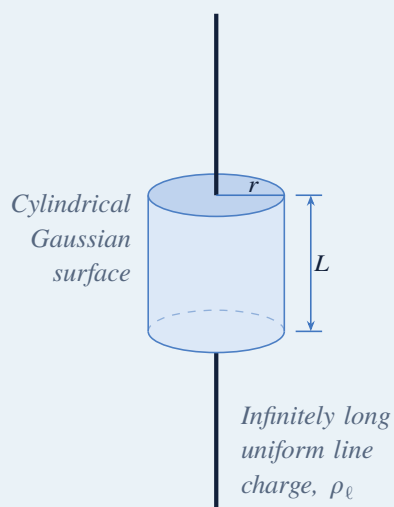
📌 Remark

Gauss's law always holds, but in using it to determine the electric field intensity the following point must be observed.

- The basis of applying Gauss's law to determine the electric field intensity lies, first, in recognizing the conditions of symmetry and, second, in choosing an appropriate surface over which the normal component of $\mathbf{E}$ due to a given charge distribution is constant (the Gaussian surface).

Example 2-4

Using Gauss's law, determine the electric field intensity of an infinitely long straight line charge of uniform density ρ_ℓ at a distance r from it.



► Solution.

$$\mathbf{E} = \mathbf{a}_r E_r$$

► Choosing a cylinder of length L and radius r as the Gaussian surface

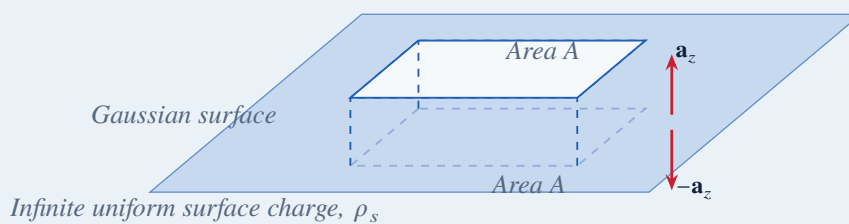
- For the top and bottom faces: $\mathbf{E} \cdot d\mathbf{s} = 0$
- For the side surface: $d\mathbf{s} = \mathbf{a}_r r d\phi dz$

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \int_0^L \int_0^{2\pi} E_r r d\phi dz = 2\pi r L E_r$$

$$Q = \rho_\ell L \quad 2\pi r L E_r = \frac{\rho_\ell L}{\epsilon_0} \quad \mathbf{E} = \mathbf{a}_r E_r = \mathbf{a}_r \frac{\rho_\ell}{2\pi \epsilon_0 r}$$

Example 2-5

Determine the electric field intensity of an infinite planar charge with a uniform surface charge density ρ_s .



► Solution.

$$\mathbf{E} = \pm E_z \mathbf{a}_z$$

- Choosing a box as the Gaussian surface
- For the top face

$$\mathbf{E} \cdot d\mathbf{s} = (\mathbf{a}_z E_z) \cdot (\mathbf{a}_z ds) = E_z ds$$

- For the bottom face

$$\mathbf{E} \cdot d\mathbf{s} = (-\mathbf{a}_z E_z) \cdot (-\mathbf{a}_z ds) = E_z ds$$

- For the side faces

$$\mathbf{E} \cdot d\mathbf{s} = 0$$

- Therefore

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = 2E_z \int_A ds = 2E_z A$$

$$Q = \rho_s A \quad 2E_z A = \frac{\rho_s A}{\epsilon_0}$$

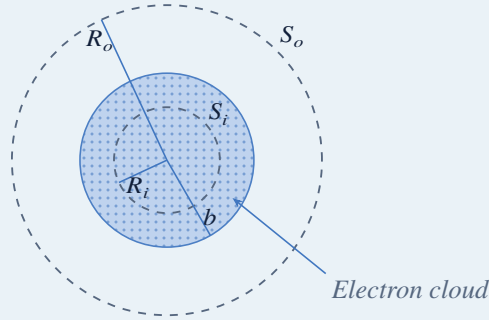
⚡ Important result

$$\mathbf{E} = \mathbf{a}_z E_z = \mathbf{a}_z \frac{\rho_s}{2\epsilon_0}, \quad z > 0$$

$$\mathbf{E} = -\mathbf{a}_z E_z = -\mathbf{a}_z \frac{\rho_s}{2\epsilon_0}, \quad z < 0$$

Example 2-6

Determine the field $\mathbf{E}$ of an electron cloud with volume charge density $\rho = -\rho_0$ in the region $0 \leq R \leq b$ and $\rho = 0$ in the region $R > b$.



► Solution.

$$\mathbf{E} = \mathbf{a}_R E_R$$

► (a) $0 \leq R \leq b$

– Choosing a sphere as the Gaussian surface

$$d\mathbf{s} = \mathbf{a}_R ds$$

$$\oint_{S_i} \mathbf{E} \cdot d\mathbf{s} = E_R \int_{S_i} ds = E_R 4\pi R^2$$

$$Q = \int_V \rho dv = -\rho_0 \int_V dv = -\rho_0 \frac{4\pi}{3} R^3 \quad \mathbf{E} = -\mathbf{a}_R \frac{\rho_0}{3\epsilon_0} R$$

► (b) $R \geq b$

– Choosing a sphere as the Gaussian surface

$$d\mathbf{s} = \mathbf{a}_R ds$$

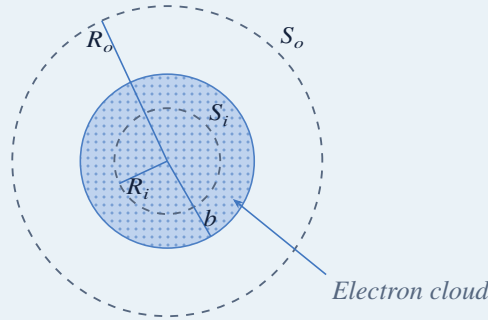
$$\oint_{S_o} \mathbf{E} \cdot d\mathbf{s} = E_R \int_{S_o} ds = E_R 4\pi R^2$$

$$Q = -\rho_0 \frac{4\pi}{3} b^3 \quad \mathbf{E} = -\mathbf{a}_R \frac{\rho_0 b^3}{3\epsilon_0 R^2}$$

Example 2-7

A spherical charge distribution of radius b has the following density. Find the electric field intensity at all points of space.

$$\rho_v = \begin{cases} \frac{\rho_0 R}{b} & 0 \leq R \leq b \\ 0 & R > b \end{cases}$$



► Solution.

$$\mathbf{E} = \mathbf{a}_R E_R$$

- (a) $0 \leq R \leq b$

– Choosing a sphere as the Gaussian surface

$$d\mathbf{s} = \mathbf{a}_R ds$$

$$\oint_{S_i} \mathbf{E} \cdot d\mathbf{s} = E_R \int_{S_i} ds = E_R 4\pi R^2$$

$$Q = \int_v \rho_v dv = \int_0^{2\pi} \int_0^\pi \int_0^R \frac{\rho_0 R}{b} R^2 \sin \theta dR d\theta d\phi = \frac{\rho_0 \pi R^4}{b}$$

$$\mathbf{E} = E_R \mathbf{a}_R = \frac{\rho_0 R^2}{4\epsilon_0 b} \mathbf{a}_R$$

- (b) $R \geq b$

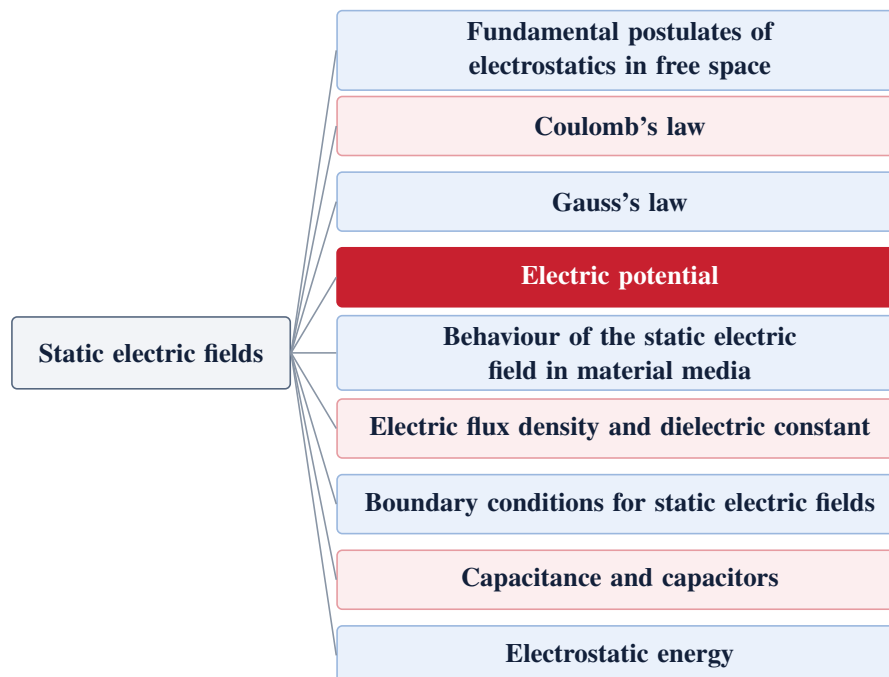
– Choosing a sphere as the Gaussian surface

$$d\mathbf{s} = \mathbf{a}_R ds$$

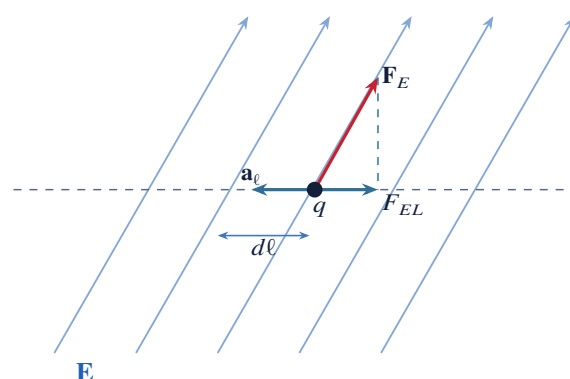
$$\oint_{S_o} \mathbf{E} \cdot d\mathbf{s} = E_R \int_{S_o} ds = E_R 4\pi R^2 \quad Q = \rho_0 \pi b^3$$

$$\mathbf{E} = E_R \mathbf{a}_R = \frac{\rho_0 b^3}{4\epsilon_0 R^2} \mathbf{a}_R$$

2.6 Electric Potential



- What is the concept of the electric potential difference between two points and of the electric potential of a point?
- How is the electric potential of a point charge, of a number of discrete point charges and of a continuous charge distribution computed?
- How can the electric field intensity be computed from the electric potential, and what is the advantage of this over computing the electric field intensity directly?
- Suppose we want to move a point charge q , located in the electric field $\mathbf{E}$, by $d\ell$.



- The force exerted on the charge by the electric field:

$$\mathbf{F}_E = q\mathbf{E}$$

- The magnitude of the component of this force along $d\ell$:

$$F_{EL} = \mathbf{F}_E \cdot \mathbf{a}_\ell = q\mathbf{E} \cdot \mathbf{a}_\ell$$

- The force that must be applied to the charge:

$$F_{\text{apply}} = -F_{EL} = -q\mathbf{E} \cdot \mathbf{a}_\ell$$

- ▶ The work done in moving the charge by $d\ell$:

$$dW = F_{\text{apply}} d\ell = -q\mathbf{E} \cdot d\ell$$

- ▶ The work done in moving the charge from point P_1 to point P_2 :

$$W = -q \int_{P_1}^{P_2} \mathbf{E} \cdot d\ell$$

$$\nabla \times \mathbf{E} = 0 \Rightarrow \mathbf{E} = -\nabla V$$

$$\nabla V \cdot \mathbf{a}_\ell = \frac{dV}{d\ell}$$

$$q = +1 \text{ C} \quad W = \int_{P_1}^{P_2} \nabla V \cdot d\ell \quad \Rightarrow \quad W = \int_{P_1}^{P_2} dV = V_2 - V_1$$

Definition | Electric potential difference

The electric potential difference between two points is the work done by an external source in moving a unit positive charge from one point to the other.

$$V_2 - V_1 = - \int_{P_1}^{P_2} \mathbf{E} \cdot d\ell$$

- ▶ It does not depend on the path of integration
- ▶ In many cases a reference point at infinity with zero potential is chosen, and the potential differences of the various points are expressed with respect to this reference point.

Important result

$$V_P = - \int_{\infty}^P \mathbf{E} \cdot d\ell$$

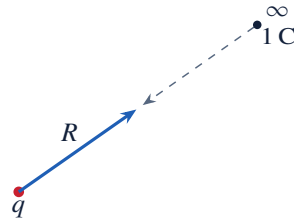
Remark | Remark 1

The minus sign in the potential relation is needed for consistency with the fact that moving against the direction of the field increases the electric potential.

Remark | Remark 2

The gradient of V is normal to the surfaces of constant V ; consequently the electric field lines are normal to the equipotential surfaces.

- ▶ The electric potential of a point charge q at a distance R from it



$$V = - \int_{\infty}^R \left(\frac{q}{4\pi\epsilon_0 R^2} \mathbf{a}_R \right) \cdot dR \mathbf{a}_R = \frac{q}{4\pi\epsilon_0 R}$$

- ▶ The electric potential of n discrete point charges

$$V = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^n \frac{q_k}{|\mathbf{R} - \mathbf{R}'_k|}$$

- ▶ $\mathbf{R}$: position vector of the field point; $\mathbf{R}'_k$: position vectors of the source points

- ▶ The electric potential of a continuous charge distribution

- Line charge distribution

$$V = \frac{1}{4\pi\epsilon_0} \int_{L'} \frac{\rho_\ell d\ell'}{|\mathbf{R} - \mathbf{R}'|}$$

- Surface charge distribution

$$V = \frac{1}{4\pi\epsilon_0} \int_{S'} \frac{\rho_s ds'}{|\mathbf{R} - \mathbf{R}'|}$$

- Volume charge distribution

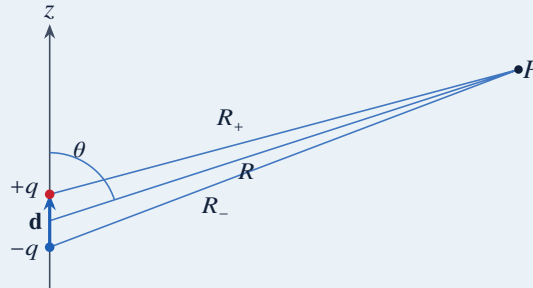
$$V = \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\rho_v dv'}{|\mathbf{R} - \mathbf{R}'|}$$

Example 2-8

Find the electric potential and the electric field intensity of an electric dipole at an arbitrary point very far from it.

► **Solution.**

$$V = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^2 \frac{q_k}{|\mathbf{R} - \mathbf{R}'_k|} = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|\mathbf{R} - \mathbf{d}/2|} + \frac{-q}{|\mathbf{R} + \mathbf{d}/2|} \right]$$



$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{R_+} - \frac{1}{R_-} \right)$$

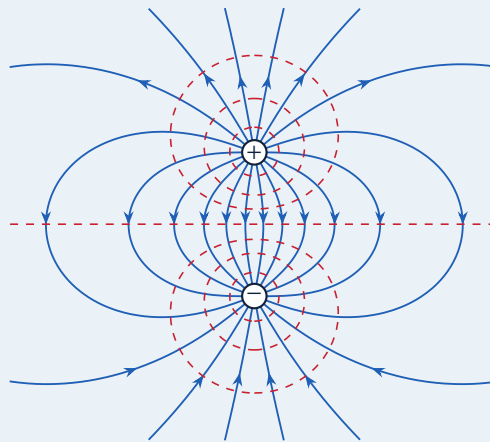
$$\frac{1}{R_+} \cong \left(R - \frac{d}{2} \cos \theta \right)^{-1} \cong R^{-1} \left(1 + \frac{d}{2R} \cos \theta \right)$$

$$\frac{1}{R_-} \cong \left(R + \frac{d}{2} \cos \theta \right)^{-1} \cong R^{-1} \left(1 - \frac{d}{2R} \cos \theta \right)$$

► $\mathbf{p} = q\mathbf{d}$: the electric dipole moment vector

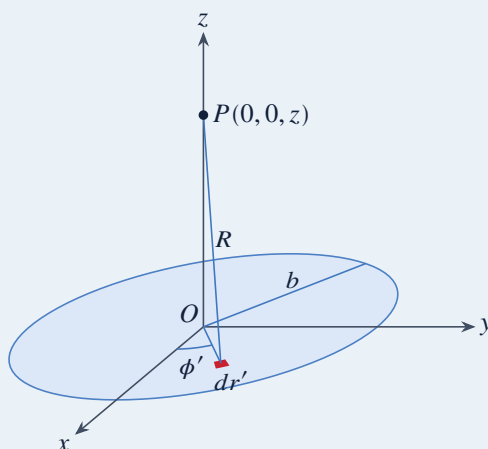
$$V = \frac{qd \cos \theta}{4\pi\epsilon_0 R^2} \quad \mathbf{d} = d\mathbf{a}_z \quad \Rightarrow \quad V = \frac{q\mathbf{d} \cdot \mathbf{a}_R}{4\pi\epsilon_0 R^2} \quad \mathbf{p} = q\mathbf{d} \quad \Rightarrow \quad V = \frac{\mathbf{p} \cdot \mathbf{a}_R}{4\pi\epsilon_0 R^2}$$

$$\begin{aligned} \mathbf{E} &= -\nabla V = -\mathbf{a}_R \frac{\partial V}{\partial R} - \mathbf{a}_\theta \frac{1}{R} \frac{\partial V}{\partial \theta} \\ &= \frac{p}{4\pi\epsilon_0 R^3} (\mathbf{a}_R 2 \cos \theta + \mathbf{a}_\theta \sin \theta) \end{aligned}$$



Example 2-9

Find the electric potential and the electric field intensity of a charged disk of radius b with surface charge density ρ_s on its axis.



► **Solution.**

$$V = \frac{1}{4\pi\epsilon_0} \int_{S'} \frac{\rho_s ds'}{|\mathbf{R} - \mathbf{R}'|}$$

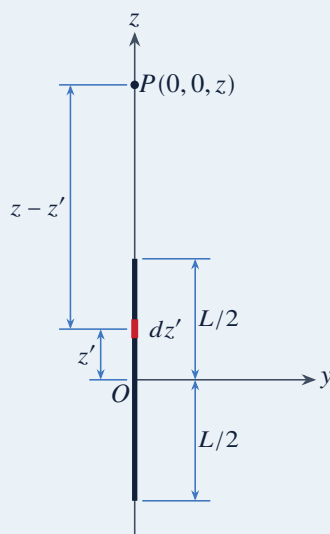
$$ds' = r' dr' d\phi' \quad \mathbf{R} = z\mathbf{a}_z \quad \mathbf{R}' = r'\mathbf{a}_{r'}$$

$$V = \frac{\rho_s}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^b \frac{r' dr' d\phi'}{\sqrt{z^2 + r'^2}} = \frac{\rho_s}{2\epsilon_0} \left[\sqrt{z^2 + b^2} - z \right] \quad z > 0$$

$$\mathbf{E} = -\nabla V = \mathbf{a}_z \frac{\rho_s}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{z^2 + b^2}} \right] \quad z > 0$$

Example 2-10

Find the electric potential and the electric field intensity of a line charge of length L and charge density ρ_ℓ along its extension.



► **Solution.**

$$V = \frac{1}{4\pi\epsilon_0} \int_{L'} \frac{\rho_\ell d\ell'}{|\mathbf{R} - \mathbf{R}'|}$$

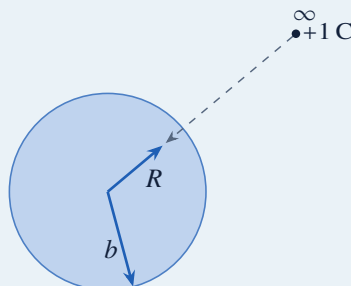
$$d\ell' = dz' \quad \mathbf{R} = z\mathbf{a}_z \quad \mathbf{R}' = z'\mathbf{a}_z$$

$$V = \frac{\rho_\ell}{4\pi\epsilon_0} \int_{-L/2}^{L/2} \frac{dz'}{z - z'} = \frac{\rho_\ell}{4\pi\epsilon_0} \ln \left[\frac{z + L/2}{z - L/2} \right] \quad z > L/2$$

$$\mathbf{E} = -\nabla V = -\mathbf{a}_z \frac{dV}{dz} = \mathbf{a}_z \frac{\rho_\ell L}{4\pi\epsilon_0 [z^2 - (L/2)^2]} \quad z > L/2$$

Example 2-11

A volume charge of density $\rho = -\rho_0$ exists in the region $0 \leq R \leq b$. What is the electric potential at an arbitrary point inside this volume?



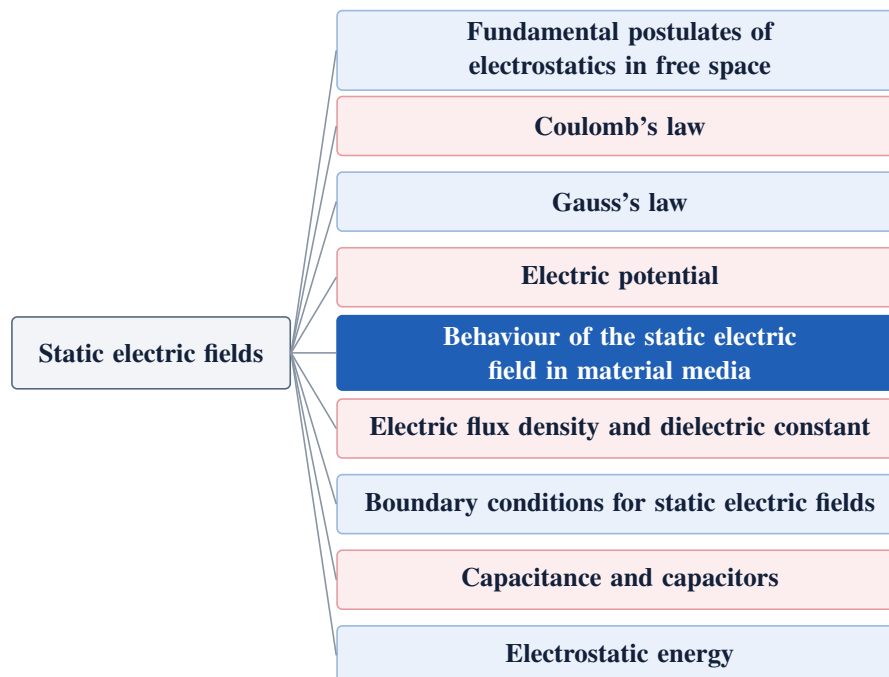
► **Solution.**

$$\mathbf{E} = -\mathbf{a}_R \frac{\rho_0}{3\epsilon_0} R, \quad 0 \leq R \leq b$$

$$\mathbf{E} = -\mathbf{a}_R \frac{\rho_0 b^3}{3\epsilon_0 R^2}, \quad R \geq b$$

$$\begin{aligned} V &= -\int_{\infty}^R \mathbf{E} \cdot d\ell = -\int_{\infty}^b \frac{-\rho_0 b^3}{3\epsilon_0 R^2} \mathbf{a}_R \cdot dR \mathbf{a}_R - \int_b^R \frac{-\rho_0 R}{3\epsilon_0} \mathbf{a}_R \cdot dR \mathbf{a}_R \\ &= -\frac{\rho_0 b^2}{3\epsilon_0} - \frac{\rho_0}{6\epsilon_0} (b^2 - R^2) \end{aligned}$$

2.7 Behaviour of the Static Electric Field in Material Media

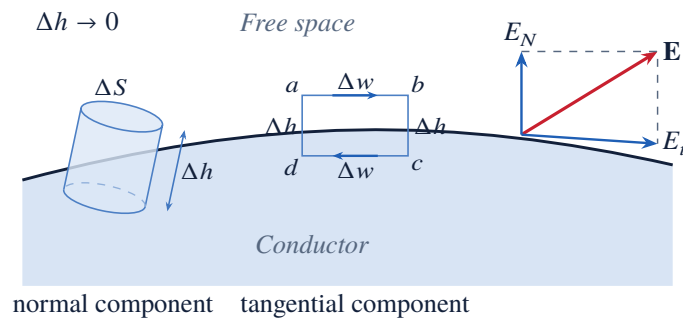


- Classification of materials according to their electrical properties
 - Conductors
 - Semiconductors
 - Insulators (dielectrics)
- In conductors the electrons of the outer shells of the atoms are held by a weak force and pass easily from one atom to another.
- But the electrons in the atoms of insulators are tightly bound to their orbits and under normal conditions cannot be detached.
- The electrical properties of semiconductors lie between those of conductors and insulators.
- How does the static electric field intensity behave in the presence of a conductor?
- How does the static electric field intensity behave in the presence of a dielectric?
- What is the concept of electric flux density, and how is it related to the electric field intensity?
- What is the concept of the dielectric constant?
- How does the static electric field intensity behave at the boundary between two different media?

2.8 Conductors in Static Electric Field

- If we release charges inside a conductor, a field forms in it that drives the charges apart and to the surface.
 - Under static conditions the charge and the electric field inside a conductor are zero.

$$\rho = 0 \quad \mathbf{E} = 0$$
 - The time needed for the charges to distribute themselves on the surface of the conductor and reach equilibrium depends on the conductivity of the conductor.
- If under static conditions the electric field on the surface of a conductor had a tangential component, it would produce a tangential force, set the charges in motion, and there would be no charge equilibrium.
 - Therefore, under static conditions the electric field on the surface of a conductor is everywhere normal to the surface. In other words, under static conditions the surface of a conductor is an equipotential surface. In fact, since $\mathbf{E} = 0$ at all points inside the conductor, the whole conductor has the same electric potential.
- **Boundary conditions at a conductor / free-space interface**



- Normal component:

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = E_n \Delta S = \frac{\rho_s \Delta S}{\epsilon_0} \quad \Rightarrow \quad E_n = \frac{\rho_s}{\epsilon_0}$$

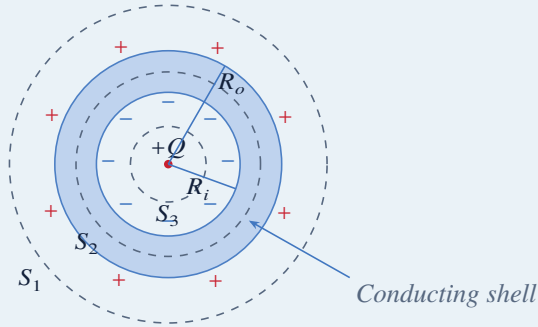
- Tangential component:

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$

$$E_t \Delta w - E_{N, \text{at } b} \frac{1}{2} \Delta h + E_{N, \text{at } a} \frac{1}{2} \Delta h = 0 \quad \Rightarrow \quad E_t = 0$$

Example 2-12

A point charge $+Q$ is at the centre of a conducting spherical shell of inner radius R_i and outer radius R_o . Find $\mathbf{E}$ and V in all space.



► Solution.

$$\mathbf{E} = E_R \mathbf{a}_R$$

► For $R > R_o$

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = E_{R1} 4\pi R^2 = \frac{Q}{\epsilon_0}$$

$$E_{R1} = \frac{Q}{4\pi\epsilon_0 R^2} \quad V_1 = -\int_{\infty}^R (E_{R1}) dR = \frac{Q}{4\pi\epsilon_0 R}$$

► For $R_i < R < R_o$

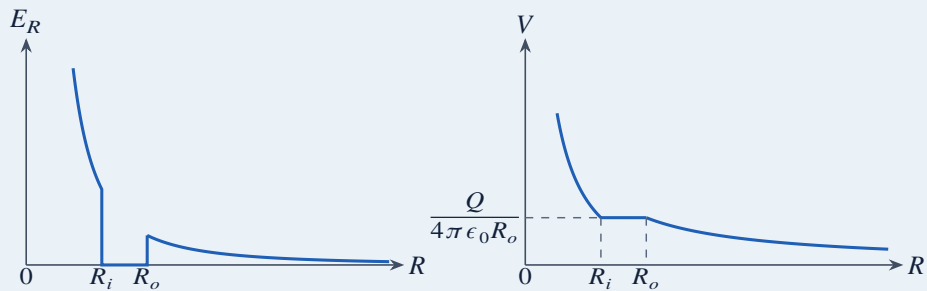
$$E_{R2} = 0 \quad V_2 = V_1|_{R=R_o} = \frac{Q}{4\pi\epsilon_0 R_o}$$

► For $R < R_i$

$$E_{R3} = \frac{Q}{4\pi\epsilon_0 R^2}$$

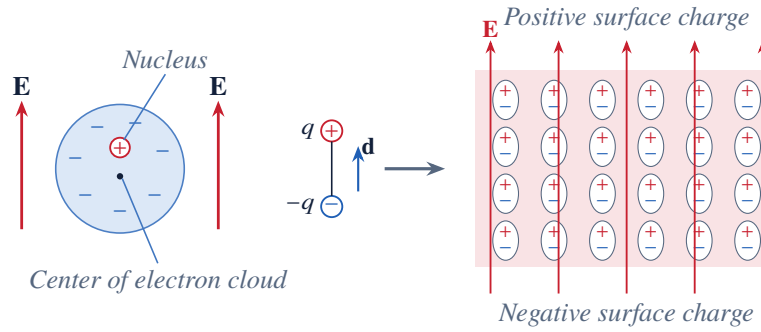
$$V_3 = -\int_{\infty}^{R_o} E_{R1} dR - \int_{R_o}^{R_i} E_{R2} dR - \int_{R_i}^R E_{R3} dR$$

$$V_3 = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{R} + \frac{1}{R_o} - \frac{1}{R_i} \right)$$



2.9 Dielectrics in Static Electric Field

- ▶ Ideal dielectrics have no free charges. As a result, their charge density and internal electric field are not zero, as they are in conductors.
- ▶ Dielectrics have **bound charges**; applying an external field to a dielectric creates electric dipoles and polarizes the dielectric material.
- ▶ The induced electric dipoles change the electric field inside and outside the dielectric material.



▶ Equivalent charge distributions in polarized dielectrics

- To analyse the macroscopic effect of the induced dipoles, we define the polarization vector $\mathbf{P}$ as follows

Definition

$$\mathbf{P} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{k=1}^{n\Delta v} \mathbf{p}_k}{\Delta v} \quad (\text{C/m}^2)$$

- ▶ n : number of dipoles per unit volume; $\mathbf{p}_k$: dipole moment
- ▶ If $d\mathbf{p}$ is the dipole moment of a small volume dv' , then

$$d\mathbf{p} = \mathbf{P} dv' \quad dV = \frac{\mathbf{P} \cdot \mathbf{a}_R}{4\pi\epsilon_0 R^2} dv' \quad V = \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\mathbf{P} \cdot \mathbf{a}_R}{R^2} dv'$$

$$V = \frac{1}{4\pi\epsilon_0} \oint_{S'} \frac{\mathbf{P} \cdot \mathbf{a}'_n}{R} ds' + \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{(-\nabla' \cdot \mathbf{P})}{R} dv'$$

- ▶ The electric potential, and hence the electric field intensity, of a polarized dielectric can be computed from the effect of two charge distributions, a surface one and a volume one, with the following densities, called the **polarization charge densities** or **bound charge densities**

Important result

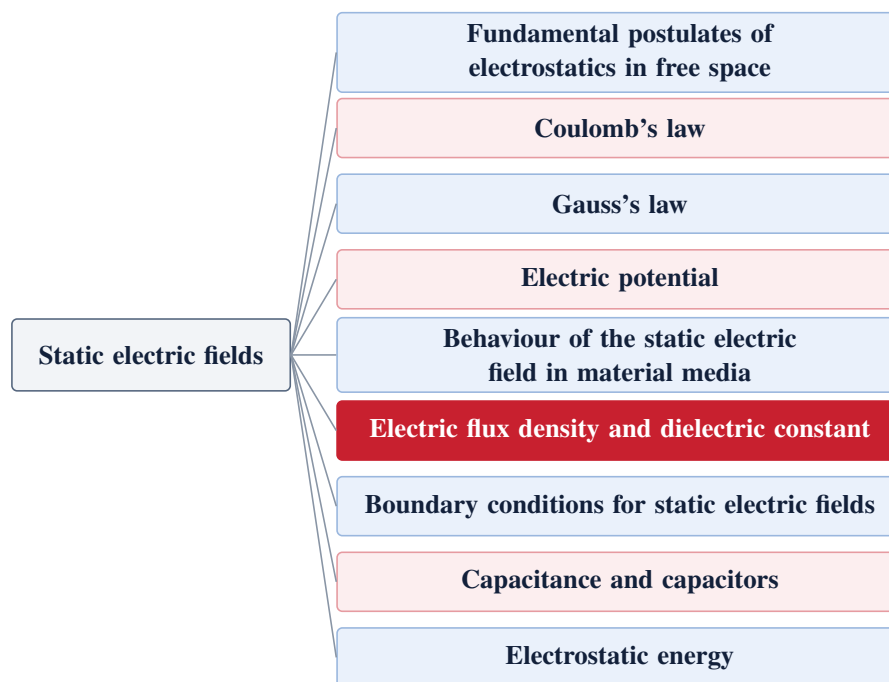
$$\begin{aligned} \rho_{ps} &= \mathbf{P} \cdot \mathbf{a}_n & \rho_p &= -\nabla \cdot \mathbf{P} \\ V &= \frac{1}{4\pi\epsilon_0} \oint_{S'} \frac{\rho_{ps}}{R} ds' + \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\rho_p}{R} dv' \end{aligned}$$

① Remark

Since we are dealing with an electrically neutral dielectric body, the total charge of the body after polarization must still be zero

$$\begin{aligned}\text{Total charge} &= \oint_S \rho_{ps} ds + \int_V \rho_p dv \\ &= \oint_S \mathbf{P} \cdot \mathbf{a}_n ds - \int_V \nabla \cdot \mathbf{P} dv = 0\end{aligned}$$

2.10 Electric Flux Density



- Since a dielectric gives rise to an equivalent volume charge density ρ_p , we expect the electric field intensity due to a given charge distribution in a dielectric to differ from the electric field intensity in free space.

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}(\rho + \rho_p)$$

$$\nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho$$

- The electric flux density or electric displacement $\mathbf{D}$ is defined as follows

≡ Definition

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad (\text{C/m}^2)$$

- Using the vector $\mathbf{D}$, the relation between the electric field and the free charge distributions in any medium can be written without having to deal with the polarization vector $\mathbf{P}$ or the polarization charge densities.

⚡ Important result

$$\nabla \cdot \mathbf{D} = \rho \quad (\text{C/m}^3)$$

- Integral form

$$\int_V \nabla \cdot \mathbf{D} dv = \int_V \rho dv$$

⚡ Important result | Another form of Gauss's law

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q \quad (\text{C})$$

- The fundamental equations of electrostatics in any medium

⚡ Important result

$$\begin{cases} \nabla \cdot \mathbf{D} = \rho \\ \nabla \times \mathbf{E} = 0 \end{cases}$$

2.11 Dielectric Constant

- The relation between the polarization vector and the electric field intensity

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$$

- χ_e : electric susceptibility
- The relation between the electric field intensity and the electric flux density

⚡ Important result

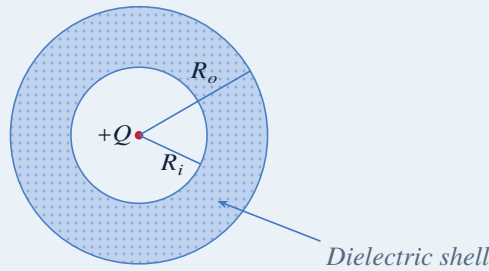
$$\begin{aligned} \mathbf{D} &= \epsilon_0(1 + \chi_e)\mathbf{E} \\ &= \epsilon_0\epsilon_r\mathbf{E} = \epsilon\mathbf{E} \quad (\text{C/m}^2) \end{aligned}$$

$$\epsilon_r = 1 + \chi_e = \frac{\epsilon}{\epsilon_0}$$

- ▶ ϵ_r : a dimensionless constant called the relative permittivity or dielectric constant
- ▶ A dielectric medium is **linear** if ϵ_r is independent of $\mathbf{E}$, **homogeneous** if ϵ_r is independent of the space coordinates, and isotropic if ϵ_r is a scalar.
 - A linear, homogeneous and isotropic medium is called a simple medium.
 - In anisotropic media the dielectric constant is a matrix. In such media $\mathbf{D}$ and $\mathbf{E}$ are not necessarily in the same direction.

Example 2-13

A point charge $+Q$ is at the centre of a dielectric spherical shell of inner radius R_i and outer radius R_o . The dielectric constant of the shell is ϵ_r . Find $\mathbf{E}$, V , $\mathbf{D}$ and $\mathbf{P}$ in all space.



Solution.

$$\mathbf{E} = E_R \mathbf{a}_R$$

- ▶ For $R > R_o$

$$E_{R1} = \frac{Q}{4\pi\epsilon_0 R^2} \quad V_1 = \frac{Q}{4\pi\epsilon_0 R}$$

$$D_{R1} = \epsilon_0 E_{R1} = \frac{Q}{4\pi R^2} \quad P_{R1} = 0$$

- ▶ For $R_i < R < R_o$

$$D_{R2} = \frac{Q}{4\pi R^2}$$

$$E_{R2} = \frac{Q}{4\pi\epsilon_0\epsilon_r R^2} = \frac{Q}{4\pi\epsilon R^2}$$

$$P_{R2} = \left(1 - \frac{1}{\epsilon_r}\right) \frac{Q}{4\pi R^2}$$

$$V_2 = - \int_{\infty}^{R_o} E_{R1} dR - \int_{R_o}^R E_{R2} dR$$

$$= V_1|_{R=R_o} - \frac{Q}{4\pi\epsilon} \int_{R_o}^R \frac{1}{R^2} dR$$

$$= \frac{Q}{4\pi\epsilon_0} \left[\left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{R_o} + \frac{1}{\epsilon_r R} \right]$$

- ▶ For $R < R_i$

$$E_{R3} = \frac{Q}{4\pi\epsilon_0 R^2} \quad D_{R3} = \frac{Q}{4\pi R^2} \quad P_{R3} = 0$$

$$\begin{aligned} V_3 &= V_2|_{R=R_i} - \int_{R_i}^R E_{R3} dR \\ &= \frac{Q}{4\pi\epsilon_0} \left[\left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{R_o} - \left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{R_i} + \frac{1}{R} \right] \end{aligned}$$

$$\begin{aligned} \rho_{ps}|_{R=R_i} &= \mathbf{P} \cdot (-\mathbf{a}_R)|_{R=R_i} = -P_{R2}|_{R=R_i} \\ &= -\left(1 - \frac{1}{\epsilon_r}\right) \frac{Q}{4\pi R_i^2} \end{aligned}$$

$$\begin{aligned} \rho_{ps}|_{R=R_o} &= \mathbf{P} \cdot \mathbf{a}_R|_{R=R_o} = P_{R2}|_{R=R_o} \\ &= \left(1 - \frac{1}{\epsilon_r}\right) \frac{Q}{4\pi R_o^2} \end{aligned}$$

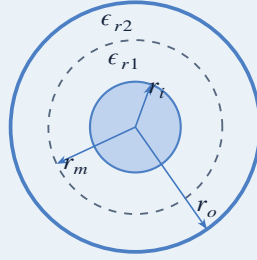
$$\begin{aligned} \rho_p &= -\nabla \cdot \mathbf{P} \\ &= -\frac{1}{R^2} \frac{\partial}{\partial R} (R^2 P_{R2}) = 0 \end{aligned}$$

2.12 Dielectric Strength

- An external electric field causes small displacements of the bound charges and creates electric dipoles.
- If the electric field is strong, it completely detaches electrons from the molecules, accelerates them and makes them collide with the molecular lattice structure.
- In this case the material may become conducting and large currents may arise.
 - This phenomenon is called **dielectric breakdown**.
- The maximum electric field intensity that a dielectric material can withstand without breakdown is called the **dielectric strength** of the material.
 - For example, the dielectric strength of air is 3 kV/mm.

Example 2-14

When a coaxial cable is used to carry electric power, the radius of the inner conductor is determined by the load current, and the overall size by the voltage and the insulating material used. Suppose that the radius of the inner conductor is 0.4 cm and that two layers, rubber with relative permittivity $\epsilon_{rr} = 3.2$ and polystyrene with relative permittivity $\epsilon_{rp} = 2.6$, are used as the insulating material. Design this cable so that it can operate at a voltage of 20 kV. To avoid breakdown due to abnormal external conditions, the maximum electric field intensity in the insulating materials must not exceed 25% of their dielectric strengths. (The dielectric strength of rubber is 25×10^6 V/m and that of polystyrene is 20×10^6 V/m.)



► Solution.

- The aim is to determine the order of the insulating layers and the values of r_o and r_m .
- Assuming that the free charge density on the surface of the inner conductor is ρ_s , Gauss's law gives

$$D_r 2\pi r L = \rho_s 2\pi r_i L \quad D_r = \rho_s \frac{r_i}{r} \quad E_r = \frac{\rho_s}{\epsilon_0 \epsilon_{r1,2}} \frac{r_i}{r}$$

$$\text{Rubber} \quad \max\{E_{rr}\} = 0.25 \times 25 \times 10^6 \text{ V/m} = \frac{\rho_s}{\epsilon_0 \epsilon_{rr}} \frac{r_i}{r_{r \min}} \quad r_{r \min} = r_i \text{ or } r_m$$

$$\text{Polystyrene} \quad \max\{E_{rp}\} = 0.25 \times 20 \times 10^6 \text{ V/m} = \frac{\rho_s}{\epsilon_0 \epsilon_{rp}} \frac{r_i}{r_{p \min}} \quad r_{p \min} = r_i \text{ or } r_m$$

$$\frac{25}{20} = \frac{\epsilon_{rp}}{\epsilon_{rr}} \times \frac{r_{p \min}}{r_{r \min}} \quad r_{p \min} > r_{r \min} \Rightarrow \begin{cases} r_{r \min} = r_i \\ r_{p \min} = r_m \end{cases}$$

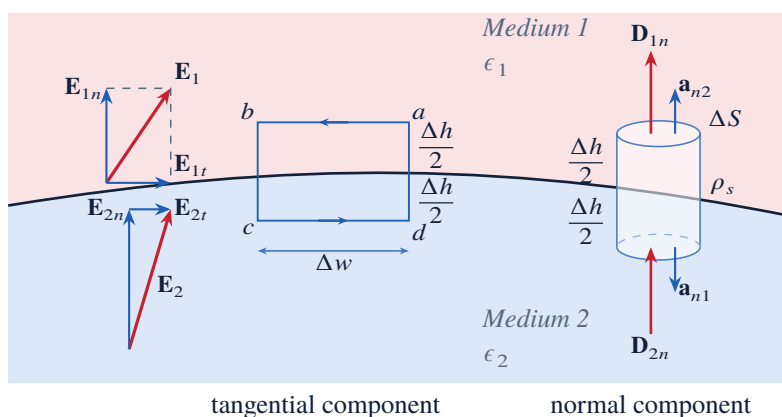
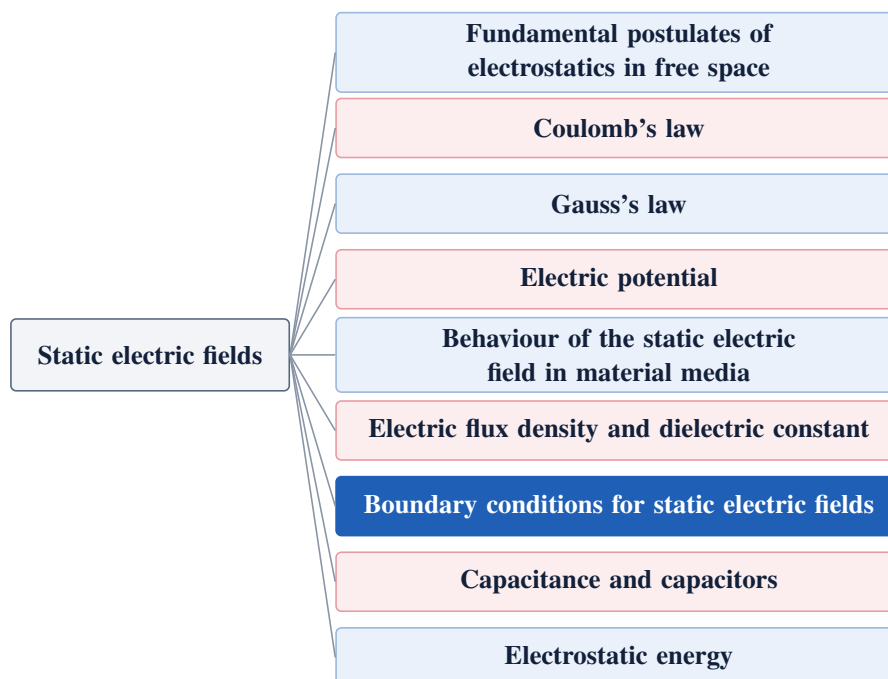
$$\frac{25}{20} = \frac{2.6}{3.2} \times \frac{r_m}{0.4} \Rightarrow r_m = 0.616 \text{ cm}$$

$$\begin{aligned} - \int_{r_o}^{r_m} E_{rp} dr - \int_{r_m}^{r_i} E_{rr} dr &= \frac{\rho_s r_i}{\epsilon_0} \left(\frac{1}{\epsilon_{rp}} \ln \left(\frac{r_o}{r_m} \right) + \frac{1}{\epsilon_{rr}} \ln \left(\frac{r_m}{r_i} \right) \right) \\ &= 20000 \text{ V} \end{aligned}$$

$$\frac{\rho_s r_i}{\epsilon_0} = 0.25 \times 25 \times 10^6 \epsilon_{rr} r_i = 8 \times 10^4$$

$$\Rightarrow r_o = 0.832 \text{ cm}$$

2.13 Boundary Conditions for Static Electric Fields



► Tangential component:

$$\oint_{ab c d a} \mathbf{E} \cdot d\boldsymbol{\ell} = \mathbf{E}_1 \cdot \Delta \mathbf{w} + \mathbf{E}_2 \cdot (-\Delta \mathbf{w}) = E_{1t} \Delta w - E_{2t} \Delta w = 0$$

⚡ Important result

$$E_{1t} = E_{2t} \qquad \frac{D_{1t}}{\epsilon_1} = \frac{D_{2t}}{\epsilon_2}$$

- Normal component:

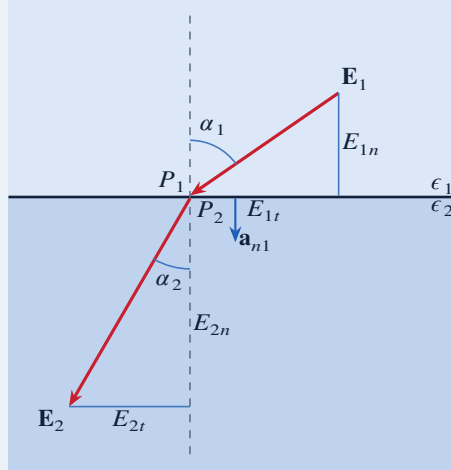
$$\begin{aligned}\oint_S \mathbf{D} \cdot d\mathbf{s} &= (\mathbf{D}_1 \cdot \mathbf{a}_{n2} + \mathbf{D}_2 \cdot \mathbf{a}_{n1}) \Delta S \\ &= \mathbf{a}_{n2} \cdot (\mathbf{D}_1 - \mathbf{D}_2) \Delta S \\ &= \rho_s \Delta S\end{aligned}$$

⚡ Important result

$$\mathbf{a}_{n2} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s \qquad \mathbf{a}_{n2} \cdot (\epsilon_1 \mathbf{E}_1 - \epsilon_2 \mathbf{E}_2) = \rho_s$$

Example 2-15

Two dielectric media with permittivities ϵ_1 and ϵ_2 are separated by a charge-free boundary as shown in the figure below. The electric field intensity in medium 1 at the point P_1 has magnitude E_1 and makes an angle α_1 with the normal. Determine the magnitude and direction of the electric field intensity at the point P_2 in region 2.



► **Solution.**

$$E_{1t} = E_{2t} \quad \Rightarrow \quad E_2 \sin \alpha_2 = E_1 \sin \alpha_1$$

$$\mathbf{a}_{n2} \cdot (\epsilon_1 \mathbf{E}_1 - \epsilon_2 \mathbf{E}_2) = \epsilon_1 E_{1n} - \epsilon_2 E_{2n} = 0 \quad \Rightarrow \quad \epsilon_2 E_2 \cos \alpha_2 = \epsilon_1 E_1 \cos \alpha_1$$

⚡ Important result

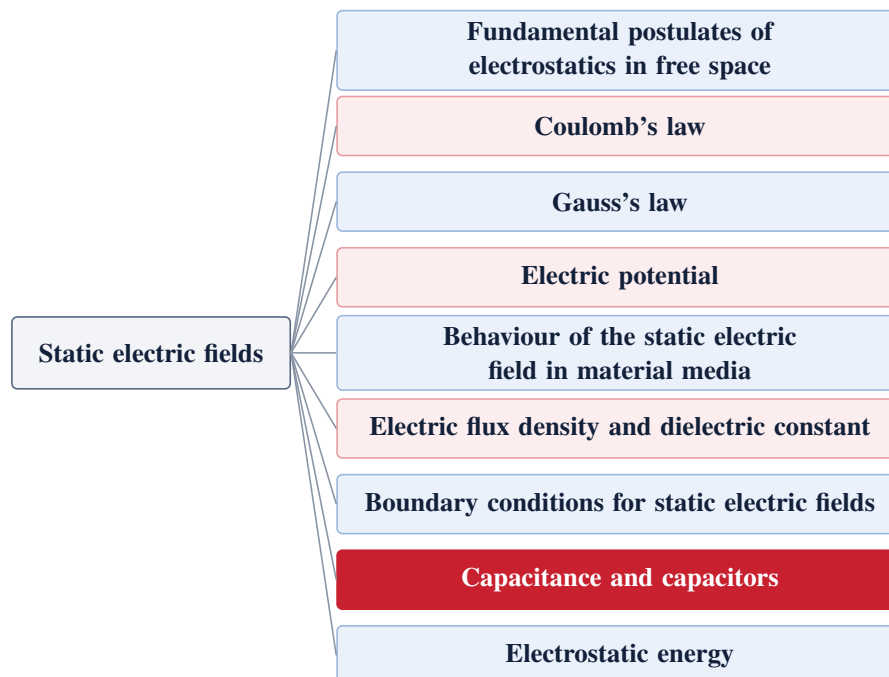
$$\frac{\tan \alpha_2}{\tan \alpha_1} = \frac{\epsilon_2}{\epsilon_1}$$

$$\begin{aligned} E_2 &= \sqrt{E_{2t}^2 + E_{2n}^2} = \sqrt{(E_2 \sin \alpha_2)^2 + (E_2 \cos \alpha_2)^2} \\ &= \left[(E_1 \sin \alpha_1)^2 + \left(\frac{\epsilon_1}{\epsilon_2} E_1 \cos \alpha_1 \right)^2 \right]^{1/2} \end{aligned}$$

⚡ Important result

$$E_2 = E_1 \left[\sin^2 \alpha_1 + \left(\frac{\epsilon_1}{\epsilon_2} \cos \alpha_1 \right)^2 \right]^{1/2}$$

2.14 Capacitance and Capacitors

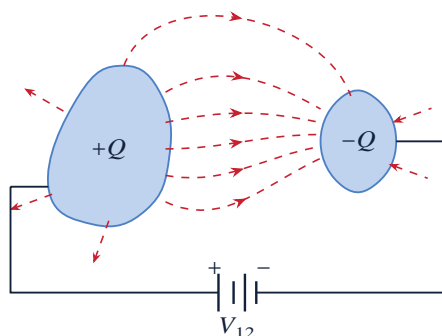


- What kind of element is a capacitor, and what is the concept of capacitance?
- How is the capacitance of various structures computed?
- How can the electric energy stored in a set of charges, or the work needed to assemble a set of charges, be found?
- How can the electric energy stored in capacitive structures be computed?
- Suppose a charge Q is placed on an arbitrary conductor. This charge distributes itself over the surface of the conductor and forms the charge density ρ_s .
- Increasing the total charge by a factor k increases the charge density ρ_s by the factor k , and consequently the potential due to this charge also increases by the factor k .
 - Therefore the ratio of Q to V remains constant

⚡ Important result

$$Q = CV$$

- ▶ The constant of proportionality (C) is called the **capacitance of the isolated conducting body**; it is the electric charge that must be added to the body to raise its electric potential by one unit.
 - Its unit is the coulomb per volt, or farad (F).
- ▶ Two conductors separated by free space or by a dielectric medium are called a capacitor.



- ▶ The capacitance of a capacitor is defined as follows

Definition

$$C = \frac{Q}{V_{12}} \quad (\text{F})$$

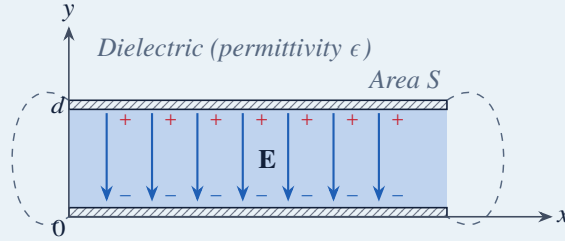
- ▶ The capacitance of a capacitor depends on its geometry and on the permittivity of the medium between the two conductors.
- ▶ Procedure for computing the capacitance of a capacitor
 - Choosing an appropriate coordinate system
 - Placing the charges $+Q$ and $-Q$ on the conductors
 - Finding $\mathbf{E}$ from Q
 - Computing the potential difference between the two conductors from the following relation, from the charge $-Q$ to the charge $+Q$

$$V = - \int_{P-Q}^{P+Q} \mathbf{E} \cdot d\ell$$

- Finding the capacitance from the ratio Q/V

Example 2-16

A parallel-plate capacitor consists of two parallel conducting plates of area S separated by d . The space between the plates is filled with a dielectric of constant permittivity ϵ . Find the capacitance of this capacitor.



► Solution.

- Using Cartesian coordinates
- Placing the charges $+Q$ and $-Q$ on the plates $\rightarrow \rho_s = \frac{Q}{S}$
- Computing the electric field intensity in the space between the two conducting plates with Gauss's law, neglecting the fringing fields

$$\mathbf{E} = -\mathbf{a}_y \frac{\rho_s}{\epsilon} = -\mathbf{a}_y \frac{Q}{\epsilon S}$$

- Computing the potential difference between the conducting plates in terms of Q

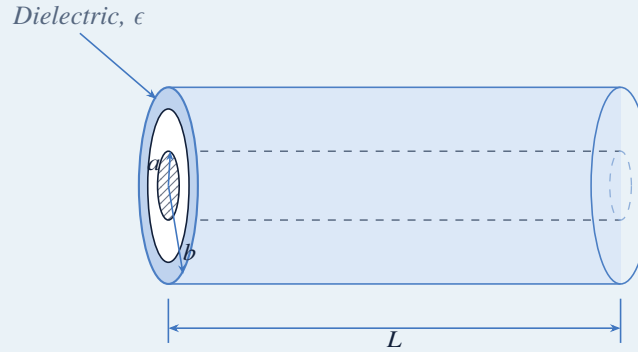
$$V_{12} = - \int_{y=0}^{y=d} \mathbf{E} \cdot d\ell = - \int_0^d \left(-\mathbf{a}_y \frac{Q}{\epsilon S} \right) \cdot (\mathbf{a}_y dy) = \frac{Q}{\epsilon S} d$$

- Determining the capacitance

$$C = \frac{Q}{V_{12}} = \epsilon \frac{S}{d}$$

Example 2-17

A cylindrical capacitor consists of an inner conductor of radius a and an outer conductor of radius b . The space between the two conductors is filled with a dielectric of constant permittivity ϵ , and the length of the capacitor is L . Find the capacitance of this capacitor.



► Solution.

- Using cylindrical coordinates
- Placing the charges $+Q$ and $-Q$ on the inner and outer conductors respectively
- Computing the electric field intensity in the space between the two conductors with Gauss's law

$$\mathbf{E} = \mathbf{a}_r E_r = \mathbf{a}_r \frac{Q}{2\pi\epsilon Lr}$$

- Computing the potential difference between the conductors in terms of Q

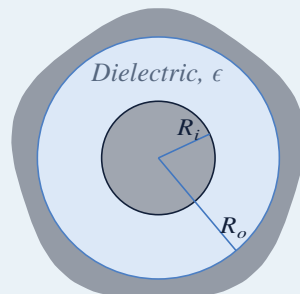
$$\begin{aligned} V_{ab} &= - \int_{r=b}^{r=a} \mathbf{E} \cdot d\ell = - \int_b^a \left(\mathbf{a}_r \frac{Q}{2\pi\epsilon Lr} \right) \cdot (\mathbf{a}_r dr) \\ &= \frac{Q}{2\pi\epsilon L} \ln\left(\frac{b}{a}\right) \end{aligned}$$

- Determining the capacitance

$$C = \frac{Q}{V_{ab}} = \frac{2\pi\epsilon L}{\ln\left(\frac{b}{a}\right)}$$

Example 2-18

A spherical capacitor consists of an inner conducting sphere of radius R_i and an outer conducting sphere of radius R_o . The space between the two conductors is filled with a dielectric of constant permittivity ϵ . Determine the capacitance of this capacitor.



► Solution.

- Using spherical coordinates
- Placing the charges $+Q$ and $-Q$ on the inner and outer conductors respectively
- Computing the electric field intensity in the space between the two conductors with Gauss's law

$$\mathbf{E} = \mathbf{a}_R E_R = \mathbf{a}_R \frac{Q}{4\pi\epsilon R^2}$$

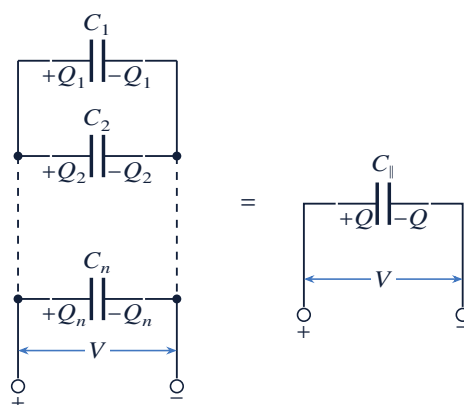
- Computing the potential difference between the conductors in terms of Q

$$V = - \int_{R_o}^{R_i} \mathbf{E} \cdot (\mathbf{a}_R dR) = - \int_{R_o}^{R_i} \frac{Q}{4\pi\epsilon R^2} dR = \frac{Q}{4\pi\epsilon} \left(\frac{1}{R_i} - \frac{1}{R_o} \right)$$

- Determining the capacitance

$$C = \frac{Q}{V} = \frac{4\pi\epsilon}{\frac{1}{R_i} - \frac{1}{R_o}}$$

► Capacitors connected in parallel



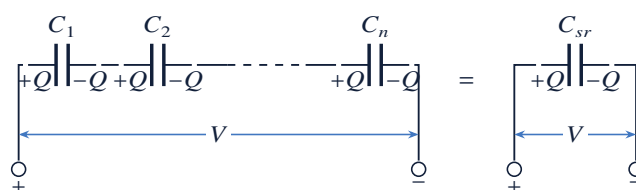
$$Q = Q_1 + Q_2 + \dots + Q_n$$

$$= C_1 V + C_2 V + \dots + C_n V = C_{\parallel} V$$

⚡ Important result

$$C_{\parallel} = C_1 + C_2 + \dots + C_n$$

► Capacitors connected in series



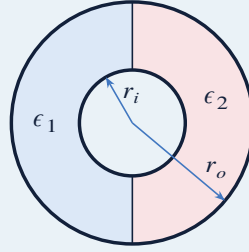
$$V = \frac{Q}{C_{sr}} = \frac{Q}{C_1} + \frac{Q}{C_2} + \dots + \frac{Q}{C_n}$$

⚡ Important result

$$\frac{1}{C_{sr}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$$

🔗 Example 2-19

A cylindrical capacitor of length L is formed by coaxial cylindrical conducting surfaces of radii r_i and r_o . Two dielectric materials of permittivities ϵ_1 and ϵ_2 fill the space between the conductors as shown in the figure below. Determine the capacitance of this capacitor.



► **Solution.**

► **First method**

- The capacitance of a cylindrical capacitor $\rightarrow C = \frac{2\pi\epsilon L}{\ln\left(\frac{r_o}{r_i}\right)}$
- Since each dielectric fills half of the space, the capacitance of each half is

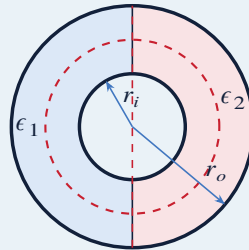
$$C_1 = \frac{\pi\epsilon_1 L}{\ln\left(\frac{r_o}{r_i}\right)} \quad C_2 = \frac{\pi\epsilon_2 L}{\ln\left(\frac{r_o}{r_i}\right)}$$

- Since the two capacitors C_1 and C_2 are in parallel, the total capacitance is

$$C = C_1 + C_2 = \frac{2\pi L}{\ln\left(\frac{r_o}{r_i}\right)} \left(\frac{\epsilon_1 + \epsilon_2}{2} \right)$$

► **Second method**

- Using cylindrical coordinates
- Placing the charges $+Q$ and $-Q$ on the inner and outer conductors respectively
- Computing the electric field intensity in the space between the two conductors with Gauss's law



$$\left. \begin{aligned} E_{r1} &= \frac{Q_1}{\pi r \epsilon_1 L} \\ E_{r2} &= \frac{Q_2}{\pi r \epsilon_2 L} \end{aligned} \right\} E_{r1} = E_{r2} \Rightarrow \left. \begin{aligned} \frac{Q_1}{\epsilon_1} &= \frac{Q_2}{\epsilon_2} \\ Q_1 + Q_2 &= Q \end{aligned} \right\} \left\{ \begin{aligned} Q_1 &= \frac{Q\epsilon_1}{\epsilon_1 + \epsilon_2} \\ Q_2 &= \frac{Q\epsilon_2}{\epsilon_1 + \epsilon_2} \end{aligned} \right.$$

- Computing the potential difference between the conductors in terms of Q

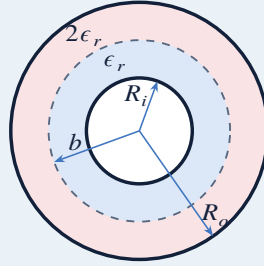
$$V = - \int_{r_o}^{r_i} \frac{Q_1}{\pi r \epsilon_1 L} dr = - \int_{r_o}^{r_i} \frac{Q}{\pi r L (\epsilon_1 + \epsilon_2)} dr = \frac{Q}{\pi L (\epsilon_1 + \epsilon_2)} \ln\left(\frac{r_o}{r_i}\right)$$

- Determining the capacitance

$$C = \frac{2\pi L}{\ln\left(\frac{r_o}{r_i}\right)} \left(\frac{\epsilon_1 + \epsilon_2}{2}\right)$$

Example 2-20

A capacitor is formed by two concentric conducting spherical shells of radii R_i and R_o . The space between the shells is filled with a dielectric of relative permittivity ϵ_r from R_i to b and a dielectric of relative permittivity $2\epsilon_r$ from b to R_o . Determine the capacitance of this capacitor.



► Solution.

► First method

- The capacitance of a spherical capacitor $\rightarrow C = \frac{Q}{V} = \frac{4\pi\epsilon}{\frac{1}{R_i} - \frac{1}{R_o}}$
- This capacitor can be regarded as two capacitors in series, as follows

$$C_1 = \frac{4\pi\epsilon_0\epsilon_r}{\frac{1}{R_i} - \frac{1}{b}} \quad C_2 = \frac{4\pi\epsilon_0(2\epsilon_r)}{\frac{1}{b} - \frac{1}{R_o}}$$

- Therefore the total capacitance is

$$C = \frac{1}{1/C_1 + 1/C_2} = \frac{4\pi\epsilon_0\epsilon_r}{\frac{1}{R_i} - \frac{1}{2b} - \frac{1}{2R_o}}$$

► Second method

- Using spherical coordinates
- Placing the charges $+Q$ and $-Q$ on the inner and outer conductors respectively
- Computing the electric field intensity in the space between the two conductors with Gauss's law

$$E_{R1} = \frac{Q}{4\pi\epsilon_0\epsilon_r R^2} \quad E_{R2} = \frac{Q}{4\pi\epsilon_0(2\epsilon_r)R^2}$$

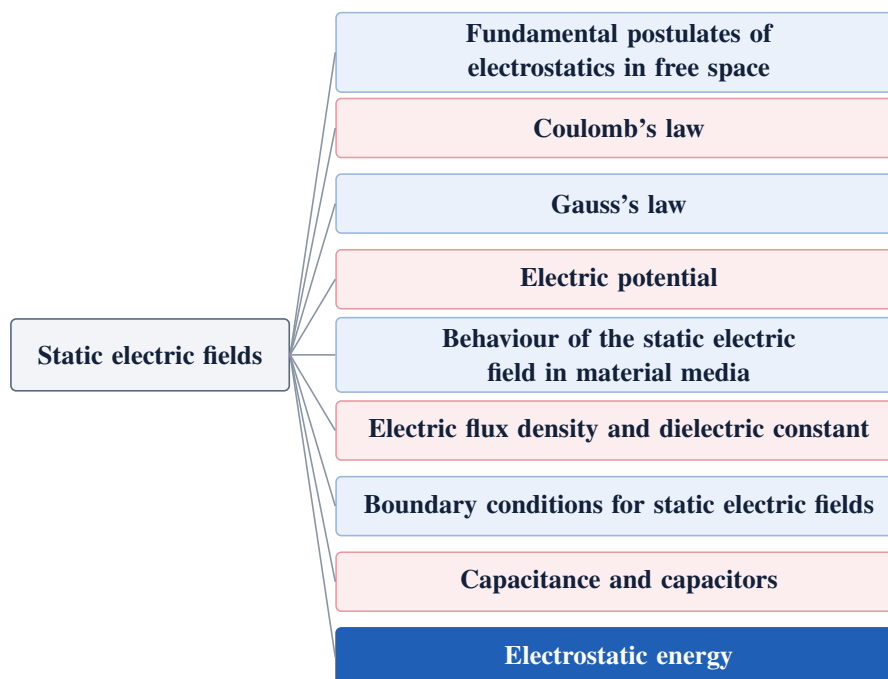
- Computing the potential difference between the conductors in terms of Q

$$V = - \int_{R_o}^b \frac{Q}{4\pi\epsilon_0(2\epsilon_r)R^2} dR - \int_b^{R_i} \frac{Q}{4\pi\epsilon_0\epsilon_r R^2} dR = \frac{Q}{4\pi\epsilon_0\epsilon_r} \left(\frac{1}{R_i} - \frac{1}{2b} - \frac{1}{2R_o} \right)$$

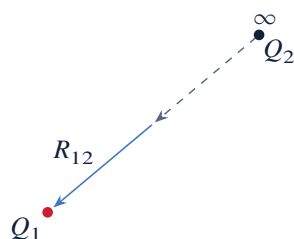
- Determining the capacitance

$$C = \frac{4\pi\epsilon_0\epsilon_r}{\frac{1}{R_i} - \frac{1}{2b} - \frac{1}{2R_o}}$$

2.15 Electrostatic Energy



- The electric potential of a point in an electric field is equal to the work needed to bring a unit positive charge from infinity (the zero-potential reference) to that point.
- To bring a charge Q_2 from infinity to the distance R_{12} from the charge Q_1 , the work needed is



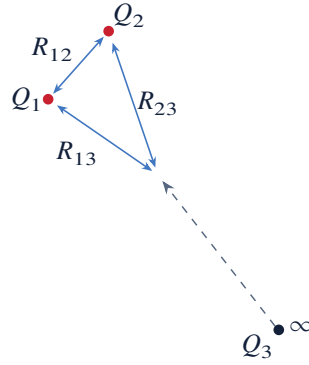
$$W_2 = Q_2 V_2 = Q_2 \frac{Q_1}{4\pi\epsilon_0 R_{12}}$$

- V_2 : the potential at the location of the charge Q_2

$$W_2 = Q_1 \frac{Q_2}{4\pi\epsilon_0 R_{12}} = Q_1 V_1$$

$$W_2 = \frac{1}{2}(Q_1 V_1 + Q_2 V_2)$$

- This work is stored in the system of two charges as potential energy.
- Now suppose another charge Q_3 is brought from infinity to a point at the distance R_{13} from Q_1 and R_{23} from Q_2 ; the additional work needed is



$$\Delta W = Q_3 V_3 = Q_3 \left(\frac{Q_1}{4\pi\epsilon_0 R_{13}} + \frac{Q_2}{4\pi\epsilon_0 R_{23}} \right)$$

- V_3 : the potential at the location of the charge Q_3
- Thus the potential energy stored in the system of three charges is

$$\begin{aligned} W_3 &= W_2 + \Delta W = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1 Q_2}{R_{12}} + \frac{Q_1 Q_3}{R_{13}} + \frac{Q_2 Q_3}{R_{23}} \right) \\ W_3 &= \frac{1}{2} \left[Q_1 \left(\frac{Q_2}{4\pi\epsilon_0 R_{12}} + \frac{Q_3}{4\pi\epsilon_0 R_{13}} \right) + Q_2 \left(\frac{Q_1}{4\pi\epsilon_0 R_{12}} + \frac{Q_3}{4\pi\epsilon_0 R_{23}} \right) \right. \\ &\quad \left. + Q_3 \left(\frac{Q_1}{4\pi\epsilon_0 R_{13}} + \frac{Q_2}{4\pi\epsilon_0 R_{23}} \right) \right] = \frac{1}{2} (Q_1 V_1 + Q_2 V_2 + Q_3 V_3) \end{aligned}$$

- Thus the potential energy of N discrete point charges at rest is

⚡ Important result

$$W_e = \frac{1}{2} \sum_{k=1}^N Q_k V_k \quad (\text{J})$$

- where V_k is the electric potential at the point of Q_k due to the other charges.
- The SI unit of energy is the joule, which is a large unit. Here another unit, the electron-volt (eV), is used.
 - One electron-volt is the energy or work needed to move an electron against a potential difference of one volt.

$$1 \text{ (eV)} = (1.60 \times 10^{-19}) \times 1 = 1.60 \times 10^{-19} \quad (\text{J})$$

- In a continuous charge distribution of density ρ the stored energy is

⚡ Important result

$$W_e = \frac{1}{2} \int_{V'} \rho V dv \quad (\text{J})$$

- ▶ where V is the potential at the point where the volume charge density is ρ , and V' is the volume of the region in which ρ exists.

Example 2-21

Find the energy required to assemble a uniformly charged sphere of radius b and volume charge density ρ .

Solution.

$$\begin{aligned}
 W_e &= \frac{\rho}{2} \int_{V'} V dv = \frac{\rho}{2} \int_0^b V 4\pi R^2 dR \\
 \mathbf{E}_{R1} &= \mathbf{a}_R \frac{Q}{4\pi\epsilon_0 R^2} = \mathbf{a}_R \frac{\rho b^3}{3\epsilon_0 R^2}, \quad R \geq b \\
 \mathbf{E}_{R2} &= \mathbf{a}_R \frac{Q_R}{4\pi\epsilon_0 R^2} = \mathbf{a}_R \frac{\rho R}{3\epsilon_0}, \quad 0 < R \leq b \\
 V &= - \int_{\infty}^R \mathbf{E} \cdot d\mathbf{R} = - \left[\int_{\infty}^b E_{R1} dR + \int_b^R E_{R2} dR \right] \\
 &= - \left[\int_{\infty}^b \frac{\rho b^3}{3\epsilon_0 R^2} dR + \int_b^R \frac{\rho R}{3\epsilon_0} dR \right] \\
 &= \frac{\rho}{3\epsilon_0} \left(b^2 + \frac{b^2}{2} - \frac{R^2}{2} \right) = \frac{\rho}{3\epsilon_0} \left(\frac{3}{2}b^2 - \frac{R^2}{2} \right) \\
 W_e &= \frac{\rho}{2} \int_0^b \frac{\rho}{3\epsilon_0} \left(\frac{3}{2}b^2 - \frac{R^2}{2} \right) 4\pi R^2 dR = \frac{4\pi\rho^2 b^5}{15\epsilon_0}
 \end{aligned}$$

- ▶ Earlier we obtained the electrostatic energy of a charge distribution in terms of the charge density and the potential.
- ▶ Sometimes it is better to express this energy in terms of field quantities.
- ▶ It can be shown that

Important result

$$W_e = \frac{1}{2} \int_{V'} \mathbf{D} \cdot \mathbf{E} dv \quad (\text{J})$$

$$W_e = \frac{1}{2} \int_{V'} \epsilon E^2 dv \quad (\text{J})$$

$$W_e = \frac{1}{2} \int_{V'} \frac{D^2}{\epsilon} dv \quad (\text{J})$$

► V' : all space

Example 2-22

Solve Example 2-21 using the expression of the energy in terms of field quantities.

► **Solution.**

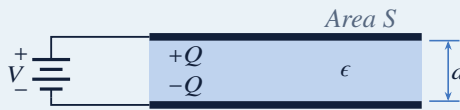
$$\mathbf{E}_{R1} = \mathbf{a}_R \frac{Q}{4\pi\epsilon_0 R^2} = \mathbf{a}_R \frac{\rho b^3}{3\epsilon_0 R^2}, \quad R \geq b$$

$$\mathbf{E}_{R2} = \mathbf{a}_R \frac{Q_R}{4\pi\epsilon_0 R^2} = \mathbf{a}_R \frac{\rho R}{3\epsilon_0}, \quad 0 < R \leq b$$

$$\begin{aligned} W_e &= \frac{1}{2} \epsilon_0 \int_{V'} |\mathbf{E}|^2 dv = \frac{1}{2} \epsilon_0 \int_0^b \left(\frac{\rho R}{3\epsilon_0} \right)^2 4\pi R^2 dR \\ &\quad + \frac{1}{2} \epsilon_0 \int_b^\infty \left(\frac{\rho b^3}{3\epsilon_0 R^2} \right)^2 4\pi R^2 dR = \frac{4\pi \rho^2 b^5}{15\epsilon_0} \end{aligned}$$

Example 2-23

In the figure below a parallel-plate capacitor of area S and plate separation d is charged by the voltage V . The permittivity of the insulator is ϵ . Find the stored energy.



► **Solution.**

$$E = \frac{V}{d}$$

$$W_e = \frac{1}{2} \int_{V'} \epsilon \left(\frac{V}{d} \right)^2 dv = \frac{1}{2} \epsilon \left(\frac{V}{d} \right)^2 (Sd) = \frac{1}{2} \left(\epsilon \frac{S}{d} \right) V^2$$

⚡ Important result

$$W_e = \frac{1}{2} CV^2 \quad (\text{J}) \quad \quad W_e = \frac{1}{2} QV \quad (\text{J}) \quad \quad W_e = \frac{Q^2}{2C} \quad (\text{J})$$

📌 Remark

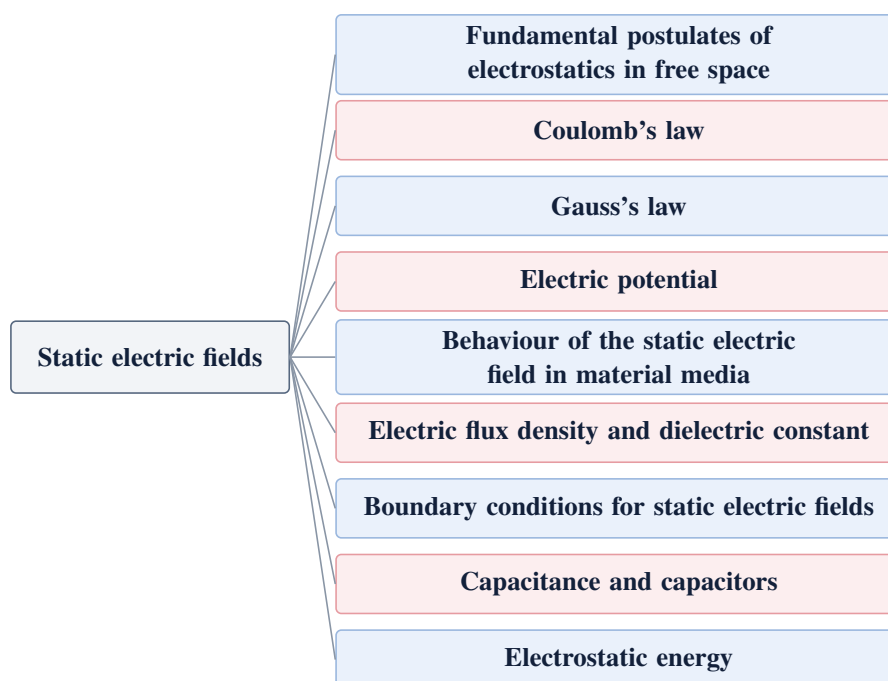
It can be proved that the above equations hold for any capacitor made of two conductors.

Example 2-24

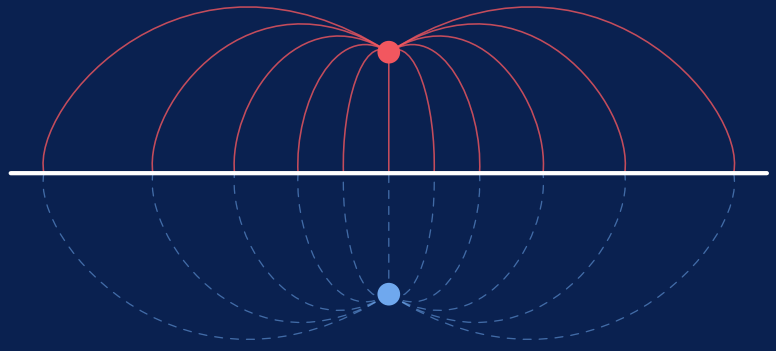
Using the energy relation, find the capacitance of a cylindrical capacitor of length L consisting of an inner conductor of radius a and an outer conductor of radius b , with permittivity ϵ .

► **Solution.**

$$\begin{aligned}\mathbf{E} &= \mathbf{a}_r E_r = \mathbf{a}_r \frac{Q}{2\pi\epsilon L r}, \quad a < r < b \\ W_e &= \frac{1}{2} \int_a^b \epsilon \left(\frac{Q}{2\pi\epsilon L r} \right)^2 (L 2\pi r dr) \\ &= \frac{Q^2}{4\pi\epsilon L} \int_a^b \frac{dr}{r} = \frac{Q^2}{4\pi\epsilon L} \ln \frac{b}{a} \\ \frac{Q^2}{2C} &= \frac{Q^2}{4\pi\epsilon L} \ln \frac{b}{a} \quad \Rightarrow \quad C = \frac{2\pi\epsilon L}{\ln \frac{b}{a}}\end{aligned}$$



3



CHAPTER

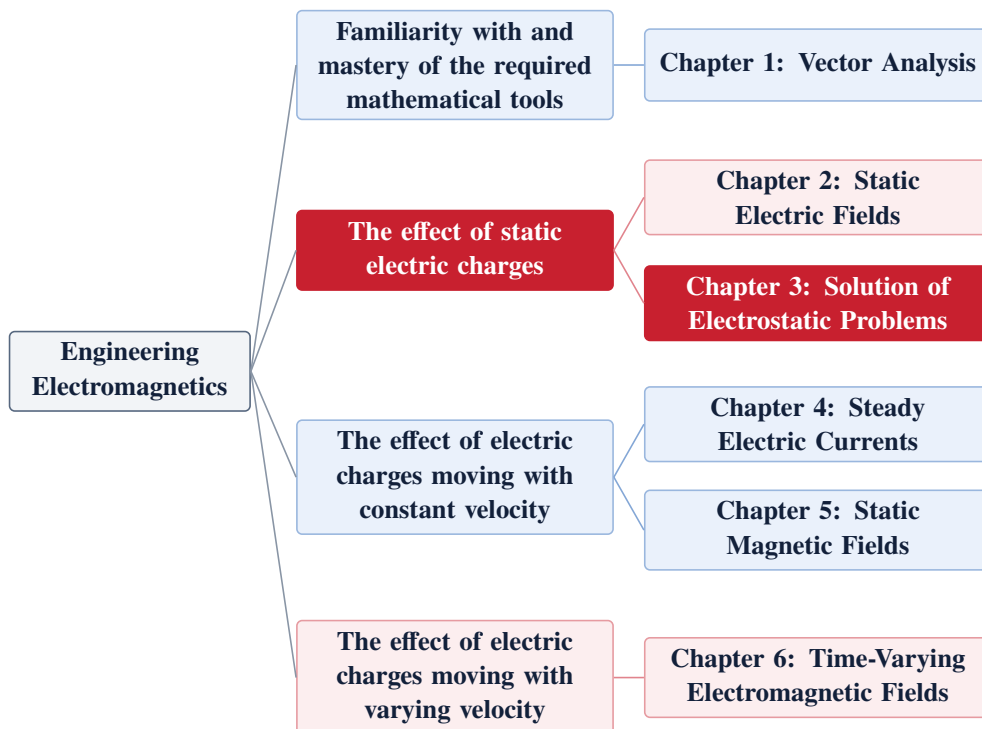
Solution of Electrostatic Problems

حل مسائل الكتريسيته ساكن

In this chapter

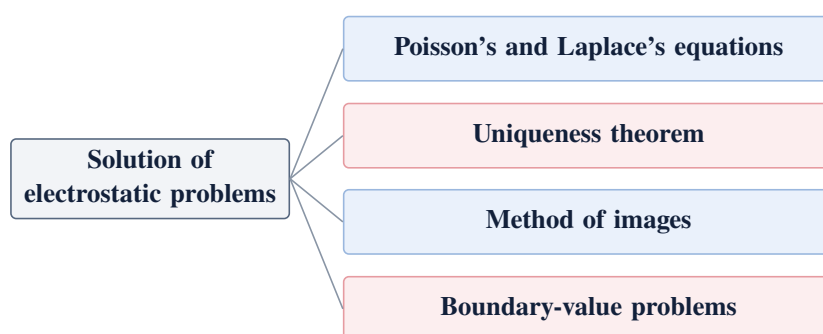
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3.1 Course Outline



3.2 Solution of Electrostatic Problems

- Electrostatic problems are problems that deal with the effects of electric charges at rest. Solving such problems usually involves determining the electric potential, the electric field intensity or the electric charge distribution.
- If the charge distribution is given, both the potential and the electric field can be computed with the relations of the previous chapter.
 - In many practical problems the exact charge distribution is not known everywhere, and finding the potential and the electric field directly with the formulas of the previous chapter is not possible. → the method of images can be used for a particular class of problems
 - In another kind of problem the potentials of all conducting bodies are known, and we want to find the potential and the electric field intensity in the surrounding space as well as the surface charge distributions on the conducting boundaries. → solving boundary-value problems



- What is the form of Poisson's and Laplace's equations that govern the electric potential, and when is each of them used?
- How can the electric potential and the electric field intensity be obtained by solving Poisson's and Laplace's equations?
- Is a solution of Poisson's equation that satisfies given boundary conditions unique?

3.3 Poisson's and Laplace's Equations

- The two fundamental equations of electrostatics in all media

$$\nabla \cdot \mathbf{D} = \rho \qquad \nabla \times \mathbf{E} = 0$$

- On the other hand we have

$$\mathbf{E} = -\nabla V$$

$$\mathbf{D} = \epsilon \mathbf{E} \quad \Rightarrow \quad \nabla \cdot \epsilon \mathbf{E} = \rho \quad \Rightarrow \quad \nabla \cdot (\epsilon \nabla V) = -\rho$$

- If the medium is homogeneous (ϵ constant)

⚡ Important result

$$\nabla^2 V = -\frac{\rho}{\epsilon}$$

- ρ : free charge density
- In the above equation, known as **Poisson's equation**, the **Laplacian** operator (the divergence of the gradient) is used.
- The Laplacian of a scalar quantity in the different coordinate systems

- Cartesian

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

- Cylindrical

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

- Spherical

$$\nabla^2 V = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

- Solving Poisson's equation in three-dimensional space with given boundary conditions is usually not a simple task.
- At points of a simple medium where there is no free charge

⚡ Important result

$$\nabla^2 V = 0$$

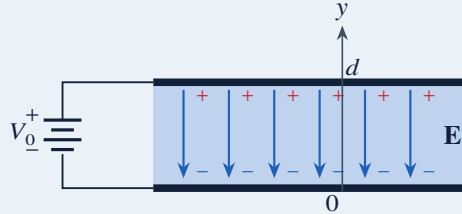
- The above equation is called **Laplace's equation**.
 - This equation governs problems involving a set of conductors held at different potentials.
 - Once V has been found from this equation, $\mathbf{E}$ and the charge distribution on the conducting surfaces are easily obtained

$$\mathbf{E} = -\nabla V \qquad \rho_s = -\epsilon E_n$$

Example 3-1

The two plates of a parallel-plate capacitor are separated by the distance d and held at the potentials 0 and V_0 as shown in the figure below. Neglecting the fringing fields, find

- (a) the potential at all points between the plates
- (b) the surface charge densities on the plates



➤ Solution.

- (a) Using Laplace's equation

$$\nabla^2 V = 0 \quad \Rightarrow \quad \frac{d^2 V}{dy^2} = 0 \quad \Rightarrow \quad \frac{dV}{dy} = C_1 \quad \Rightarrow \quad V = C_1 y + C_2$$

- Boundary conditions:

$$\left. \begin{array}{l} \text{At } y = 0, \quad V = 0 \\ \text{At } y = d, \quad V = V_0 \end{array} \right\} \Rightarrow V = \frac{V_0}{d} y$$

- (b)

$$\mathbf{E} = -\nabla V \quad \Rightarrow \quad \mathbf{E} = -\mathbf{a}_y \frac{dV}{dy} = -\mathbf{a}_y \frac{V_0}{d}$$

$$E_n = \mathbf{a}_n \cdot \mathbf{E} = \frac{\rho_s}{\epsilon}$$

- For the lower plate

$$\mathbf{a}_n = \mathbf{a}_y, \quad E_{n\ell} = -\frac{V_0}{d}, \quad \rho_{s\ell} = -\frac{\epsilon V_0}{d}$$

- For the upper plate

$$\mathbf{a}_n = -\mathbf{a}_y, \quad E_{nu} = \frac{V_0}{d}, \quad \rho_{su} = \frac{\epsilon V_0}{d}$$

Example 3-2

Solving Poisson's and Laplace's equations for V , determine the field $\mathbf{E}$ inside and outside a spherical electron cloud with uniform volume charge density $\rho = -\rho_0$ for $0 \leq R \leq b$ and $\rho = 0$ for $R > b$.

► Solution.

- Inside the electron cloud $\rightarrow$ using Poisson's equation

$$\nabla^2 V = -\frac{\rho}{\epsilon} \Rightarrow \frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{dV_i}{dR} \right) = \frac{\rho_0}{\epsilon_0} \Rightarrow \frac{d}{dR} \left(R^2 \frac{dV_i}{dR} \right) = \frac{\rho_0}{\epsilon_0} R^2 \Rightarrow R^2 \frac{dV_i}{dR} = \frac{\rho_0}{3\epsilon_0} R^3 + C_1$$

$$\frac{dV_i}{dR} = \frac{\rho_0}{3\epsilon_0} R + \frac{C_1}{R^2} \Rightarrow \mathbf{E}_i = -\nabla V_i = -\mathbf{a}_R \left(\frac{dV_i}{dR} \right)$$

- Since the field at $R = 0$ cannot be infinite

$$C_1 = 0 \Rightarrow \mathbf{E}_i = -\mathbf{a}_R \frac{\rho_0}{3\epsilon_0} R, \quad 0 \leq R \leq b$$

- Outside the electron cloud $\rightarrow$ using Laplace's equation

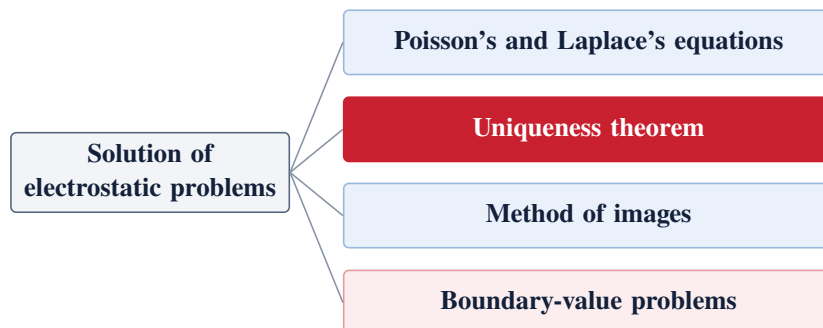
$$\nabla^2 V = 0 \Rightarrow \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{dV_o}{dR} \right) = 0 \Rightarrow \frac{dV_o}{dR} = \frac{C_2}{R^2}$$

$$\mathbf{E}_o = -\nabla V_o = -\mathbf{a}_R \frac{dV_o}{dR} = -\mathbf{a}_R \frac{C_2}{R^2}$$

- Because the medium is continuous, the fields $\mathbf{E}_i$ and $\mathbf{E}_o$ are equal at $R = b$

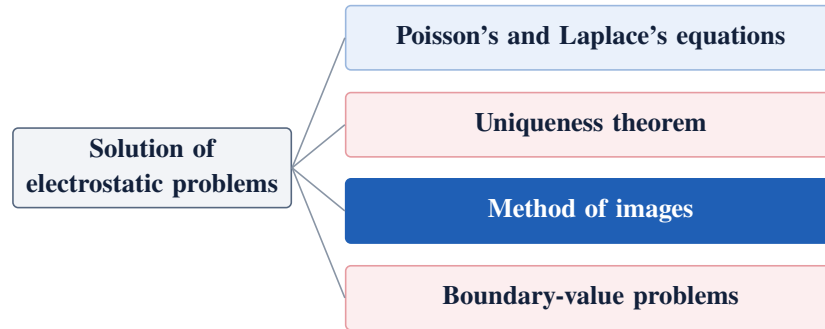
$$\frac{C_2}{b^2} = \frac{\rho_0}{3\epsilon_0} b \Rightarrow C_2 = \frac{\rho_0 b^3}{3\epsilon_0} \Rightarrow \mathbf{E}_o = -\mathbf{a}_R \frac{\rho_0 b^3}{3\epsilon_0 R^2}, \quad R \geq b$$

3.4 Uniqueness Theorem

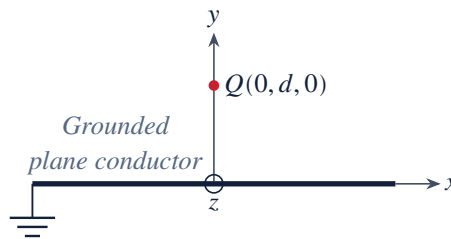


- A solution of Poisson's equation (and, as a special case, of Laplace's equation) that satisfies the given boundary conditions is the unique solution of that equation.
- The consequence of this theorem is that a solution of any electrostatic problem that satisfies the boundary conditions is the only possible solution, regardless of the method by which it was obtained.

3.5 Method of Images



- What is the method of images, and how can it be used to solve a particular class of electrostatic problems?
- In a class of electromagnetic problems, solving Laplace's equation while satisfying the boundary conditions is very difficult. In these problems, however, the boundary conditions can be satisfied with the help of **image charges**.
 - This method, which consists of replacing the boundary surfaces by equivalent image charges, is called the **method of images**.
- As an example consider the following problem. We want to find the potential at all points $y > 0$.



- Gauss's law cannot be used to compute the field and then the potential, because a Gaussian surface cannot be determined.
- Nor can the potential be computed directly, because the presence of the positive charge Q induces negative charges on the surface of the conductor, giving a surface charge density ρ_s . We have

$$V(x, y, z) = \frac{Q}{4\pi\epsilon_0\sqrt{x^2 + (y-d)^2 + z^2}} + \frac{1}{4\pi\epsilon_0} \int_S \frac{\rho_s}{R_1} ds$$

- The difficulty is that ρ_s must be found first, and even when ρ_s is known the surface integral is difficult to evaluate.
- Solving Laplace's equation to obtain the solution is also impossible or very difficult. In fact the solution of Laplace's equation must satisfy the following conditions

$$V(x, 0, z) = 0$$

$$V \rightarrow \frac{Q}{4\pi\epsilon_0 R}, \text{ as } R \rightarrow 0$$

- At points very far from the charge Q the potential must be zero

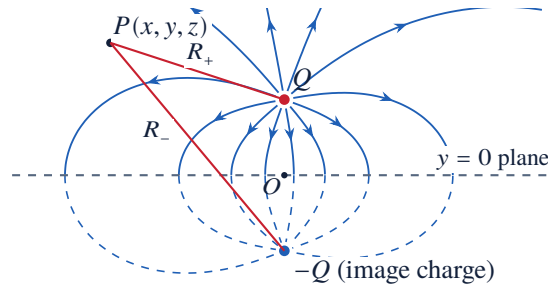
$$V(x, y, z) = V(-x, y, z)$$

$$V(x, y, z) = V(x, y, -z)$$

- Such a problem can be solved easily with the method of images.

➤ Point Charge and Conducting Plane

- We remove the conducting plane and replace it by an image point charge $-Q$ at $y = -d$



⚡ Important result

$$V(x, y, z) = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{R_+} - \frac{1}{R_-} \right)$$

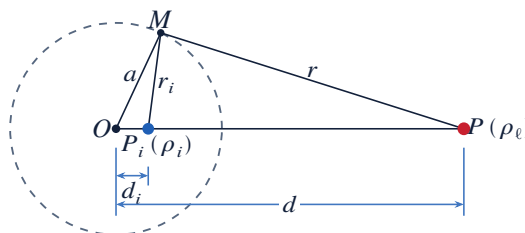
$$R_+ = [x^2 + (y - d)^2 + z^2]^{1/2}, \quad R_- = [x^2 + (y + d)^2 + z^2]^{1/2}$$

- It can easily be shown that this solution satisfies Laplace's equation and the boundary conditions. Therefore it is a solution of the problem and, by the uniqueness theorem, also its only solution.
- Once the potential is known, computing the field and the surface charge distribution on the conductor is easy.

① Remark

Note that this method cannot be used to compute the potential in the region $y < 0$.

➤ Line Charge and Parallel Conducting Cylinder



- ▶ The aim is to find the potential outside the conducting cylinder.
- ▶ The image must be a parallel line charge inside the cylinder that makes the cylinder surface $r = a$ an equipotential surface. The aim is to find d_i and ρ_i .
- ▶ We assume

⚡ Important result

$$\rho_i = -\rho_\ell$$

- ▶ The potential at the distance r from a line charge of density ρ_ℓ , with the reference at r_0 , is

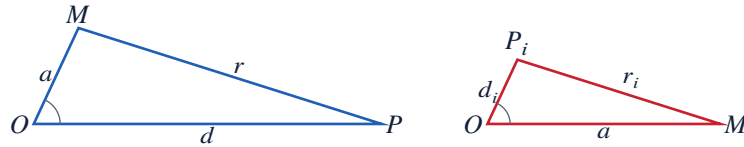
$$V = - \int_{r_0}^r E_r dr = - \frac{\rho_\ell}{2\pi\epsilon_0} \int_{r_0}^r \frac{1}{r} dr = \frac{\rho_\ell}{2\pi\epsilon_0} \ln \frac{r_0}{r}$$

- ▶ The potential at the point M on the surface of the cylinder is

$$\begin{aligned} V_M &= \frac{\rho_\ell}{2\pi\epsilon_0} \ln \frac{r_0}{r} - \frac{\rho_\ell}{2\pi\epsilon_0} \ln \frac{r_0}{r_i} \\ &= \frac{\rho_\ell}{2\pi\epsilon_0} \ln \frac{r_i}{r} \end{aligned}$$

- ▶ Therefore the equipotential surfaces are specified by

$$\frac{r_i}{r} = \text{Constant}$$



- ▶ If the condition $\frac{d_i}{a} = \frac{a}{d}$ holds, then by the similarity of the two triangles OMP_i and OPM the ratio $\frac{r_i}{r}$ is always constant. Therefore

⚡ Important result

$$d_i = \frac{a^2}{d}$$

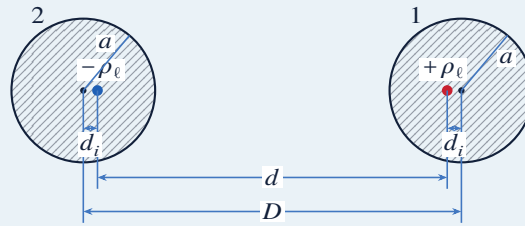
- ▶ Thus the line charge $-\rho_\ell$ can replace the conducting cylindrical surface, and the potential and the electric field at any point outside the cylinder can be found.

📌 Remark

Note that this method cannot be used to compute the potential in the region inside the conducting cylinder.

Example 3-3

Find the capacitance per unit length between two very long parallel circular conducting wires of radius a separated by D .



► Solution.

- We place the charges $+Q$ and $-Q$ on conductors 1 and 2 respectively.
- With the method of images, the equipotential surfaces of the wires can be replaced by two line charges $+\rho_\ell$ and $-\rho_\ell$.

$$V_2 = \frac{\rho_\ell}{2\pi\epsilon_0} \ln \frac{a}{d}$$

$$V_1 = -\frac{\rho_\ell}{2\pi\epsilon_0} \ln \frac{a}{d}$$

$$C = \frac{\rho_\ell}{V_1 - V_2} = \frac{\pi\epsilon_0}{\ln(d/a)}$$

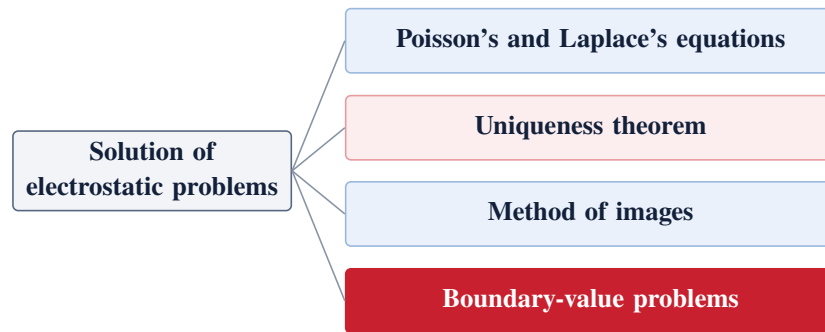
$$d = D - d_i = D - \frac{a^2}{d} \quad \Rightarrow \quad d = \frac{1}{2} \left(D + \sqrt{D^2 - 4a^2} \right)$$

$$C = \frac{\pi\epsilon_0}{\ln \left[(D/2a) + \sqrt{(D/2a)^2 - 1} \right]} \quad (\text{F/m})$$

⚡ Important result

$$C = \frac{\pi\epsilon_0}{\cosh^{-1}(D/2a)} \quad (\text{F/m})$$

3.6 Boundary-Value Problems



- The method of images applies to problems involving free charges near conducting boundaries of simple geometry.
- If the problem involves a set of conductors held at given potentials, with no isolated free charges, it cannot be solved by the method of images, and Laplace's equation must be solved.
- Laplace's equation is a partial differential equation. Problems governed by partial differential equations with given boundary conditions are called **boundary-value problems**.
- Boundary-value problems for potential functions are divided into classes
 - Dirichlet problems: the value of the potential is specified at all points of the boundaries.
 - Neumann problems: the normal derivative of the potential is specified on all boundaries.
 - Mixed boundary-value problems: the potential is specified on some of the boundaries and the normal derivative of the potential on the rest.
- How are electrostatic boundary-value problems solved in the different coordinate systems to obtain the potential distribution?

➤ Cartesian Coordinates

- Laplace's equation in Cartesian coordinates

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

- Using the method of separation of variables

$$\begin{aligned}
 V(x, y, z) &= X(x)Y(y)Z(z) \\
 Y(y)Z(z)\frac{d^2 X(x)}{dx^2} + X(x)Z(z)\frac{d^2 Y(y)}{dy^2} + X(x)Y(y)\frac{d^2 Z(z)}{dz^2} &= 0 \\
 \underbrace{\frac{1}{X(x)}\frac{d^2 X(x)}{dx^2}}_{-k_x^2} + \underbrace{\frac{1}{Y(y)}\frac{d^2 Y(y)}{dy^2}}_{-k_y^2} + \underbrace{\frac{1}{Z(z)}\frac{d^2 Z(z)}{dz^2}}_{-k_z^2} &= 0 \\
 k_x^2 + k_y^2 + k_z^2 &= 0
 \end{aligned}$$

⚡ Important result

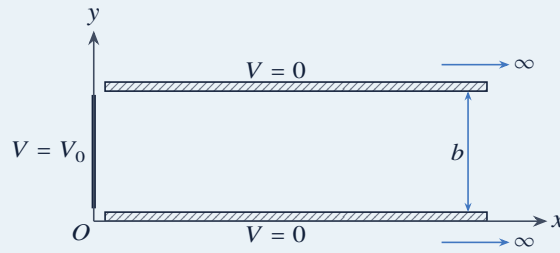
$$\frac{d^2X(x)}{dx^2} + k_x^2 X(x) = 0 \quad \frac{d^2Y(y)}{dy^2} + k_y^2 Y(y) = 0 \quad \frac{d^2Z(z)}{dz^2} + k_z^2 Z(z) = 0$$

Possible Solutions of $X''(x) + k_x^2 X(x) = 0$

k_x^2	k_x	$X(x)$	Exponential forms of $X(x)$
0	0	$A_0x + B_0$	
+	k	$A_1 \sin kx + B_1 \cos kx$	$C_1 e^{jkx} + D_1 e^{-jkx}$
-	jk	$A_2 \sinh kx + B_2 \cosh kx$	$C_2 e^{kx} + D_2 e^{-kx}$

Example 3-4

Two grounded, semi-infinite, parallel conducting plates are separated by the distance d . A third plate, perpendicular to both and insulated from them, is held at the potential V_0 . Determine the potential distribution in the region enclosed by the plates.



► Solution.

► Boundary conditions

$$(1) \quad V(x, y, z) = V(x, y) \quad (2) \quad V(0, y) = V_0 \quad (3) \quad V(\infty, y) = 0 \quad (4) \quad V(x, 0) = 0$$

$$(5) \quad V(x, b) = 0$$

$$(1) \Rightarrow \frac{\partial V}{\partial z} = 0 \Rightarrow Z(z) = B_0$$

$$\left. \begin{aligned} k_z^2 &= -\frac{1}{Z(z)} \frac{d^2 Z(z)}{dz^2} = 0 \\ k_x^2 + k_y^2 + k_z^2 &= 0 \end{aligned} \right\} \quad k_y^2 = -k_x^2 = k^2 \quad (k \text{ real})$$

$$k_x = jk \Rightarrow X(x) = C_2 e^{kx} + D_2 e^{-kx} \xrightarrow{(3)} X(x) = D_2 e^{-kx}$$

$$k_y = k \Rightarrow Y(y) = A_1 \sin ky + B_1 \cos ky \xrightarrow{(4)} Y(y) = A_1 \sin ky$$

$$\begin{aligned} V_n(x, y) &= (B_0 D_2 A_1) e^{-kx} \sin ky \\ &= C_n e^{-kx} \sin ky \end{aligned}$$

$$\xrightarrow{(5)} V_n(x, b) = C_n e^{-kb} \sin kb = 0 \Rightarrow \sin kb = 0 \Rightarrow kb = n\pi$$

$$k = \frac{n\pi}{b}, \quad n = 1, 2, 3, \dots$$

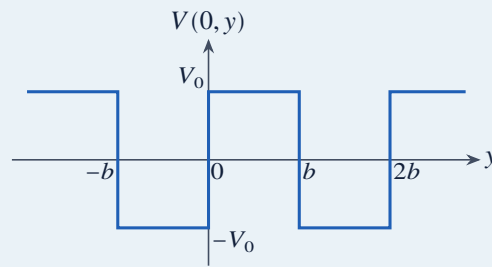
$$V_n(x, y) = C_n e^{-n\pi x/b} \sin \frac{n\pi}{b} y$$

- ▶ The above relation satisfies Laplace's equation but by itself cannot satisfy the second boundary condition at $x = 0$ for all y from 0 to b .
- ▶ Since Laplace's equation is a linear partial differential equation, a combination of V_n of the above form for different values of n is also a solution.

$$V(x, y) = \sum_{n=1}^{\infty} V_n(x, y)$$

$$\begin{aligned} \stackrel{(2)}{\Rightarrow} V(0, y) &= \sum_{n=1}^{\infty} V_n(0, y) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi}{b} y \\ &= V_0, \quad 0 < y < b \end{aligned}$$

- ▶ The above relation is the Fourier series expansion of a periodic odd function whose value on the interval $0 < y < b$ is V_0 .



Remark | Reminder

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \left(\frac{2n\pi}{T} t \right) + b_n \sin \left(\frac{2n\pi}{T} t \right) \right]$$

$$\begin{aligned} b_n &= \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin \left(\frac{2n\pi}{T} t \right) dt & a_n &= \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos \left(\frac{2n\pi}{T} t \right) dt & a_0 \\ &= \frac{1}{T} \int_{-T/2}^{T/2} f(t) dt \end{aligned}$$

$$\begin{aligned} C_n &= \frac{2}{2b} \left[\int_0^b V_0 \sin \left(\frac{2n\pi}{2b} y \right) dy + \int_b^{2b} -V_0 \sin \left(\frac{2n\pi}{2b} y \right) dy \right] \\ &= \frac{V_0}{n\pi} (-2 \cos n\pi + 2) = \begin{cases} \frac{4V_0}{n\pi} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases} \end{aligned}$$

⚡ Important result

$$\begin{aligned}
 V(x, y) &= \sum_{n=1}^{\infty} C_n e^{-n\pi x/b} \sin \frac{n\pi}{b} y \\
 &= \frac{4V_0}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{1}{n} e^{-n\pi x/b} \sin \frac{n\pi}{b} y, \\
 n &= 1, 3, 5, \dots, \quad x > 0 \quad \text{and} \quad 0 < y < b
 \end{aligned}$$

► Cylindrical Coordinates

- Laplace's equation in cylindrical coordinates

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

- Assuming the length is large compared with the radius, in cylindrical geometry the potential is independent of z .

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} = 0$$

- Using the method of separation of variables

$$\begin{aligned}
 V(r, \phi) &= R(r)\Phi(\phi) \\
 \underbrace{\frac{r}{R(r)} \frac{d}{dr} \left[r \frac{dR(r)}{dr} \right]}_{k^2} + \underbrace{\frac{1}{\Phi(\phi)} \frac{d^2 \Phi(\phi)}{d\phi^2}}_{-k^2} &= 0
 \end{aligned}$$

- The differential equation in ϕ

⚡ Important result

$$\frac{d^2 \Phi(\phi)}{d\phi^2} + k^2 \Phi(\phi) = 0$$

► **When $k \neq 0$**

– In circular cylindrical configurations the potential functions, and hence $\Phi(\phi)$, are periodic in ϕ , and the hyperbolic functions are not used.

► If the range of ϕ is not restricted, k must be an integer ($k = n$)

$$\Phi(\phi) = A_\phi \sin n\phi + B_\phi \cos n\phi$$

► The differential equation in r (Euler's equation)

$$r^2 \frac{d^2 R(r)}{dr^2} + r \frac{dR(r)}{dr} - n^2 R(r) = 0$$

$$R(r) = A_r r^n + B_r r^{-n}$$

⚡ **Important result**

$$V_n(r, \phi) = r^n (A_n \sin n\phi + B_n \cos n\phi) + r^{-n} (A'_n \sin n\phi + B'_n \cos n\phi), \quad n \neq 0$$

► When the region of interest includes $r = 0$, the terms containing r^{-n} cannot exist, and when the region of interest includes infinity, the terms containing r^n cannot exist.

► **When $k = 0$**

$$\frac{d^2 \Phi(\phi)}{d\phi^2} = 0$$

$$\Phi(\phi) = A_0 \phi + B_0 \quad \Rightarrow \quad \Phi(\phi) = B_0$$

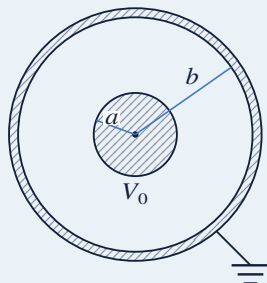
► If there is no circumferential variation

$$\frac{d}{dr} \left[r \frac{dR(r)}{dr} \right] = 0$$

$$R(r) = C_0 \ln r + D_0$$

Example 3-5

Consider a very long coaxial cable. The inner conductor has radius a and is held at the potential V_0 . The outer conductor has radius b and is grounded. Find the potential distribution in the space between the two conductors.



► Solution.

- Since the cable is long, there is no dependence on z .
- By symmetry, there is no dependence on ϕ ($k = 0$).

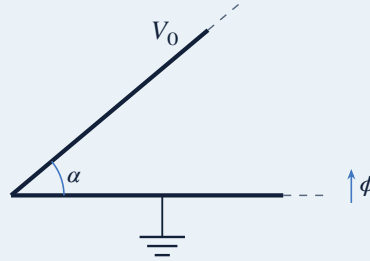
$$V(r) = C_1 \ln r + C_2$$

$$V(b) = 0, \quad V(a) = V_0$$

$$C_1 = -\frac{V_0}{\ln(b/a)}, \quad C_2 = \frac{V_0 \ln b}{\ln(b/a)} \quad \Rightarrow \quad V(r) = \frac{V_0}{\ln(b/a)} \ln \left(\frac{b}{r} \right)$$

Example 3-6

Two semi-infinite conducting plates are held at the potentials V_0 and zero as shown in the figure below. Assuming that the plates are very long along the z -axis, find the potential distribution in all space.



► Solution.

- Since the plates are long, there is no dependence on z .
- There is no dependence on r ($k = 0$).

$$\Phi(\phi) = A_0\phi + B_0$$

$$V(\phi) = \begin{cases} 0 & \phi = 0, 2\pi \\ V_0 & \phi = \alpha \end{cases}$$

- $0 \leq \phi \leq \alpha$

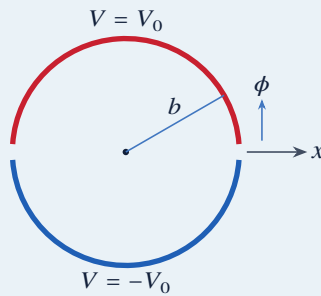
$$V(\phi) = \frac{V_0}{\alpha}\phi$$

- $\alpha \leq \phi \leq 2\pi$

$$V(\phi) = \frac{V_0}{2\pi - \alpha}(2\pi - \phi)$$

Example 3-7

A very long, thin conducting cylindrical tube of radius b is split into two parts. The upper half is held at the potential V_0 and the lower half at the potential $-V_0$. Find the potential distribution inside and outside the tube.



► Solution.

- Since the tube is long, there is no dependence on z .

$$V_n(r, \phi) = r^n(A_n \sin n\phi + B_n \cos n\phi) + r^{-n}(A'_n \sin n\phi + B'_n \cos n\phi)$$

$$V(b, \phi) = \begin{cases} V_0 & \text{for } 0 < \phi < \pi \\ -V_0 & \text{for } \pi < \phi < 2\pi \end{cases}$$

- Inside the tube ($r < b$)
- Since the region of interest includes $r = 0$, the terms containing r^{-n} cannot exist.
 - Since $V(r, \phi)$ is an odd function of ϕ , therefore

$$V_n(r, \phi) = A_n r^n \sin n\phi$$

- The above term alone cannot satisfy the boundary conditions. Therefore

$$\begin{aligned} V(r, \phi) &= \sum_{n=1}^{\infty} V_n(r, \phi) \\ &= \sum_{n=1}^{\infty} A_n r^n \sin n\phi \end{aligned}$$

$$\sum_{n=1}^{\infty} A_n b^n \sin n\phi = \begin{cases} V_0 & \text{for } 0 < \phi < \pi \\ -V_0 & \text{for } \pi < \phi < 2\pi \end{cases}$$

- The above relation is the Fourier series expansion of a periodic odd function whose value is V_0 on the interval $0 < \phi < \pi$ and $-V_0$ on the interval $\pi < \phi < 2\pi$.

$$A_n = \begin{cases} \frac{4V_0}{n\pi b^n} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases} \quad V(r, \phi) = \frac{4V_0}{\pi} \sum_{n=\text{odd}} \frac{1}{n} \left(\frac{r}{b}\right)^n \sin n\phi, \quad r < b$$

- Outside the tube ($r > b$)
- Since the region of interest includes $r \rightarrow \infty$, the terms containing r^n cannot exist.
 - $V(r, \phi)$ is an odd function of ϕ .
 - A single term cannot satisfy the boundary conditions. Therefore

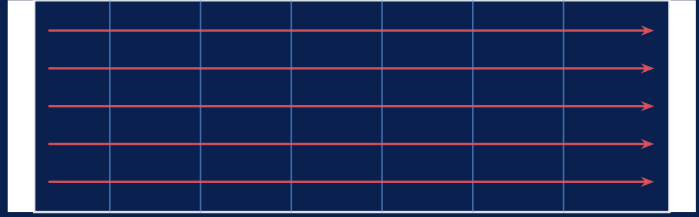
$$\begin{aligned} V(r, \phi) &= \sum_{n=1}^{\infty} V_n(r, \phi) & V(b, \phi) &= \sum_{n=1}^{\infty} B'_n b^{-n} \sin n\phi \\ &= \sum_{n=1}^{\infty} B'_n r^{-n} \sin n\phi & &= \begin{cases} V_0 & \text{for } 0 < \phi < \pi \\ -V_0 & \text{for } \pi < \phi < 2\pi \end{cases} \end{aligned}$$

$$B'_n = \begin{cases} \frac{4V_0 b^n}{n\pi} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

⚡ Important result

$$V(r, \phi) = \frac{4V_0}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{1}{n} \left(\frac{b}{r}\right)^n \sin n\phi, \quad r > b$$

4



CHAPTER

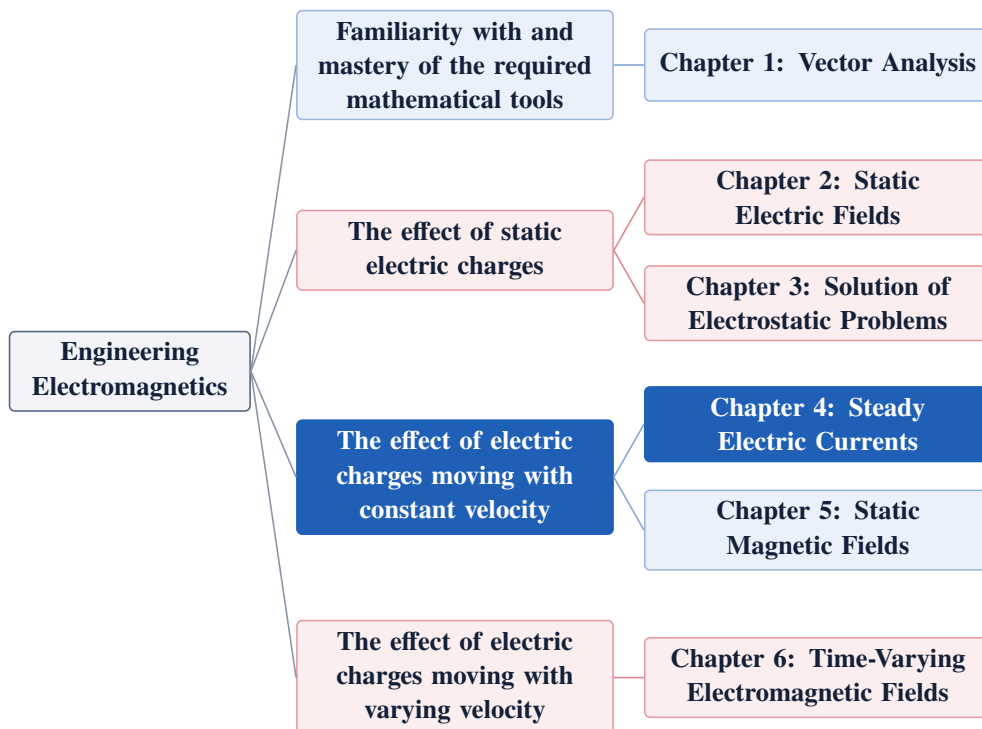
Steady Electric Currents

جریان‌های الکتریکی دائم

In this chapter

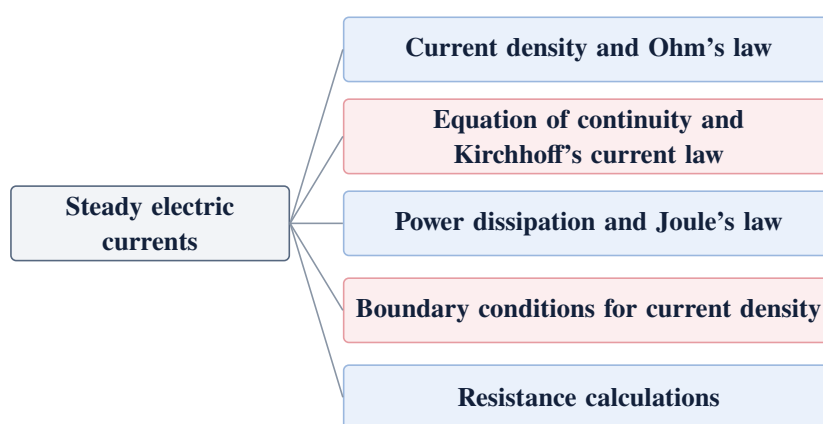
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4.1 Course Outline



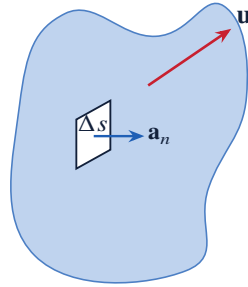
4.2 Steady Electric Currents

- In the previous chapters we dealt with electrostatic problems due to electric charges at rest.
 - In this chapter we turn to moving electric charges, which produce electric current.
- Electric currents due to the motion of free charges are divided into three kinds
 - **Conduction currents:** produced in conductors and semiconductors by the drift motion of electrons or holes.
 - **Electrolytic currents:** due to the motion of positive and negative ions
 - **Convection currents:** due to the motion of electrons or ions in vacuum
- In this chapter we concentrate on conduction currents, which are governed by Ohm's law.



- What is the concept of the **volume current density**, and what is the relation for the **convection current density**?
- What is the relation for the **conduction current density**?
- What is the point form of **Ohm's law**?
- What is the **equation of continuity**, and how is **Kirchhoff's current law** derived from it?
- What is **Joule's law**, which gives the power dissipated by a current flowing in a conducting body?

4.3 Current Density and Ohm's Law



- Consider the steady motion of charge carriers, each with charge q , moving with velocity $\mathbf{u}$ across a small surface element Δs
- If N is the number of charge carriers per unit volume, the charge passing through the surface Δs in the time Δt is

$$\Delta Q = Nq\mathbf{u} \cdot \mathbf{a}_n \Delta s \Delta t \quad (\text{C})$$

- Since current is the time rate of change of charge, we have

$$\Delta I = \frac{\Delta Q}{\Delta t} = Nq\mathbf{u} \cdot \mathbf{a}_n \Delta s = Nq\mathbf{u} \cdot \Delta \mathbf{s} \quad (\text{A})$$

- We define the volume current density, in amperes per square metre, as follows

$$\mathbf{J} = Nq\mathbf{u} \quad (\text{A/m}^2)$$

⚡ Important result | Convection current density

$$\mathbf{J} = \rho\mathbf{u} \quad (\text{A/m}^2)$$

- Therefore

$$\Delta I = \mathbf{J} \cdot \Delta \mathbf{s}$$

- Thus the total current flowing through an arbitrary surface S is

⚡ Important result

$$I = \int_S \mathbf{J} \cdot d\mathbf{s} \quad (\text{A})$$

- Conduction currents are the result of the drift motion of charge carriers under the influence of an applied electric field.
- For most conducting materials the average drift velocity is directly proportional to the electric field intensity.

– For metallic conductors we have

$$\mathbf{u} = -\mu_e \mathbf{E} \quad (\text{m/s})$$

- μ_e : electron mobility, in $\text{m}^2/\text{V} \cdot \text{s}$

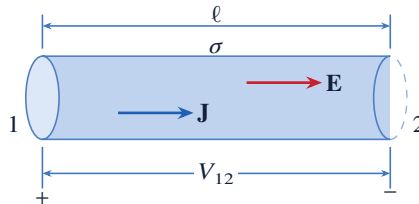
$$\mathbf{J} = -\rho_e \mu_e \mathbf{E}$$

- $\rho_e = -Ne$: the volume charge density of the drifting electrons, a negative quantity

⚡ Important result | Point form of Ohm's law

$$\mathbf{J} = \sigma \mathbf{E} \quad (\text{A/m}^2)$$

- $\sigma = -\rho_e \mu_e$: a fundamental parameter of the medium, called the conductivity
- The unit of conductivity is $\text{A}/\text{V} \cdot \text{m}$, or siemens per metre. The reciprocal of conductivity is called resistivity, in $\Omega \cdot \text{m}$.
- Obtaining the relation between the voltage and the current of a piece of homogeneous material of conductivity σ , length ℓ and uniform cross-section S from the point form of Ohm's law



$$\left. \begin{aligned} V_{12} = E\ell &\Rightarrow E = \frac{V_{12}}{\ell} \\ I = \int_S \mathbf{J} \cdot d\mathbf{s} = JS &\Rightarrow J = \frac{I}{S} \end{aligned} \right\} \frac{I}{S} = \sigma \frac{V_{12}}{\ell}$$

$$V_{12} = \left(\frac{\ell}{\sigma S} \right) I = RI$$

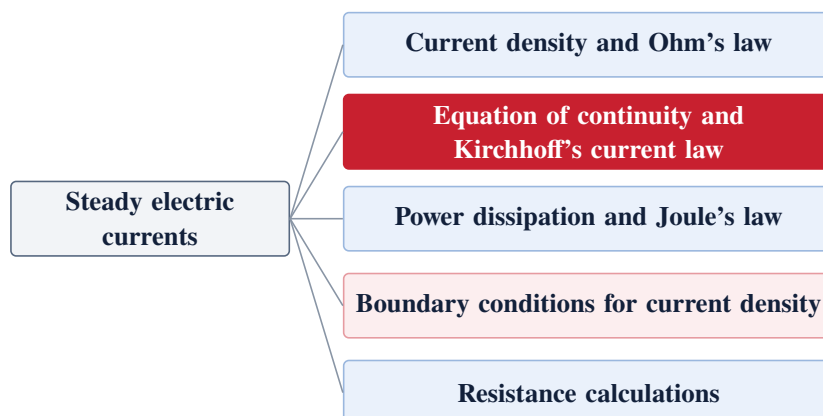
⚡ Important result

$$R = \frac{\ell}{\sigma S} \quad (\Omega)$$

$$G = \frac{1}{R} = \sigma \frac{S}{\ell} \quad (\text{S})$$

- G : conductance, the reciprocal of resistance, in mhos or siemens

4.4 Equation of Continuity and Kirchhoff's Current Law



- The principle of conservation of charge is one of the fundamental postulates of physics.
 - Electric charge can be neither created nor destroyed.
- Consider an arbitrary volume V bounded by the surface S , containing a net charge Q . If a net current I flows out of the region through the surface, the charge in the volume must decrease at a rate equal to the current. Conversely, if a net current flows into the region through the surface, the charge in the volume must increase at a rate equal to the current.
- The current leaving the region is the total outward flux of the current density vector through the surface S

$$I = \oint_S \mathbf{J} \cdot d\mathbf{s} = -\frac{dQ}{dt} = -\frac{d}{dt} \int_V \rho \, dv$$

$$\int_V \nabla \cdot \mathbf{J} \, dv = - \int_V \frac{\partial \rho}{\partial t} \, dv$$

- Since the above equation must hold regardless of the choice of V

⚡ Important result | Equation of continuity

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} \quad (\text{A/m}^3)$$

- If there are no time variations

$$\nabla \cdot \mathbf{J} = 0$$

⚡ Important result | Kirchhoff's current law

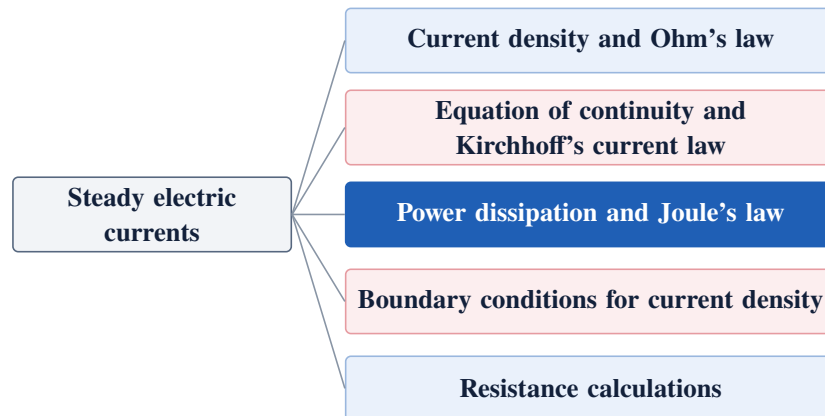
$$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0$$

- ▶ We said earlier that charges placed inside a conductor move to its surface, so that at equilibrium $\rho = 0$ inside the conductor. We now want to prove this statement

$$\left. \begin{array}{l} \sigma \nabla \cdot \mathbf{E} = -\frac{\partial \rho}{\partial t} \\ \nabla \cdot \mathbf{E} = \rho / \epsilon \end{array} \right\} \quad \frac{\partial \rho}{\partial t} + \frac{\sigma}{\epsilon} \rho = 0 \quad \Rightarrow \quad \rho = \rho_0 e^{-(\sigma/\epsilon)t} \quad (\text{C/m}^3)$$

- ▶ ρ_0 : initial charge density
- ▶ The above equation states that the charge density at a given point decreases exponentially with time.
- ▶ The initial density decays to $1/e$, or 36.8%, of its value in a time equal to $\tau = \epsilon/\sigma$.
 - For a good conductor such as copper, $\tau = 1.52 \times 10^{-19}$ s.
 - For a good insulator this time may be hours or days.

4.5 Power Dissipation and Joule's Law



- ▶ Under the influence of an electric field, the free electrons in a conductor move and collide with the atoms of the crystal lattice. Energy is thus transferred from the electric field to the vibrating atoms.
- ▶ The work done by the electric field in moving a charge q through the distance $\Delta \ell$ is

$$\Delta W = \mathbf{F} \cdot \Delta \ell = q\mathbf{E} \cdot \Delta \ell$$

- ▶ This work corresponds to the power

$$p = \lim_{\Delta t \rightarrow 0} \frac{\Delta w}{\Delta t} = q\mathbf{E} \cdot \mathbf{u}$$

- ▶ $\mathbf{u}$: the velocity of the charge carriers

$$dp = \rho dv \mathbf{E} \cdot \mathbf{u}$$

$$dP = \mathbf{E} \cdot \mathbf{J} dv$$

- For a given volume V the total power converted into heat is

⚡ Important result | Joule's law

$$P = \int_V \mathbf{E} \cdot \mathbf{J} dv \quad (\text{W})$$

- In a conductor of constant cross-section we have

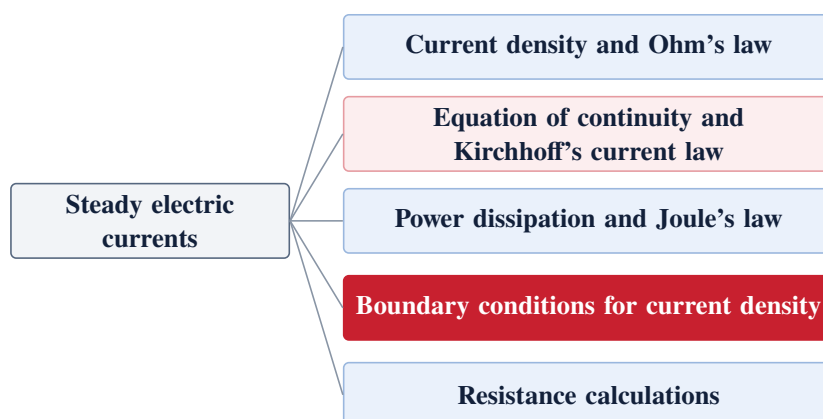
$$P = \int_L E d\ell \int_S J ds = VI$$

⚡ Important result

$$P = I^2 R \quad (\text{W})$$

- which is the familiar relation for ohmic power and gives the heat dissipated in the resistance R .

4.6 Boundary Conditions for Current Density



- When a current crosses the interface between two media of different conductivities obliquely, the current density vector changes in both direction and magnitude.
- The governing equations for the steady current density are

Governing Equations for Steady Current Density

Differential Form	Integral Form
$\nabla \cdot \mathbf{J} = 0$	$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0$
$\nabla \times \left(\frac{\mathbf{J}}{\sigma} \right) = 0$	$\oint_C \frac{1}{\sigma} \mathbf{J} \cdot d\boldsymbol{\ell} = 0$

- The boundary conditions are as follows

⚡ Important result

$$\nabla \cdot \mathbf{J} = 0 \quad \Rightarrow \quad J_{1n} = J_{2n} \quad (\text{A/m}^2)$$

$$\nabla \times (\mathbf{J}/\sigma) = 0 \quad \Rightarrow \quad \frac{J_{1t}}{J_{2t}} = \frac{\sigma_1}{\sigma_2}$$

- When a steady current crosses the boundary between two different lossy dielectrics

$$J_{1n} = J_{2n} \rightarrow \sigma_1 E_{1n} = \sigma_2 E_{2n}$$

$$D_{1n} - D_{2n} = \rho_s \rightarrow \epsilon_1 E_{1n} - \epsilon_2 E_{2n} = \rho_s$$

$$\rho_s = \left(\epsilon_1 \frac{\sigma_2}{\sigma_1} - \epsilon_2 \right) E_{2n} = \left(\epsilon_1 - \epsilon_2 \frac{\sigma_1}{\sigma_2} \right) E_{1n}$$

- $E_{1n} = \mathbf{a}_{n2} \cdot \mathbf{E}_1$ and $E_{2n} = \mathbf{a}_{n2} \cdot \mathbf{E}_2$
- If medium 2 is a much better conductor than medium 1

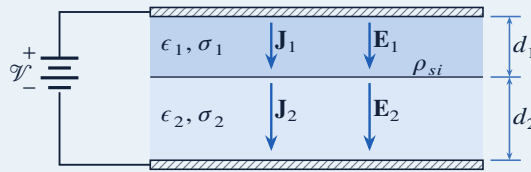
$$\rho_s = \epsilon_1 E_{1n} = D_{1n}$$

- which is the boundary condition of a conductor.

Example 4-1

A potential difference $\mathcal{V}$ is applied to a parallel-plate capacitor of area S . The space between the conducting plates is filled with two lossy dielectric materials of thicknesses d_1 and d_2 , permittivities ϵ_1 and ϵ_2 , and conductivities σ_1 and σ_2 . Find

- (a) the electric field intensity in the two media
- (b) the current density in the two media
- (c) the surface charge densities on the conducting plates and at the interface of the two dielectrics



➤ Solution.

- (a)

$$J_1 = J_2 \quad \Rightarrow \quad \begin{aligned} \mathcal{V} &= E_1 d_1 + E_2 d_2 \\ \sigma_1 E_1 &= \sigma_2 E_2 \end{aligned}$$

$$E_1 = \frac{\sigma_2 \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2} \quad E_2 = \frac{\sigma_1 \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2}$$

- (b)

$$J_1 = J_2 = \sigma_1 E_1 = \sigma_2 E_2 = \frac{\sigma_1 \sigma_2 \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2}$$

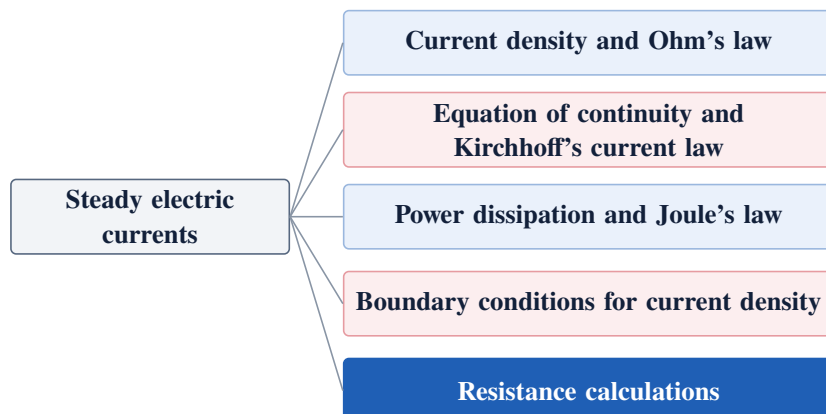
- (c)

$$\mathbf{a}_n \cdot \mathbf{E}_1 = \frac{\rho_{s1}}{\epsilon_1} \quad \Rightarrow \quad \rho_{s1} = \epsilon_1 E_1 = \frac{\epsilon_1 \sigma_2 \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2}$$

$$\mathbf{a}_n \cdot \mathbf{E}_2 = \frac{\rho_{s2}}{\epsilon_2} \quad \Rightarrow \quad \rho_{s2} = -\epsilon_2 E_2 = -\frac{\epsilon_2 \sigma_1 \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2}$$

$$\rho_{si} = \left(\epsilon_1 - \epsilon_2 \frac{\sigma_1}{\sigma_2} \right) E_{1n} \quad \Rightarrow \quad \begin{aligned} \rho_{si} &= \left(\epsilon_2 \frac{\sigma_1}{\sigma_2} - \epsilon_1 \right) \frac{\sigma_2 \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2} \\ &= \frac{(\epsilon_2 \sigma_1 - \epsilon_1 \sigma_2) \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2} \quad (\text{C/m}^2) \end{aligned}$$

4.7 Resistance Calculations



- How is the **leakage resistance** of capacitive structures computed?
- How is the resistance of various structures computed?
- The capacitance between two conductors separated by a dielectric medium is

$$C = \frac{Q}{V} = \frac{\oint_S \mathbf{D} \cdot d\mathbf{s}}{-\int_L \mathbf{E} \cdot d\boldsymbol{\ell}} = \frac{\oint_S \epsilon \mathbf{E} \cdot d\mathbf{s}}{-\int_L \mathbf{E} \cdot d\boldsymbol{\ell}}$$

- The surface integral is taken over a surface enclosing the positive conductor, and the line integral from the negative conductor to the positive conductor.
- If the dielectric medium is lossy, a current flows from the positive to the negative conductor. The resistance between the conductors (the leakage resistance) is

$$R = \frac{V}{I} = \frac{-\int_L \mathbf{E} \cdot d\boldsymbol{\ell}}{\oint_S \mathbf{J} \cdot d\mathbf{s}} = \frac{-\int_L \mathbf{E} \cdot d\boldsymbol{\ell}}{\oint_S \sigma \mathbf{E} \cdot d\mathbf{s}}$$

- If the permittivity and the conductivity have the same space dependence ($\epsilon(x, y, z) = k\sigma(x, y, z)$), or if the dielectric medium is homogeneous (ϵ and σ independent of the space coordinates)

⚡ Important result

$$RC = \frac{C}{G} = \frac{\epsilon}{\sigma}$$

Example 4-2

Find the leakage resistance per unit length

- (a) between the inner and outer conductors of a coaxial cable with inner-conductor radius a , outer-conductor radius b and a dielectric of conductivity σ
- (b) of a two-wire transmission line consisting of two wires of radius a separated by D , in a medium of conductivity σ

➤ Solution.

- (a) The capacitance per unit length of a coaxial cable

$$C_1 = \frac{2\pi\epsilon}{\ln(b/a)} \quad (\text{F/m})$$

- The leakage resistance per unit length

$$R_1 = \frac{\epsilon}{\sigma} \left(\frac{1}{C_1} \right) = \frac{1}{2\pi\sigma} \ln \left(\frac{b}{a} \right) \quad (\Omega \cdot \text{m})$$

- (b) The capacitance per unit length of a two-wire transmission line

$$C'_1 = \frac{\pi\epsilon}{\cosh^{-1} \left(\frac{D}{2a} \right)} \quad (\text{F/m})$$

- The leakage resistance per unit length

$$\begin{aligned} R'_1 &= \frac{\epsilon}{\sigma} \left(\frac{1}{C'_1} \right) = \frac{1}{\pi\sigma} \cosh^{-1} \left(\frac{D}{2a} \right) \\ &= \frac{1}{\pi\sigma} \ln \left[\frac{D}{2a} + \sqrt{\left(\frac{D}{2a} \right)^2 - 1} \right] \quad (\Omega \cdot \text{m}) \end{aligned}$$

- In certain situations electrostatic and steady-current problems are not exactly analogous.
- **First method** for computing the resistance of a piece of conducting material between two surfaces or terminals

- Choosing an appropriate coordinate system
- Applying a potential difference V_0 between the two terminals
- Finding the electric field intensity (if the material is homogeneous, by solving Laplace's equation for the electric potential and then computing the electric field intensity)
- Finding the total current

$$I = \int_S \mathbf{J} \cdot d\mathbf{s} = \int_S \sigma \mathbf{E} \cdot d\mathbf{s}$$

- S : the cross-section through which the current flows
- The resistance is V_0/I

- **Second method** for computing the resistance of a piece of conducting material between two surfaces or terminals (when J can easily be determined from I)

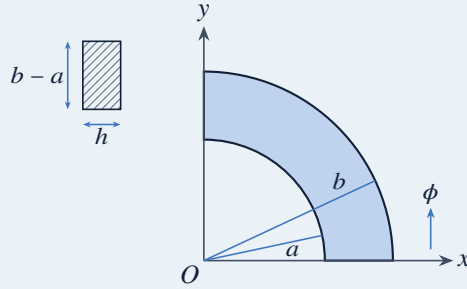
- Choosing an appropriate coordinate system
- Passing a current I between the two terminals
- Finding J from I
- Computing the electric field intensity $\mathbf{E} = \mathbf{J}/\sigma$ and the potential difference V_0

$$V_0 = - \int \mathbf{E} \cdot d\boldsymbol{\ell}$$

- The integration is carried out from the terminal at the lower potential to the terminal at the higher potential.
- The resistance is V_0/I

Example 4-3

Consider a conducting material of uniform thickness h and conductivity σ in the shape shown in the figure below. Find the resistance between its two end faces.



► Solution.

► Choosing cylindrical coordinates

$$V = 0 \quad \text{at} \quad \phi = 0, \quad V = V_0 \quad \text{at} \quad \phi = \pi/2$$

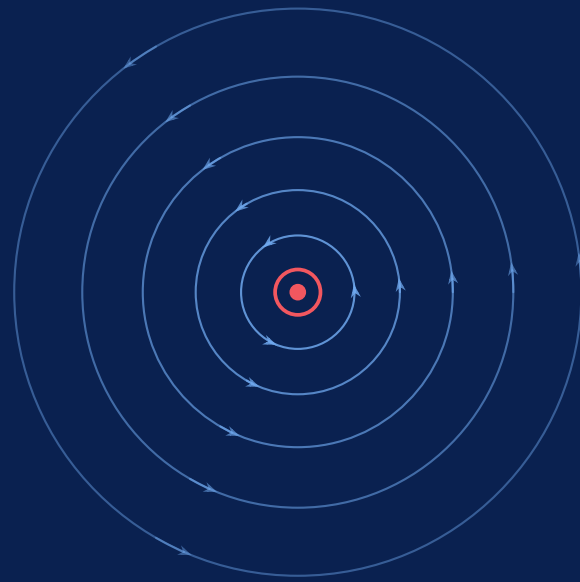
$$\frac{d^2 V}{d\phi^2} = 0 \quad \Rightarrow \quad V = c_1 \phi + c_2 \quad \Rightarrow \quad V = \frac{2V_0}{\pi} \phi$$

$$\begin{aligned} \mathbf{J} &= \sigma \mathbf{E} = -\sigma \nabla V \\ &= -\mathbf{a}_\phi \sigma \frac{\partial V}{r \partial \phi} = -\mathbf{a}_\phi \frac{2\sigma V_0}{\pi r} \end{aligned}$$

$$\begin{aligned} I &= \int_S \mathbf{J} \cdot d\mathbf{s} = \int_0^h \int_a^b \left(-\mathbf{a}_\phi \frac{2\sigma V_0}{\pi r} \right) \cdot (-\mathbf{a}_\phi dr dz) \\ &= \frac{2\sigma h V_0}{\pi} \ln \frac{b}{a} \end{aligned}$$

$$R = \frac{V_0}{I} = \frac{\pi}{2\sigma h \ln(b/a)}$$

5



CHAPTER

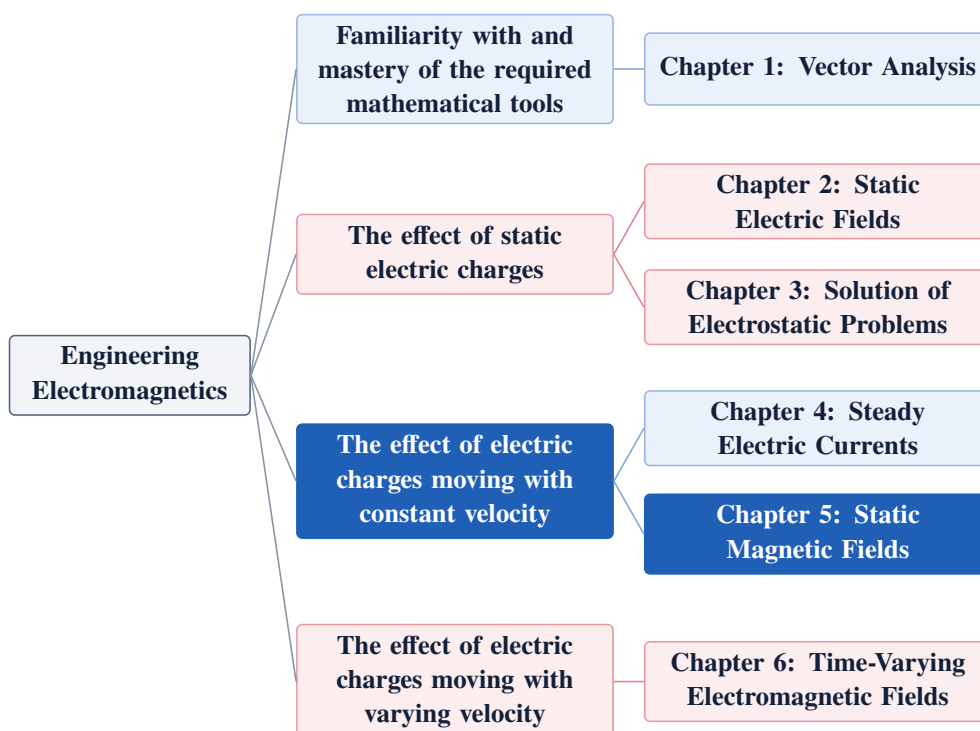
Static Magnetic Fields

میدان‌های مغناطیسی ساکن

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5.1 Course Outline



5.2 Static Magnetic Fields

- When a small test charge is placed in an electric field $\mathbf{E}$, the following force acts on it

$$\mathbf{F}_e = q\mathbf{E} \quad (\text{N})$$

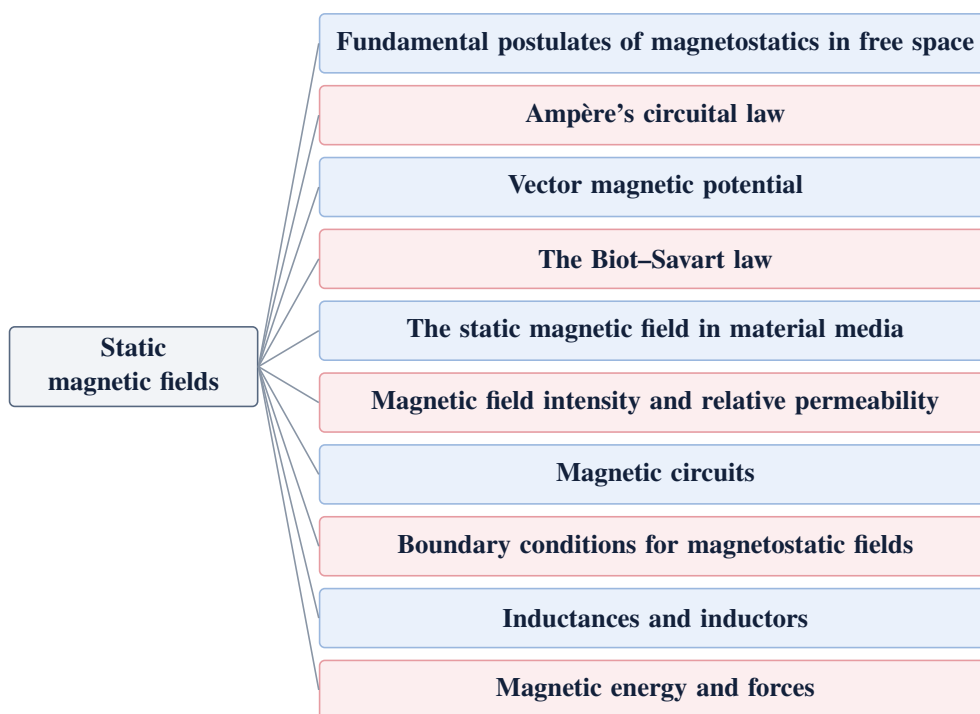
- When this test charge moves in a magnetic field, another force acts on it, equal to

$$\mathbf{F}_m = q\mathbf{u} \times \mathbf{B} \quad (\text{N})$$

- $\mathbf{B}$: the magnetic flux density, in webers per square metre or teslas
- So the total electromagnetic force on the charge q is

⚡ **Important result** | Lorentz's force equation

$$\mathbf{F} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B}) \quad (\text{N})$$



- What are the fundamental postulates of magnetostatics, from which the other relations and laws of the field can be derived?
- How is **Ampère's circuital law** derived from these postulates?
- How can the magnetic flux density of a current be found with Ampère's circuital law?

5.3 Fundamental Postulates of Magnetostatics in Free Space

- The two fundamental postulates of magnetostatics in free space

⚡ Important result

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

- $\mu_0 = 4\pi \times 10^{-7}$ (H/m): the permeability of free space
- $\mathbf{J}$: the volume current density (A/m²)
- Since the divergence of the curl of any vector field is zero, it follows that

$$\nabla \cdot \mathbf{J} = 0$$

- There is no magnetic analogue of the electric charge density.
- The integral form of the first relation → integrate over an arbitrary volume V and apply the divergence theorem

⚡ **Important result** | Law of conservation of magnetic flux

$$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$$

- This equation is called the **law of conservation of magnetic flux**.
 - There are no magnetic flow sources, and the magnetic flux lines always close upon themselves.
- The integral form of the second relation → integrate over an arbitrary open surface S and apply Stokes's theorem

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} = \mu_0 \int_S \mathbf{J} \cdot d\mathbf{s}$$

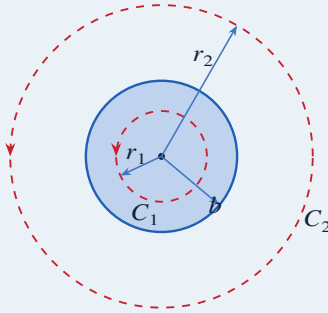
⚡ **Important result** | Ampère's circuital law

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I$$

- C : the contour bounding the surface S (the contour C and the current I follow the right-hand rule)
- I : the current passing through the surface S
- This equation is a form of **Ampère's circuital law**.
 - The circulation of the magnetic flux density in free space around any closed path is equal to μ_0 times the total current flowing through the surface bounded by the path.
- Ampère's circuital law is useful for finding the magnetic flux density caused by a current I when there is a closed path C around the current such that $\mathbf{B}$ is constant on the path.

Example 5-1

An infinitely long, straight conductor with a circular cross-section of radius b carries a current I . Determine the magnetic flux density both inside and outside the conductor.



► Solution.

- If the conductor lies along the z -axis, $\mathbf{B}$ is in the ϕ direction and its magnitude is constant along any circular path about the z -axis.

– Inside the conductor

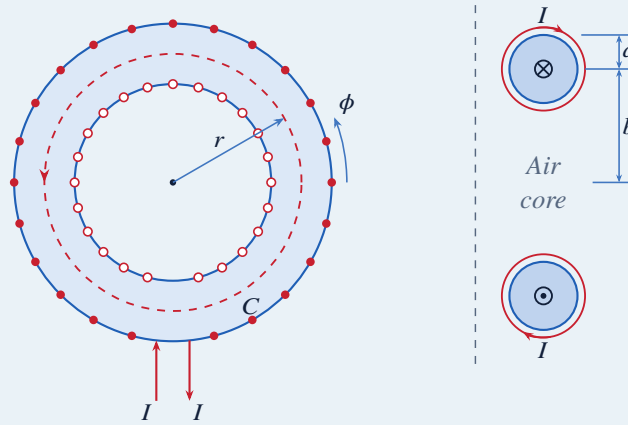
$$\left. \begin{aligned} \mathbf{B}_1 &= \mathbf{a}_\phi B_{\phi 1}, & d\ell &= \mathbf{a}_\phi r_1 d\phi \\ \oint_{C_1} \mathbf{B}_1 \cdot d\ell &= \int_0^{2\pi} B_{\phi 1} r_1 d\phi = 2\pi r_1 B_{\phi 1} \\ I_1 &= \frac{\pi r_1^2}{\pi b^2} I = \left(\frac{r_1}{b}\right)^2 I \end{aligned} \right\} \mathbf{B}_1 = \mathbf{a}_\phi B_{\phi 1} = \mathbf{a}_\phi \frac{\mu_0 r_1 I}{2\pi b^2}$$

- Outside the conductor

$$\begin{aligned} \mathbf{B}_2 &= \mathbf{a}_\phi B_{\phi 2}, & d\ell &= \mathbf{a}_\phi r_2 d\phi \\ \oint_{C_2} \mathbf{B}_2 \cdot d\ell &= 2\pi r_2 B_{\phi 2} \\ \mathbf{B}_2 &= \mathbf{a}_\phi B_{\phi 2} = \mathbf{a}_\phi \frac{\mu_0 I}{2\pi r_2} \end{aligned}$$

Example 5-2

Determine the magnetic flux density inside a closely wound toroidal coil with an air core, having N turns of wire carrying a current I .



► Solution.

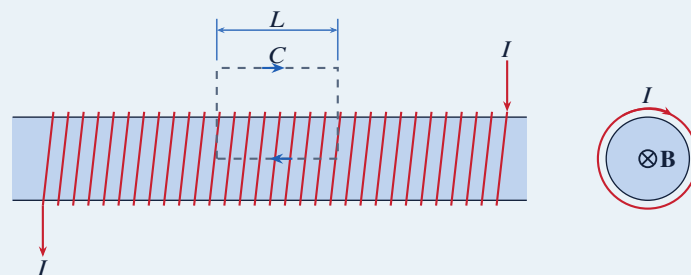
- Cylindrical symmetry ensures that $\mathbf{B}$ has only a ϕ component and that its magnitude is constant along any circular path about the axis of the coil.

$$\oint \mathbf{B} \cdot d\ell = 2\pi r B_\phi = \mu_0 N I$$

$$\mathbf{B} = \mathbf{a}_\phi B_\phi = \mathbf{a}_\phi \frac{\mu_0 N I}{2\pi r}, \quad (b - a) < r < (b + a)$$

Example 5-3

Determine the magnetic flux density inside an infinitely long solenoid with an air core, having n turns of wire per unit length carrying a current I .



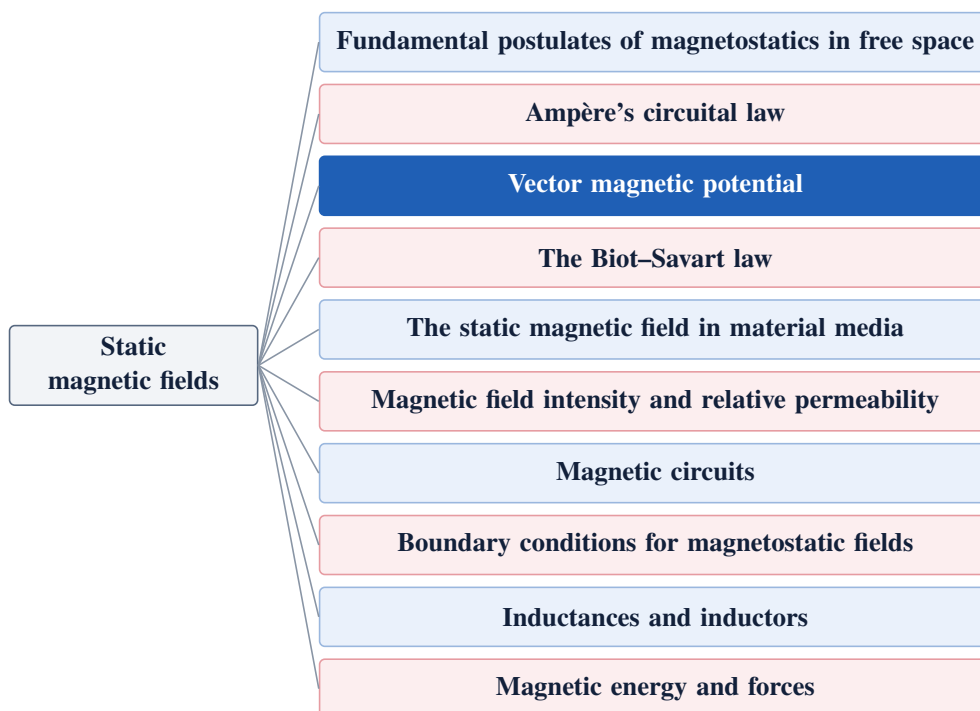
► Solution.

- Outside the solenoid the magnetic field is zero.
- By the symmetry of the structure, $\mathbf{B}$ is parallel to the axis of the solenoid and its magnitude is constant along the solenoid.

$$BL = \mu_0 n L I$$

$$B = \mu_0 n I$$

5.4 Vector Magnetic Potential



- What is the vector magnetic potential, where is it used, and how is it calculated?
- What is the Biot–Savart law, and how is it used to calculate the magnetic flux density?

$$\nabla \cdot \mathbf{B} = 0 \quad \Rightarrow \quad \boxed{\mathbf{B} = \nabla \times \mathbf{A} \quad (\text{T})}$$

- $\mathbf{A}$ is called the **vector magnetic potential**. Its unit is the weber per metre.
 - If $\mathbf{A}$ can be found for a current distribution, $\mathbf{B}$ follows easily (exactly as with V and $\mathbf{E}$).
- A vector is defined by its curl and its divergence. To specify the divergence of $\mathbf{A}$ we proceed as follows

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad \Rightarrow \quad \nabla \times \nabla \times \mathbf{A} = \mu_0 \mathbf{J}$$

- We define the **Laplacian of a vector quantity** as

$$\nabla^2 \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times \nabla \times \mathbf{A}$$

- For example, the Laplacian of $\mathbf{A}$ in Cartesian coordinates is

$$\nabla^2 \mathbf{A} = \mathbf{a}_x \nabla^2 A_x + \mathbf{a}_y \nabla^2 A_y + \mathbf{a}_z \nabla^2 A_z$$

- Therefore

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}$$

- To simplify this relation we choose

$$\nabla \cdot \mathbf{A} = 0 \quad \Rightarrow \quad \boxed{\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}}$$

- This last relation is the **vector Poisson's equation**.
- In Cartesian coordinates

$$\nabla^2 A_x = -\mu_0 J_x, \quad \nabla^2 A_y = -\mu_0 J_y, \quad \nabla^2 A_z = -\mu_0 J_z$$

- Each of these three equations is mathematically the same as Poisson's equation in electrostatics.
- Earlier we had

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad \Rightarrow \quad V = \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\rho}{|\mathbf{R} - \mathbf{R}'|} dv'$$

- In the same way we obtain

⚡ Important result

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\mathbf{J}}{|\mathbf{R} - \mathbf{R}'|} dv'$$

- So $\mathbf{A}$ can be found from the current density with this relation, and $\mathbf{B}$ then follows by taking its curl.
- The physical meaning of the vector magnetic potential

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$$

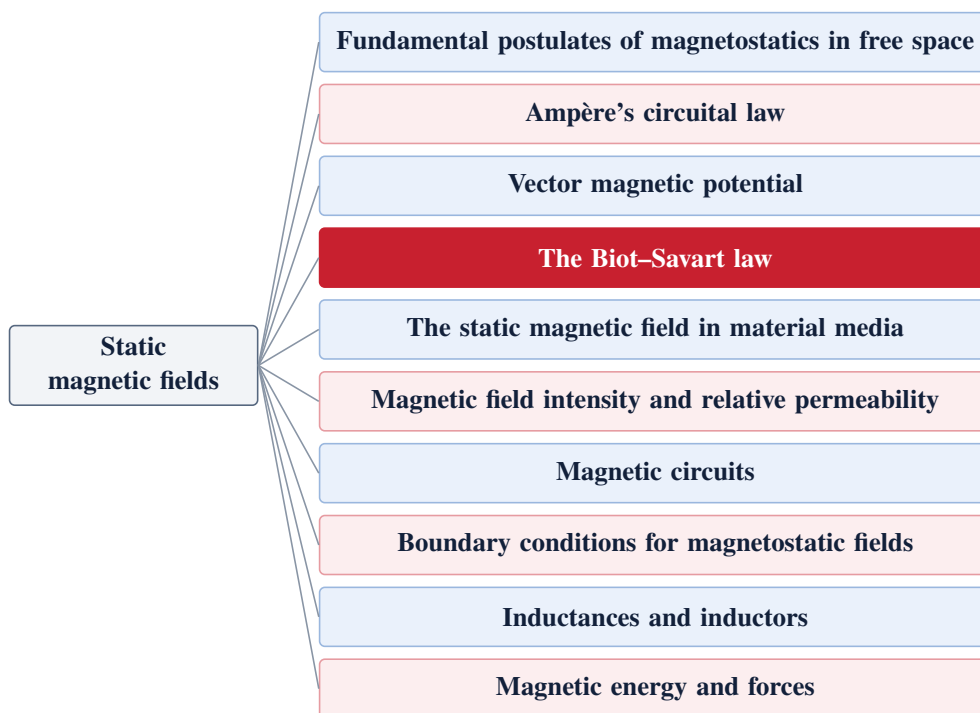
- Φ : the magnetic flux, in webers

⚡ Important result

$$\Phi = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell} \quad (\text{Wb})$$

- The line integral of $\mathbf{A}$ around any closed path C equals the total magnetic flux passing through the surface bounded by the path.

5.5 The Biot–Savart Law



- For a thin wire carrying a current I , with cross-section S , we have

$$\mathbf{J} dv' = JS d\ell' = I d\ell' \quad \Rightarrow \quad \mathbf{A} = \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\ell'}{|\mathbf{R} - \mathbf{R}'|}$$

- It can be shown that

$$\mathbf{B} = \nabla \times \mathbf{A} = \nabla \times \left[\frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\ell'}{|\mathbf{R} - \mathbf{R}'|} \right]$$

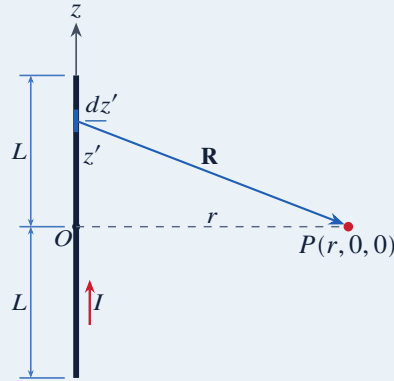
⚡ Important result | Biot–Savart law

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\ell' \times (\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3}$$

- This equation is called the **Biot–Savart law**.
- In general the Biot–Savart law is more difficult to use than Ampère's circuital law. But when no closed path can be found on which $\mathbf{B}$ is constant, Ampère's circuital law cannot be used to determine $\mathbf{B}$.

Example 5-4

A direct current I flows in a straight wire of length $2L$. Find the magnetic flux density at a point at a distance r from the wire in the bisecting plane.



► Solution.

- Currents exist only in closed circuits. The wire in this example must therefore be part of a current-carrying loop. Since we know nothing about the rest of the circuit, Ampère's circuital law cannot be used.

► First method

$$\left. \begin{aligned} \mathbf{A} &= \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\ell'}{|\mathbf{R} - \mathbf{R}'|} \\ d\ell' &= dz' \mathbf{a}_z \\ \mathbf{R} &= r \mathbf{a}_r \\ \mathbf{R}' &= z' \mathbf{a}_z \end{aligned} \right\} \quad \begin{aligned} \mathbf{A} &= \mathbf{a}_z \frac{\mu_0 I}{4\pi} \int_{-L}^L \frac{dz'}{\sqrt{z'^2 + r^2}} \\ &= \mathbf{a}_z \frac{\mu_0 I}{4\pi} \ln \frac{\sqrt{L^2 + r^2} + L}{\sqrt{L^2 + r^2} - L} \end{aligned}$$

$$\mathbf{B} = \nabla \times \mathbf{A} = \nabla \times (\mathbf{a}_z A_z) = \mathbf{a}_r \frac{1}{r} \frac{\partial A_z}{\partial \phi} - \mathbf{a}_\phi \frac{\partial A_z}{\partial r}$$

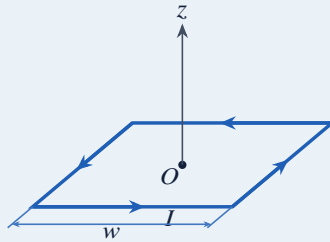
$$\begin{aligned} \mathbf{B} &= -\mathbf{a}_\phi \frac{\partial}{\partial r} \left[\frac{\mu_0 I}{4\pi} \ln \frac{\sqrt{L^2 + r^2} + L}{\sqrt{L^2 + r^2} - L} \right] \\ &= \mathbf{a}_\phi \frac{\mu_0 I L}{2\pi r \sqrt{L^2 + r^2}} \end{aligned}$$

► Second method

$$\left. \begin{aligned} \mathbf{B} &= \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\ell' \times (\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3} \\ \mathbf{R} - \mathbf{R}' &= r \mathbf{a}_r - z' \mathbf{a}_z \\ d\ell' \times (\mathbf{R} - \mathbf{R}') &= dz' \mathbf{a}_z \times (r \mathbf{a}_r - z' \mathbf{a}_z) = r dz' \mathbf{a}_\phi \end{aligned} \right\} \quad \begin{aligned} \mathbf{B} &= \int d\mathbf{B} = \mathbf{a}_\phi \frac{\mu_0 I}{4\pi} \int_{-L}^L \frac{r dz'}{(z'^2 + r^2)^{3/2}} \\ &= \mathbf{a}_\phi \frac{\mu_0 I L}{2\pi r \sqrt{L^2 + r^2}} \end{aligned}$$

Example 5-5

Find the magnetic flux density at the centre of a square loop of side w carrying a current I .



► Solution.

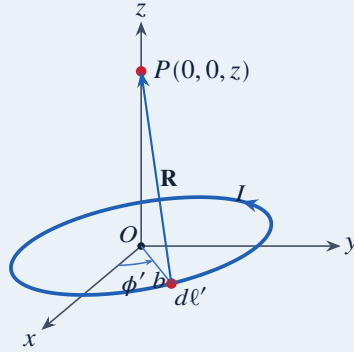
- Assume the loop lies in the xy -plane. $\mathbf{B}$ at the centre of the loop is four times the magnetic flux density caused by one side of length w . Therefore
- The magnitude of the magnetic flux density of a straight wire of length $2L$ at a distance r from it

$$B = \frac{\mu_0 I L}{2\pi r \sqrt{L^2 + r^2}} \quad L = \frac{w}{2}, \quad r = \frac{w}{2}$$

$$\mathbf{B} = \mathbf{a}_z \frac{\mu_0 I}{\sqrt{2}\pi w} \times 4 = \mathbf{a}_z \frac{2\sqrt{2}\mu_0 I}{\pi w}$$

Example 5-6

Find the magnetic flux density on the axis of a circular loop of radius b carrying a current I .



► Solution.

$$\left. \begin{aligned} \mathbf{B} &= \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\ell' \times (\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3} \\ d\ell' &= b d\phi' \mathbf{a}_{\phi'}, \quad \mathbf{R} = z\mathbf{a}_z, \quad \mathbf{R}' = b\mathbf{a}_{r'} \\ \mathbf{R} - \mathbf{R}' &= z\mathbf{a}_z - b\mathbf{a}_{r'} \\ d\ell' \times (\mathbf{R} - \mathbf{R}') &= b d\phi' \mathbf{a}_{\phi'} \times (z\mathbf{a}_z - b\mathbf{a}_{r'}) \\ &= bz d\phi' \mathbf{a}_{r'} + b^2 d\phi' \mathbf{a}_z \end{aligned} \right\} \quad \mathbf{B} = \frac{\mu_0 I}{4\pi} \int_0^{2\pi} \mathbf{a}_z \frac{b^2 d\phi'}{(z^2 + b^2)^{3/2}}$$

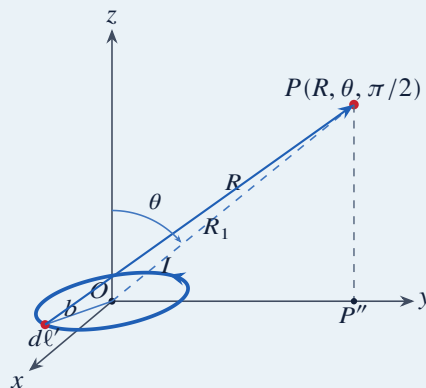
► It is evident from the geometry that the r -component of the magnetic flux density is zero.

⚡ Important result

$$\mathbf{B} = \mathbf{a}_z \frac{\mu_0 I b^2}{2(z^2 + b^2)^{3/2}} \quad (\text{T})$$

Example 5-7

Find the magnetic flux density at a distant point of a magnetic dipole of radius b carrying a current I .



► **Solution.**

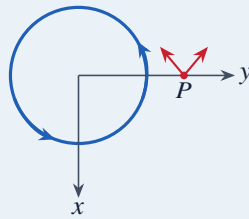
$$\mathbf{A} = \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\ell'}{|\mathbf{R} - \mathbf{R}'|}$$

- Because of the symmetry of the geometry, the magnetic field is independent of the angle ϕ . For simplicity, the field point is therefore placed in the yz -plane.

$$d\ell' = b d\phi' \mathbf{a}_{\phi'} = b d\phi' (-\mathbf{a}_x \sin \phi' + \mathbf{a}_y \cos \phi')$$

$$\mathbf{A} = -\mathbf{a}_x \frac{\mu_0 I}{4\pi} \int_0^{2\pi} \frac{b \sin \phi'}{|\mathbf{R} - \mathbf{R}'|} d\phi' + \mathbf{a}_y \frac{\mu_0 I}{4\pi} \int_0^{2\pi} \frac{b \cos \phi'}{|\mathbf{R} - \mathbf{R}'|} d\phi'$$

- It is evident from the geometry that the y -component of $\mathbf{A}$ is zero.



$$\begin{aligned} \mathbf{R} &= R\mathbf{a}_R = R(\sin \theta \cos(\pi/2)\mathbf{a}_x + \sin \theta \sin(\pi/2)\mathbf{a}_y + \cos \theta \mathbf{a}_z) \\ &= R(\sin \theta \mathbf{a}_y + \cos \theta \mathbf{a}_z) \end{aligned}$$

$$\mathbf{R}' = b\mathbf{a}_{r'} = b(\cos \phi' \mathbf{a}_x + \sin \phi' \mathbf{a}_y)$$

$$\mathbf{R} - \mathbf{R}' = -b \cos \phi' \mathbf{a}_x + (R \sin \theta - b \sin \phi') \mathbf{a}_y + R \cos \theta \mathbf{a}_z$$

$$\begin{aligned} |\mathbf{R} - \mathbf{R}'|^2 &= R^2 + b^2 - 2bR \sin \theta \sin \phi' \\ &= R^2 \left(1 + \frac{b^2}{R^2} - \frac{2b}{R} \sin \theta \sin \phi' \right) \end{aligned}$$

$$\begin{aligned} \frac{1}{|\mathbf{R} - \mathbf{R}'|} &= \frac{1}{R} \left(1 + \frac{b^2}{R^2} - \frac{2b}{R} \sin \theta \sin \phi' \right)^{-1/2} \\ &\cong \frac{1}{R} \left(1 - \frac{2b}{R} \sin \theta \sin \phi' \right)^{-1/2} \cong \frac{1}{R} \left(1 + \frac{b}{R} \sin \theta \sin \phi' \right) \end{aligned}$$

$$\begin{aligned} \mathbf{A} &= -\mathbf{a}_x \frac{\mu_0 I b}{4\pi R} \int_0^{2\pi} \left(1 + \frac{b}{R} \sin \theta \sin \phi' \right) \sin \phi' d\phi' \\ &= -\mathbf{a}_x \frac{\mu_0 I b^2}{4R^2} \sin \theta \end{aligned}$$

- At an arbitrary field point, the vector magnetic potential is then

$$\mathbf{A} = \mathbf{a}_\phi \frac{\mu_0 I b^2}{4R^2} \sin \theta \quad \Rightarrow \quad \mathbf{A} = \mathbf{a}_\phi \frac{\mu_0 (I\pi b^2)}{4\pi R^2} \sin \theta$$

- The magnetic dipole moment vector

$$\mathbf{m} = \mathbf{a}_z I\pi b^2 = \mathbf{a}_z IS = \mathbf{a}_z m$$

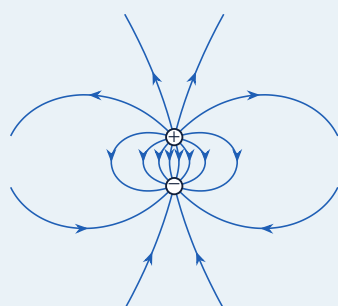
⚡ Important result

$$\mathbf{A} = \frac{\mu_0 \mathbf{m} \times \mathbf{a}_R}{4\pi R^2}$$

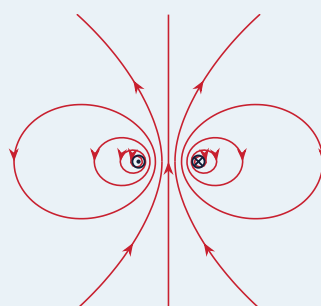
$$\mathbf{B} = \nabla \times \mathbf{A} \quad \Rightarrow \quad \mathbf{B} = \frac{\mu_0 I b^2}{4R^3} (\mathbf{a}_R 2 \cos \theta + \mathbf{a}_\theta \sin \theta)$$

⚡ Important result

$$\mathbf{B} = \frac{\mu_0 m}{4\pi R^3} (\mathbf{a}_R 2 \cos \theta + \mathbf{a}_\theta \sin \theta)$$

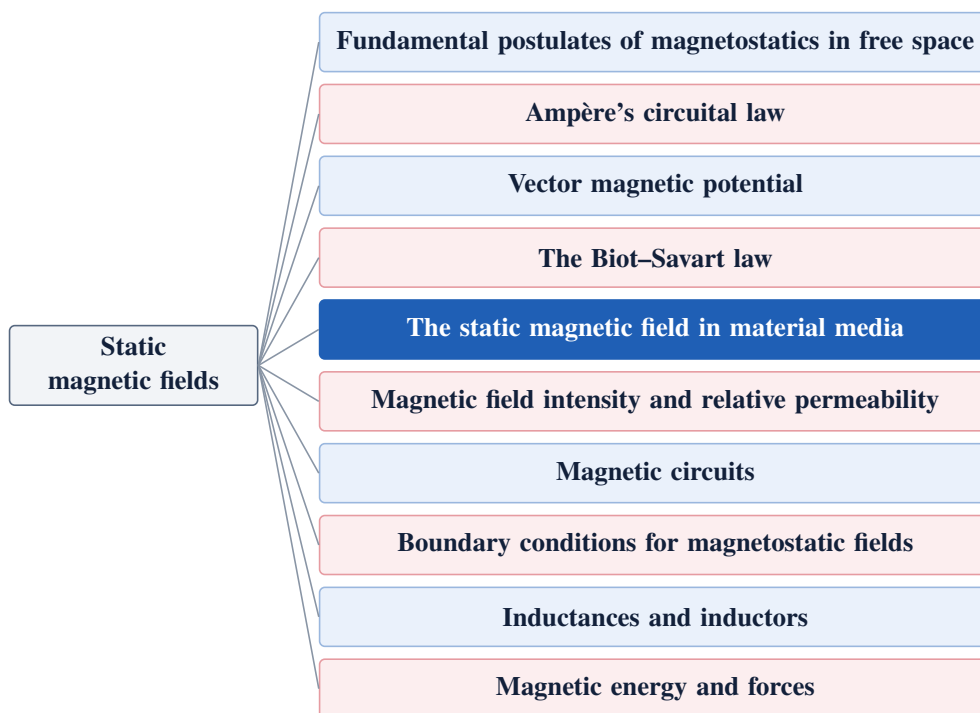


Electric dipole



Magnetic dipole

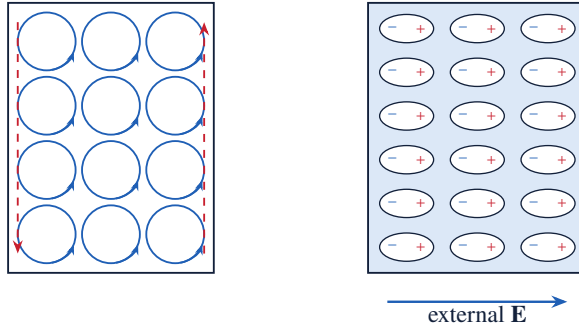
5.6 The Static Magnetic Field in Material Media



- All materials consist of atoms with a positively charged nucleus and a number of negatively charged electrons orbiting around it.
- The motion of the electrons produces circulating currents and hence microscopic magnetic dipoles.
- In the absence of an external magnetic field, the magnetic dipoles of the atoms of most materials (except permanent magnets) have random orientations, and no net magnetic moment results. When an external magnetic field is applied, the magnetic moments of the orbiting electrons align.
- How do different materials behave when exposed to a magnetic field, and how can this behaviour be analysed?

► Magnetization and Equivalent Current Densities

⊙ $\mathbf{M}$, out of the paper



- The magnetization vector (and the polarization vector in electrostatics)

$$\mathbf{M} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{k=1}^{n\Delta v} \mathbf{m}_k}{\Delta v} \quad (\text{A/m}) \qquad \mathbf{P} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{k=1}^{n\Delta v} \mathbf{p}_k}{\Delta v} \quad (\text{C/m}^2)$$

- The vector magnetic potential of a magnetized material (and the electric potential of a polarized material)

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\nabla' \times \mathbf{M}}{R} dv' + \frac{\mu_0}{4\pi} \oint_{S'} \frac{\mathbf{M} \times \mathbf{a}'_n}{R} ds'$$

$$V = \frac{1}{4\pi\epsilon_0} \oint_{S'} \frac{\mathbf{P} \cdot \mathbf{a}'_n}{R} ds' + \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{(-\nabla' \cdot \mathbf{P})}{R} dv'$$

- The equivalent magnetization surface and volume current densities

⚡ Important result

$$\mathbf{J}_{ms} = \mathbf{M} \times \mathbf{a}_n, \qquad \mathbf{J}_m = \nabla \times \mathbf{M}$$

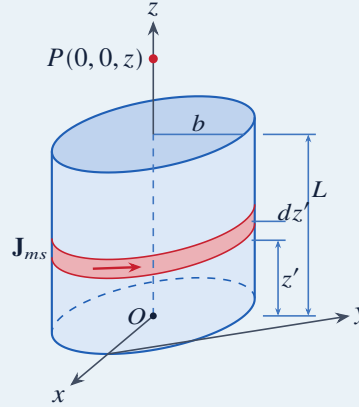
- The equivalent polarization surface and volume charge densities

$$\rho_{ps} = \mathbf{P} \cdot \mathbf{a}_n, \qquad \rho_p = -\nabla \cdot \mathbf{P}$$

- The problem of finding the magnetic flux density $\mathbf{B}$ caused by a given volume density of magnetic dipole moment $\mathbf{M}$ becomes one of finding the equivalent magnetization current densities, then $\mathbf{A}$, and then $\mathbf{B}$.

Example 5-8

Determine the magnetic flux density on the axis of a magnetized cylinder. The cylinder has a radius b , a length L and a magnetization vector $\mathbf{M} = \mathbf{a}_z M_0$.



► Solution.

$$\mathbf{J}_m = \nabla \times \mathbf{M} = 0$$

- The equivalent surface current density on the side wall

$$\mathbf{J}_{ms} = \mathbf{M} \times \mathbf{a}_n = \mathbf{a}_z M_0 \times \mathbf{a}_r = \mathbf{a}_\phi M_0$$

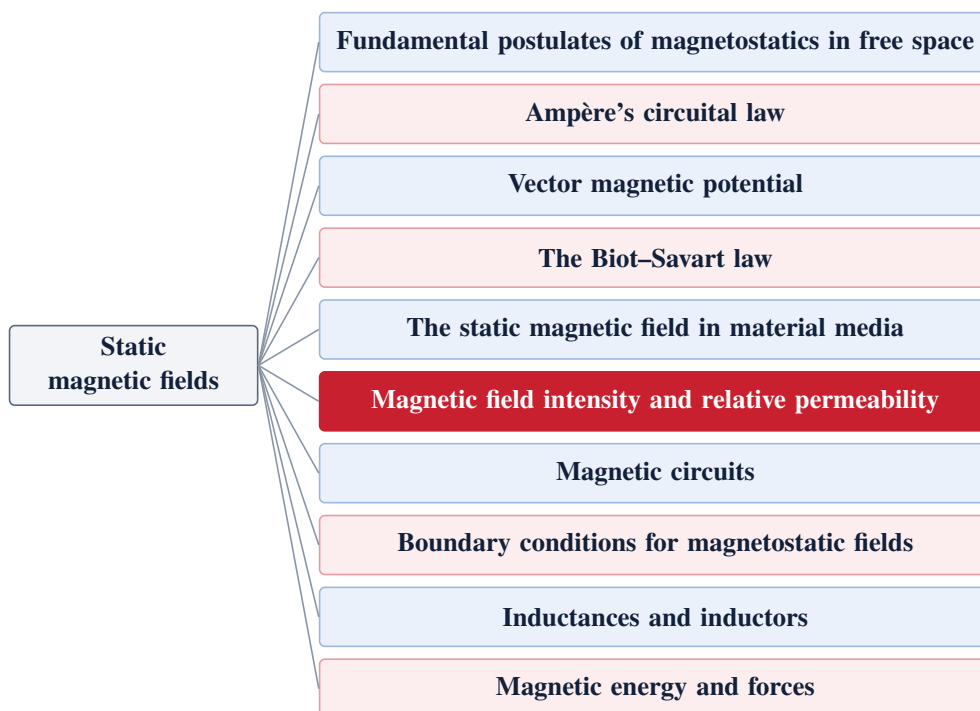
- The equivalent surface current density on the top and bottom faces is zero.

$$\left. \begin{aligned} \mathbf{B} &= \frac{\mu_0}{4\pi} \int_{S'} \frac{\mathbf{J}_{ms} \times (\mathbf{R} - \mathbf{R}')}{|\mathbf{R} - \mathbf{R}'|^3} ds' \\ ds' &= b d\phi' dz' \\ \mathbf{R} &= z\mathbf{a}_z \\ \mathbf{R}' &= b\mathbf{a}_{r'} + z'\mathbf{a}_z \\ \mathbf{R} - \mathbf{R}' &= (z - z')\mathbf{a}_z - b\mathbf{a}_{r'} \\ \mathbf{J}_{ms} \times (\mathbf{R} - \mathbf{R}') &= M_0 \mathbf{a}_\phi \times ((z - z')\mathbf{a}_z - b\mathbf{a}_{r'}) \\ &= M_0(z - z')\mathbf{a}_{r'} + bM_0\mathbf{a}_z \end{aligned} \right\}$$

- It is evident from the geometry that the r -component of the magnetic flux density is zero.

$$\begin{aligned} \mathbf{B} &= \int d\mathbf{B} = \mathbf{a}_z \int_0^L \frac{\mu_0 M_0 b^2 dz'}{2[(z - z')^2 + b^2]^{3/2}} \\ &= \mathbf{a}_z \frac{\mu_0 M_0}{2} \left[\frac{z}{\sqrt{z^2 + b^2}} - \frac{z - L}{\sqrt{(z - L)^2 + b^2}} \right] \end{aligned}$$

5.7 Magnetic Field Intensity and Relative Permeability



► Magnetic Field Intensity

- Since an external magnetic field aligns the internal dipole moments and induces a magnetic moment in a magnetic material, we expect the resulting magnetic flux density in the presence of the magnetic material to differ from its value in free space.
- Taking the equivalent volume current density in the material into account, we have

$$\frac{1}{\mu_0} \nabla \times \mathbf{B} = \mathbf{J} + \mathbf{J}_m = \mathbf{J} + \nabla \times \mathbf{M} \quad \Rightarrow \quad \nabla \times \left(\frac{\mathbf{B}}{\mu_0} - \mathbf{M} \right) = \mathbf{J}$$

- We define the magnetic field intensity as

⚡ Important result

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M} \quad (\text{A/m})$$

- The vector $\mathbf{H}$ lets us write the curl equation relating the magnetic field and the distribution of free currents in any medium without using $\mathbf{M}$ or $\mathbf{J}_m$

⚡ Important result

$$\nabla \times \mathbf{H} = \mathbf{J} \quad (\text{A/m}^2)$$

- **J**: the density of the free current
- The integral form

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{s} = \int_S \mathbf{J} \cdot d\mathbf{s} \quad \Rightarrow \quad \oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I \quad (\text{A})$$

- This last relation is another form of Ampère's circuital law.
- The two fundamental governing equations for magnetostatics in any medium

⚡ Important result

$$\begin{cases} \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{H} = \mathbf{J} \end{cases}$$

‣ Relative Permeability

- The relation between the magnetization vector and the magnetic field intensity

$$\mathbf{M} = \chi_m \mathbf{H}$$

- χ_m : the magnetic susceptibility
- The relation between the magnetic field intensity and the magnetic flux density

⚡ Important result

$$\begin{aligned} \mathbf{B} &= \mu_0 (1 + \chi_m) \mathbf{H} \\ &= \mu_0 \mu_r \mathbf{H} = \mu \mathbf{H} \quad (\text{Wb/m}^2) \end{aligned}$$

$$\mu_r = 1 + \chi_m = \frac{\mu}{\mu_0}$$

- μ_r : a dimensionless quantity called the relative permeability of the medium
- The relations of electrostatics and of magnetostatics are dual to each other

Electrostatics	Magnetostatics
E	B
D	H
ϵ	$\frac{1}{\mu}$
P	−M
ρ	J
V	A
$\cdot$	$\times$
$\times$	$\cdot$

► Behaviour of Magnetic Materials

- Magnetic materials can be roughly classified into three main groups according to their relative permeability
 - **Diamagnetic materials:** $\mu_r < 1$ (χ_m is a very small negative number)
 - In these materials the net magnetic moment is zero in the absence of an applied external magnetic field. → copper, lead, mercury, silver and gold → $\chi_m \approx -10^{-5}$
 - **Paramagnetic materials:** $\mu_r > 1$ (χ_m is a very small positive number)
 - In these materials there is a net average magnetic moment in the absence of an applied external magnetic field. → aluminium, magnesium and tungsten → $\chi_m \approx 10^{-5}$
 - **Ferromagnetic materials:** $\mu_r \gg 1$ (χ_m is a large positive number)
 - In these materials the magnetic moment is many times larger than in paramagnetic materials. → cobalt, nickel and iron
 - The relation between $\mathbf{B}$ and $\mathbf{H}$ in a ferromagnetic material is nonlinear.

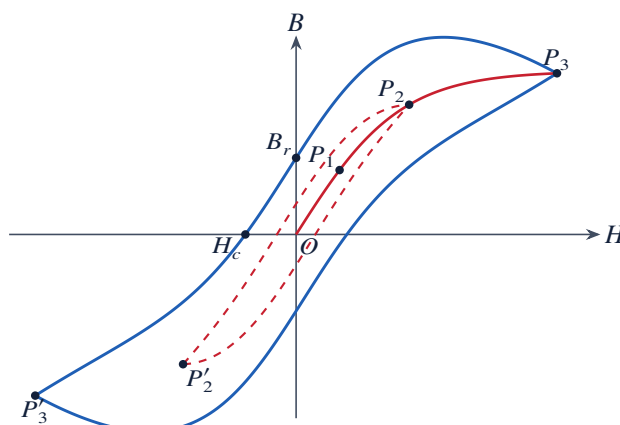
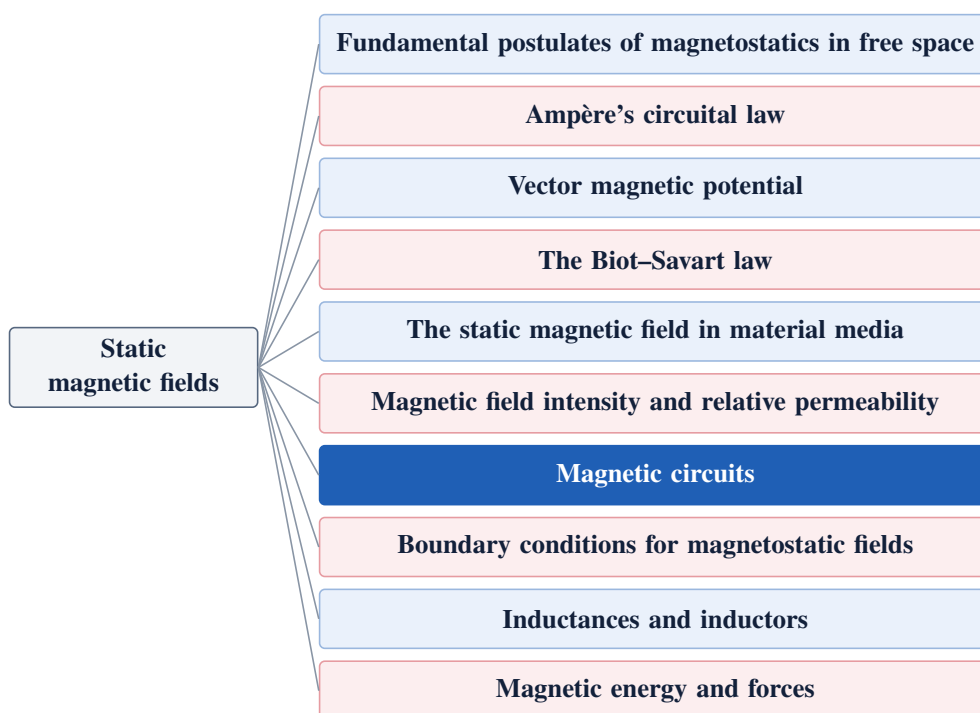


Figure 5.3 | The hysteresis phenomenon

5.8 Magnetic Circuits



- As with electric circuits, it is sometimes necessary to calculate the magnetic flux and the magnetic field intensity produced in a magnetic circuit by current-carrying windings.
- To analyse magnetic circuits we have

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{H} = \mathbf{J}$$

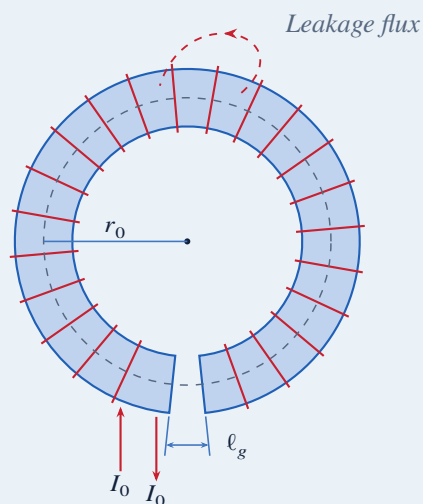
$$\oint_C \mathbf{H} \cdot d\ell = NI = \mathcal{V}_m$$

- $\mathcal{V}_m$: the magnetomotive force, in A. It is the analogue of the electromotive force in electric circuits.

Example 5-9

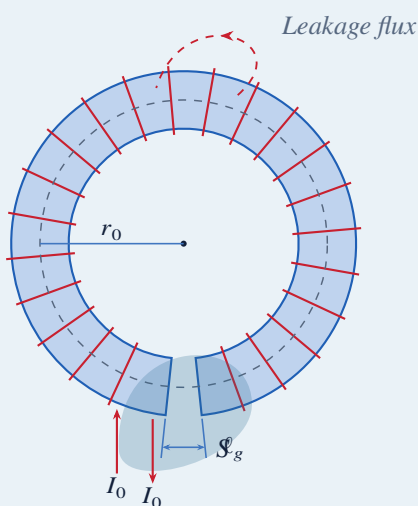
Assume that N turns of wire are wound around a toroidal core of a magnetic material with permeability μ . The core has a mean radius r_0 , a circular cross-section of radius a and a narrow air gap of length ℓ_g . A steady current I_0 flows in the wire. Determine

- (a) the magnetic flux density in the magnetic core,
- (b) the magnetic field intensity in the magnetic core,
- (c) the magnetic field intensity in the air gap.



► **Solution.**

- **First assumption:** the leakage flux is neglected



$$\nabla \cdot \mathbf{B} = 0 \quad \Rightarrow \quad \oint_S \mathbf{B} \cdot d\mathbf{s} = 0$$

- The flux in the core and in the air gap is the same.
- **Second assumption:** the fringing of the flux in the air gap is neglected.
- The flux density in the core and in the air gap is the same.

$$\mathbf{B}_f = \mathbf{B}_g = \mathbf{a}_\phi B_f$$

$$\mathbf{H}_f = \mathbf{a}_\phi \frac{B_f}{\mu}, \quad \mathbf{H}_g = \mathbf{a}_\phi \frac{B_f}{\mu_0}$$

$$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = NI_0 \quad \Rightarrow \quad \frac{B_f}{\mu} (2\pi r_0 - \ell_g) + \frac{B_f}{\mu_0} \ell_g = NI_0$$

- (a)

$$\mathbf{B}_f = \mathbf{a}_\phi \frac{\mu_0 \mu N I_0}{\mu_0 (2\pi r_0 - \ell_g) + \mu \ell_g}$$

► (b)

$$\mathbf{H}_f = \mathbf{a}_\phi \frac{\mu_0 N I_0}{\mu_0 (2\pi r_0 - \ell_g) + \mu \ell_g}$$

► (c)

$$\mathbf{H}_g = \mathbf{a}_\phi \frac{\mu N I_0}{\mu_0 (2\pi r_0 - \ell_g) + \mu \ell_g}$$

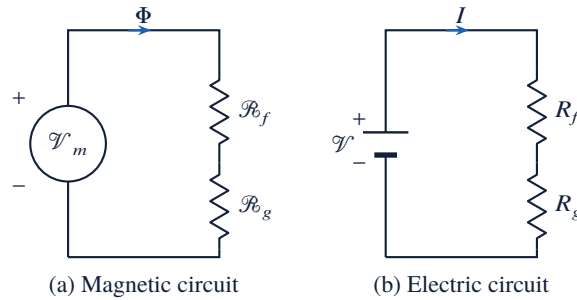
- If the radius of the cross-section of the core is much smaller than the mean radius of the toroid, the magnetic flux density in the core is approximately constant, and

$$\Phi \cong BS \Rightarrow \Phi = \frac{N I_0}{(2\pi r_0 - \ell_g)/\mu S + \ell_g/\mu_0 S} \Rightarrow \Phi = \frac{\mathcal{V}_m}{\mathcal{R}_f + \mathcal{R}_g}$$

⚡ Important result

$$\mathcal{R}_f = \frac{2\pi r_0 - \ell_g}{\mu S} = \frac{\ell_f}{\mu S}, \quad \mathcal{R}_g = \frac{\ell_g}{\mu_0 S}$$

- $\mathcal{R}_f$ and $\mathcal{R}_g$: reluctances, in 1/H
 ► This magnetic circuit is analogous to an electric circuit as follows



$$\Phi = \frac{\mathcal{V}_m}{\mathcal{R}_f + \mathcal{R}_g}, \quad I = \frac{\mathcal{V}}{R_f + R_g}$$

Magnetic Circuits	Electric Circuits
mmf, $\mathcal{V}_m (= NI)$	emf, $\mathcal{V}$
magnetic flux, Φ	electric current, I
reluctance, $\mathcal{R}$	resistance, R
permeability, μ	conductivity, σ

- In analogy to Kirchhoff's voltage law in electric circuits, in magnetic circuits we have

⚡ Important result

$$\sum_j N_j I_j = \sum_k \mathcal{R}_k \Phi_k$$

- So around any closed path in a magnetic circuit, the algebraic sum of the ampere-turns equals the algebraic sum of the products of the reluctances and the fluxes.
- In analogy to Kirchhoff's current law in electric circuits, in magnetic circuits we have

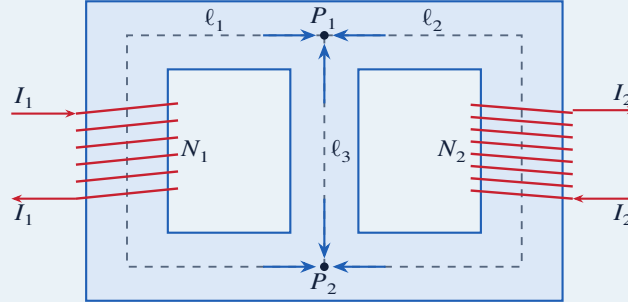
⚡ Important result

$$\sum_j \Phi_j = 0$$

- which states that the algebraic sum of all the magnetic fluxes flowing out of a junction in a magnetic circuit is zero.
- Despite this simple analogy, an exact analysis of magnetic circuits is very difficult, for the following reasons
 - the leakage fluxes,
 - the fringing effects, which make the magnetic flux lines spread out and bulge in the air gap,
 - the dependence of the permeability of ferromagnetic materials on the magnetic field intensity, that is, the nonlinear relation between **B** and **H**.

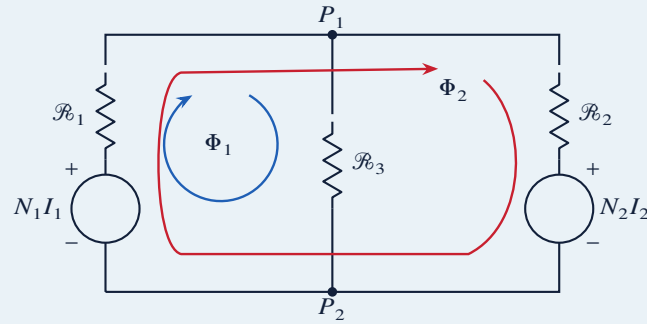
Example 5-10

Consider the magnetic circuit shown below. Steady currents I_1 and I_2 flow in windings of N_1 and N_2 turns respectively. The core has a cross-sectional area S_c and a permeability μ . Determine the magnetic flux in the middle leg.



► Solution.

- The equivalent circuit



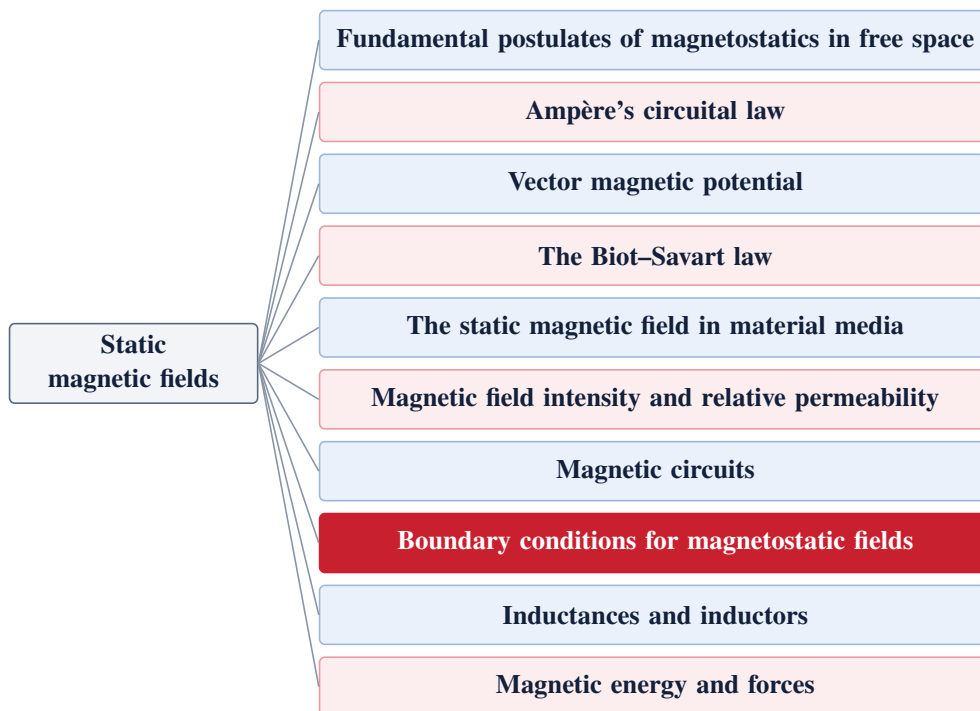
$$\mathcal{R}_1 = \frac{\ell_1}{\mu S_c}, \quad \mathcal{R}_2 = \frac{\ell_2}{\mu S_c}, \quad \mathcal{R}_3 = \frac{\ell_3}{\mu S_c}$$

$$\text{Loop 1 :} \quad N_1 I_1 = (\mathcal{R}_1 + \mathcal{R}_3) \Phi_1 + \mathcal{R}_1 \Phi_2$$

$$\text{Loop 2 :} \quad N_1 I_1 - N_2 I_2 = \mathcal{R}_1 \Phi_1 + (\mathcal{R}_1 + \mathcal{R}_2) \Phi_2$$

$$\Phi_1 = \frac{\mathcal{R}_2 N_1 I_1 + \mathcal{R}_1 N_2 I_2}{\mathcal{R}_1 \mathcal{R}_2 + \mathcal{R}_1 \mathcal{R}_3 + \mathcal{R}_2 \mathcal{R}_3}$$

5.9 Boundary Conditions for Magnetostatic Fields



- The boundary condition for the normal components

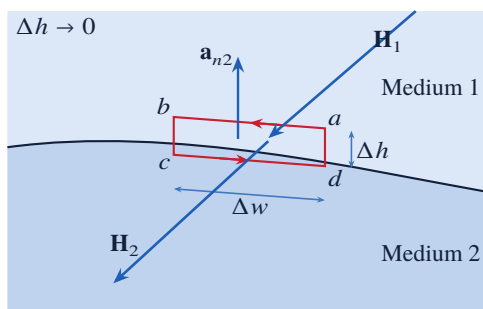
$$\nabla \cdot \mathbf{B} = 0 \quad \Rightarrow \quad B_{1n} = B_{2n} \quad (\text{T})$$

- In linear media

⚡ Important result

$$\mu_1 H_{1n} = \mu_2 H_{2n}$$

- The boundary condition for the tangential components



$$\oint_C \mathbf{H} \cdot d\ell = I$$

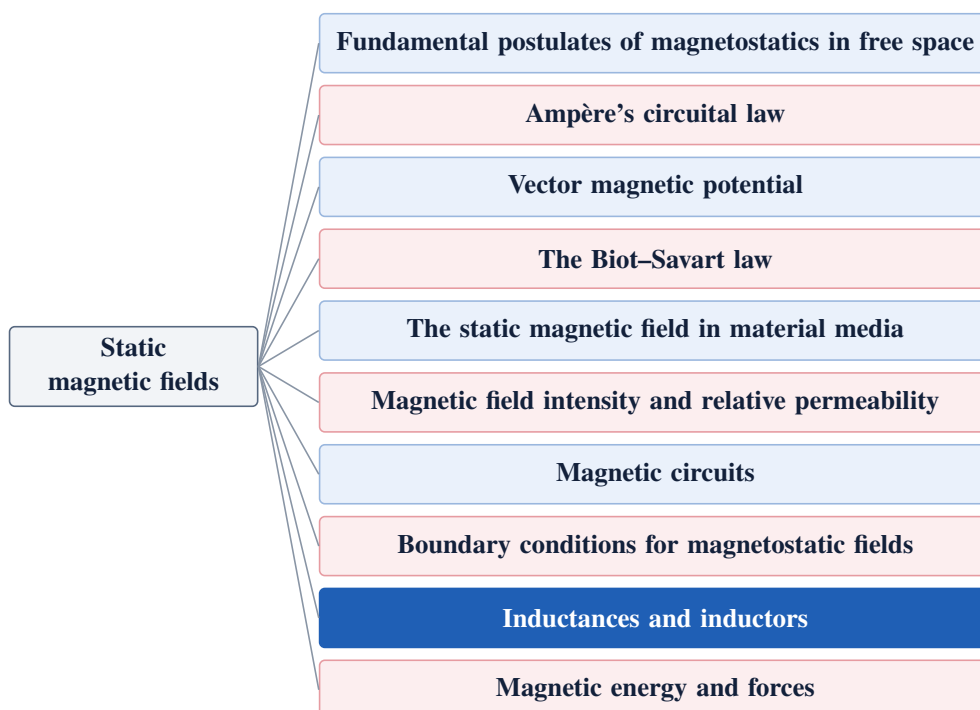
$$\oint_{ab c d a} \mathbf{H} \cdot d\ell = \mathbf{H}_1 \cdot \Delta \mathbf{w} + \mathbf{H}_2 \cdot (-\Delta \mathbf{w}) = J_{sn} \Delta w$$

$$H_{1t} - H_{2t} = J_{sn} \quad (\text{A/m})$$

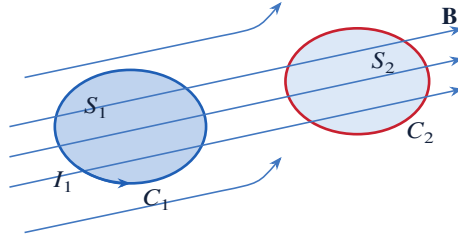
⚡ Important result

$$\mathbf{a}_{n2} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s \quad (\text{A/m})$$

5.10 Inductances and Inductors



- What kind of element is an inductor, and what is inductance?
- How is the inductance of different structures calculated?



- Consider two closed loops. If a current I_1 flows in C_1 , a magnetic field $\mathbf{B}_1$ is produced. Some of the magnetic flux due to $\mathbf{B}_1$ links C_2 . This mutual flux is

$$\Phi_{12} = \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}_2 \quad (\text{Wb})$$

- If C_2 has N_2 turns, the flux linkage is

$$\Lambda_{12} = N_2 \Phi_{12} \quad (\text{Wb})$$

- From the Biot–Savart law, $\mathbf{B}_1$, and hence Φ_{12} , is proportional to I_1 . Therefore

$$\Lambda_{12} = L_{12} I_1 \quad (\text{Wb}) \quad \Rightarrow \quad \boxed{L_{12} = \frac{\Lambda_{12}}{I_1} \quad (\text{H})}$$

- The proportionality constant L_{12} is called the **mutual inductance** between the loops C_1 and C_2 ; its unit is the henry (H). The mutual inductance between two circuits is the magnetic flux linkage with one circuit per unit current in the other.
- The total flux linkage with C_1 caused by I_1 is

$$\Lambda_{11} = N_1 \Phi_{11} \quad \Rightarrow \quad \boxed{L_{11} = \frac{\Lambda_{11}}{I_1}}$$

- The **self-inductance** of a loop is the magnetic flux linkage of the loop per unit current in the loop itself.
- The self-inductance of a circuit and the mutual inductance between two circuits depend on the geometrical shape and on the permeability of the medium. In a linear medium the self- and mutual inductances do not depend on the currents.
- A conductor shaped suitably to provide a certain amount of self-inductance is called an inductor.
- Just as a capacitor can store electric energy, an inductor can store magnetic energy.
- The steps for determining the **self-inductance** of an inductor are
 - choose a suitable coordinate system,
 - assume a current I in the wire,
 - find $\mathbf{B}$ from I ,
 - find the flux linking each turn

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$$

- S : the surface over which $\mathbf{B}$ exists and which links the assumed current
 - find the flux linkage,
 - find L

$$L = \Lambda / I$$

- The steps for determining the **mutual inductance** between two circuits are
 - choose a suitable coordinate system,
 - assume a current I_1 in the wire,
 - find $\mathbf{B}_1$ from I_1 ,
 - find the mutual flux

$$\Phi_{12} = \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}_2$$

- find the flux linkage

$$\Lambda_{12} = N_2 \Phi_{12}$$

- find L_{12}

$$L_{12} = \frac{\Lambda_{12}}{I_1} \quad (\text{H})$$

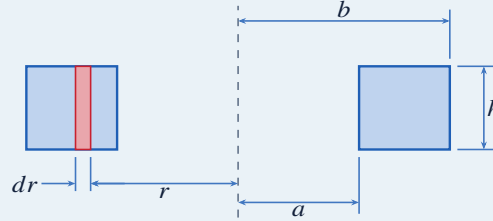
- It can be shown that

⚡ Important result

$$L_{12} = L_{21}$$

Example 5-11

Assume that N turns of wire are tightly wound on a toroidal frame of rectangular cross-section, as shown below. Assuming that the permeability of the medium is μ_0 , find the self-inductance of the toroidal coil.



► Solution.

- Choose cylindrical coordinates.
- Assume a current I in the coil.
- Find $\mathbf{B}$ with Ampère's circuital law

$$\mathbf{B} = \mathbf{a}_\phi B_\phi, \quad d\ell = \mathbf{a}_\phi r d\phi$$

$$\oint_C \mathbf{B} \cdot d\ell = \int_0^{2\pi} B_\phi r d\phi = 2\pi r B_\phi$$

$$2\pi r B_\phi = \mu_0 N I \quad \Rightarrow \quad B_\phi = \frac{\mu_0 N I}{2\pi r}$$

- Find Φ and Λ

$$\begin{aligned} \Phi &= \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S \left(\mathbf{a}_\phi \frac{\mu_0 N I}{2\pi r} \right) \cdot (\mathbf{a}_\phi h dr) \\ &= \frac{\mu_0 N I h}{2\pi} \int_a^b \frac{dr}{r} = \frac{\mu_0 N I h}{2\pi} \ln \frac{b}{a} \quad \Rightarrow \quad \Lambda = \frac{\mu_0 N^2 I h}{2\pi} \ln \frac{b}{a} \end{aligned}$$

- Find L

$$L = \frac{\Lambda}{I} = \frac{\mu_0 N^2 h}{2\pi} \ln \frac{b}{a}$$

Example 5-12

Find the inductance per unit length of a very long solenoid with an air core and n turns per unit length.

► Solution.

$$B = \mu_0 n I$$

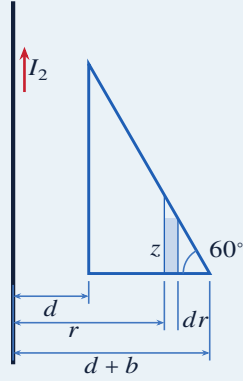
$$\Phi = BS = \mu_0 n S I$$

$$\Lambda' = n\Phi = \mu_0 n^2 S I$$

$$L' = \mu_0 n^2 S \quad (\text{H/m})$$

Example 5-13

Find the mutual inductance between a triangular conducting loop and a very long straight wire.



► Solution.

- Assume a current I_2 in the long wire and find the $\mathbf{B}$ it produces

$$\mathbf{B}_2 = \mathbf{a}_\phi \frac{\mu_0 I_2}{2\pi r}$$

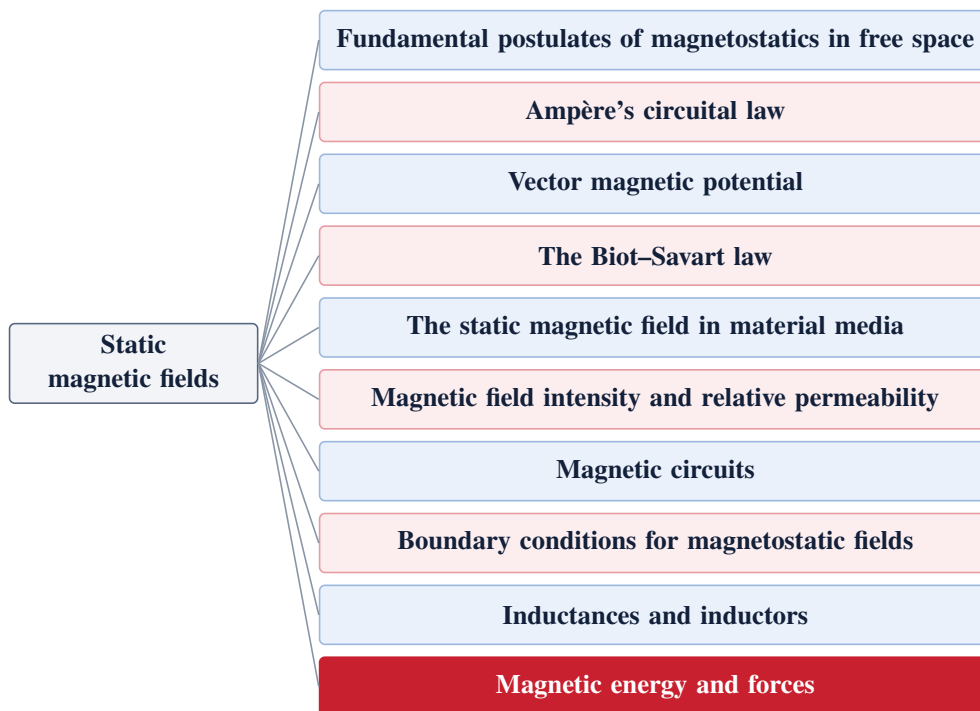
- Find the flux linkage

$$\begin{aligned} \Lambda_{21} &= \int_{S_1} \mathbf{B}_2 \cdot d\mathbf{s}_1 \\ &= \int_d^{d+b} \int_0^{[(d+b)-r] \tan 60^\circ} \frac{\mu_0 I_2}{2\pi r} dz dr \\ \Lambda_{21} &= -\frac{\sqrt{3}\mu_0 I_2}{2\pi} \int_d^{d+b} \frac{1}{r} [r - (d+b)] dr \\ &= \frac{\sqrt{3}\mu_0 I_2}{2\pi} \left[(d+b) \ln \left(1 + \frac{b}{d} \right) - b \right] \end{aligned}$$

⚡ Important result

$$L_{21} = \frac{\Lambda_{21}}{I_2} = \frac{\sqrt{3}\mu_0}{2\pi} \left[(d+b) \ln \left(1 + \frac{b}{d} \right) - b \right] \quad (\text{H})$$

5.11 Magnetic Energy and Forces



► Magnetic Energy

- Work is required to assemble a group of charges in place, and this work is stored as **electric energy**.
- In the same way we expect that work must be done to establish currents in conducting loops, and that this work is stored as **magnetic energy**.
- Consider a closed loop with self-inductance L_1 and initial current $i_1 = 0$.
 - A generator is connected to the loop and increases the current i_1 from zero to I_1 .
 - In this process a changing magnetic flux passes through the loop. So by Faraday's law of induction an electromotive force is induced in the loop

$$v_1 = \frac{d\Phi_1}{dt} = L_1 \frac{di_1}{dt}$$

- By Lenz's law this electromotive force opposes the change of flux and current. Work must therefore be done to overcome it. This work is stored as magnetic energy

$$W_1 = \int v_1 i_1 dt = L_1 \int_0^{I_1} i_1 di_1 = \frac{1}{2} L_1 I_1^2$$

- Now consider two closed loops carrying the currents i_1 and i_2 . The initial currents are zero, and they are increased to I_1 and I_2 respectively.
 - To find the work required (or the magnetic energy stored), first keep $i_2 = 0$ and increase i_1 from zero to I_1 . This requires the work W_1 .
 - Then keep i_1 at I_1 and increase i_2 from zero to I_2 . Two kinds of work must be done.
 - the work W_{22} to increase i_2 from zero to I_2

$$W_{22} = \frac{1}{2} L_2 I_2^2$$

- Because of the mutual coupling, part of the magnetic flux due to i_2 links the first loop. So by Faraday's law of induction an electromotive force is induced in the first loop.

$$v_{21} = \frac{d\Phi_{21}}{dt} = L_{21} \frac{di_2}{dt}$$

- By Lenz's law this electromotive force opposes the change of the flux linking the first loop, through a change in the current of the first loop. Work must therefore be done to overcome it

$$W_{21} = \int v_{21} I_1 dt = L_{21} I_1 \int_0^{I_2} di_2 = L_{21} I_1 I_2$$

- The total work done, or the total magnetic energy stored, is

$$\begin{aligned} W_2 &= \frac{1}{2} L_1 I_1^2 + L_{21} I_1 I_2 + \frac{1}{2} L_2 I_2^2 \\ &= \frac{1}{2} \sum_{j=1}^2 \sum_{k=1}^2 L_{jk} I_j I_k \end{aligned}$$

- In general, for N current loops

$$W_m = \frac{1}{2} \sum_{j=1}^N \sum_{k=1}^N L_{jk} I_j I_k \quad (\text{J})$$

- The flux linking the k th loop

$$\Phi_k = \sum_{j=1}^N L_{jk} I_j$$

⚡ Important result

$$W_m = \frac{1}{2} \sum_{k=1}^N I_k \Phi_k \quad (\text{J})$$

- It can be shown that the magnetic energy stored by a continuous current distribution in a volume is

⚡ Important result

$$W_m = \frac{1}{2} \int_{V'} \mathbf{A} \cdot \mathbf{J} dv' \quad (\text{J})$$

- V' : the region containing $\mathbf{J}$, which may extend over all space
- It can be shown that the stored magnetic energy in terms of the field quantities is

⚡ Important result

$$W_m = \frac{1}{2} \int_{V'} \mathbf{H} \cdot \mathbf{B} \, dv' \quad (\text{J})$$

$$W_m = \frac{1}{2} \int_{V'} \frac{B^2}{\mu} \, dv' \quad (\text{J})$$

$$W_m = \frac{1}{2} \int_{V'} \mu H^2 \, dv' \quad (\text{J})$$

- V' : all space

① Remark

It is often easier to determine the self-inductance from the stored magnetic energy in terms of $\mathbf{H}$ or $\mathbf{B}$ than from the flux linkage

$$W_m = \frac{1}{2} \sum_{j=1}^N \sum_{k=1}^N L_{jk} I_j I_k$$

- For an inductor carrying a current I , with inductance L

$$W_m = \frac{1}{2} L I^2$$

$$L = \frac{2W_m}{I^2} \quad (\text{H})$$

Example 5-14

Using the stored magnetic energy, determine the inductance per unit length of an air coaxial transmission line with a solid inner conductor of radius a and a very thin outer conductor of inner radius b .

► Solution.

- The energy stored per unit length in the inner conductor

$$B_{\phi 1} = \frac{\mu_0 I r}{2\pi a^2} \quad \Rightarrow \quad \begin{aligned} W'_{m1} &= \frac{1}{2\mu_0} \int_0^a \int_0^{2\pi} B_{\phi 1}^2 r \, dr \, d\phi \\ &= \frac{\mu_0 I^2}{4\pi a^4} \int_0^a r^3 \, dr = \frac{\mu_0 I^2}{16\pi} \quad (\text{J/m}) \end{aligned}$$

- The energy stored per unit length in the region between the two conductors

$$B_{\phi 2} = \frac{\mu_0 I}{2\pi r} \quad \Rightarrow \quad \begin{aligned} W'_{m2} &= \frac{1}{2\mu_0} \int_a^b \int_0^{2\pi} B_{\phi 2}^2 r \, dr \, d\phi \\ &= \frac{\mu_0 I^2}{4\pi} \int_a^b \frac{1}{r} \, dr = \frac{\mu_0 I^2}{4\pi} \ln \frac{b}{a} \quad (\text{J/m}) \end{aligned}$$

- Therefore

$$\begin{aligned} L' &= \frac{2}{I^2} (W'_{m1} + W'_{m2}) \\ &= \frac{\mu_0}{8\pi} + \frac{\mu_0}{2\pi} \ln \frac{b}{a} \quad (\text{H/m}) \end{aligned}$$

Example 5-15

For a very long solenoid with cross-section S , n turns per unit length and an air core, find the inductance per unit length.

► Solution.

$$\begin{aligned} B &= \mu_0 n I \quad \Rightarrow \quad W_m = \frac{1}{2} \int_{V'} \frac{B^2}{\mu} \, dv' = \frac{1}{2\mu_0} \int \mu_0^2 n^2 I^2 \, dv \\ &= \frac{1}{2} \mu_0 n^2 I^2 S = \frac{1}{2} L I^2 \quad \Rightarrow \quad L' = \mu_0 n^2 S \quad \text{H/m} \end{aligned}$$

► Magnetic Forces and Torques

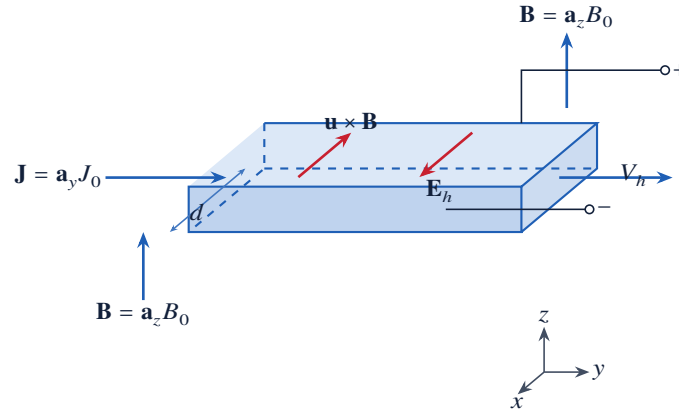
- When a charge q moves with velocity $\mathbf{u}$ in a magnetic field of flux density $\mathbf{B}$, a magnetic force $\mathbf{F}_m$ acts on it.

$$\mathbf{F}_m = q\mathbf{u} \times \mathbf{B} \quad (\text{N})$$

- Topics
 - the Hall effect,
 - forces and torques on current-carrying conductors.

► The Hall Effect

- Consider a conducting material of rectangular cross-section $d \times b$ in a uniform magnetic field $\mathbf{B} = \mathbf{a}_z B_0$. A uniform direct current flows in the y direction.



$$\mathbf{J} = \mathbf{a}_y J_0 = Nq\mathbf{u}$$

- The charge carriers are electrons, and q is negative

$$\mathbf{u} = -\mathbf{a}_y u_0$$

- The magnetic force tends to move the electrons in the $+x$ direction and creates a transverse electric field. The process continues until the transverse field is strong enough to stop the drift of the electrons.
- In the steady state the net force on the charge carriers is zero.

$$\begin{aligned} \mathbf{E}_h + \mathbf{u} \times \mathbf{B} &= 0 \quad \Rightarrow \quad \mathbf{E}_h = -\mathbf{u} \times \mathbf{B} \quad \Rightarrow \quad \mathbf{E}_h = -(-\mathbf{a}_y u_0) \times \mathbf{a}_z B_0 \\ &= \mathbf{a}_x u_0 B_0 \end{aligned}$$

- $\mathbf{E}_h$: the Hall field
- The Hall voltage

$$V_h = - \int_0^d E_h dx = u_0 B_0 d$$

- The Hall coefficient

$$E_x / J_y B_z = 1 / Nq$$

- This phenomenon is called the **Hall effect**.
- The Hall effect is used to measure magnetic fields.

► Forces and Torques on Current-Carrying Conductors

- The magnetic force on a differential element of a closed circuit with contour C , cross-section S and current I , in a magnetic field of flux density $\mathbf{B}$



$$d\mathbf{F}_m = q\mathbf{u} \times \mathbf{B}$$

$$\begin{aligned} d\mathbf{F}_m &= -NeS|d\ell|\mathbf{u} \times \mathbf{B} \\ &= +NeS|\mathbf{u}| d\ell \times \mathbf{B} \end{aligned}$$

- N : the number of electrons per unit volume, and e : the charge of an electron
- $d\ell$ is opposite to $\mathbf{u}$, and $NeS|\mathbf{u}| = I$

⚡ Important result

$$d\mathbf{F}_m = I d\ell \times \mathbf{B} \quad (\text{N})$$

- The magnetic force on a closed circuit with contour C , cross-section S and current I , in a magnetic field of flux density $\mathbf{B}$

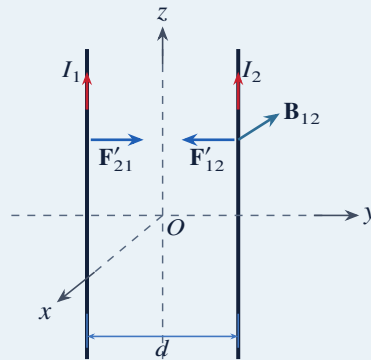
⚡ Important result

$$\mathbf{F}_m = I \oint_C d\ell \times \mathbf{B} \quad (\text{N})$$

► $d\ell$: in the direction of the current

Example 5-16

Determine the force per unit length between two very long parallel conducting wires carrying the currents I_1 and I_2 in the same direction, as shown below.



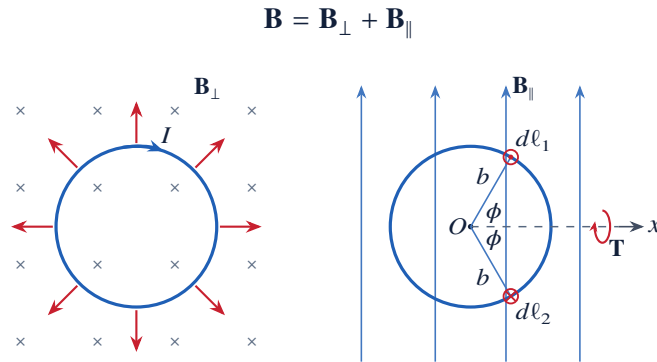
► **Solution.**

$$\mathbf{F}'_{12} = I_2 \oint_{C_2} d\ell_2 \times \mathbf{B}_{12} = I_2 \int_0^1 dz \mathbf{a}_z \times \left(-\mathbf{a}_x \frac{\mu_0 I_1}{2\pi d} \right)$$

⚡ **Important result** | An attractive force

$$\mathbf{F}'_{12} = -\mathbf{a}_y \frac{\mu_0 I_1 I_2}{2\pi d} \quad (\text{N/m})$$

- Consider a circular loop of radius b carrying a current I in a uniform magnetic field of flux density $\mathbf{B}$



- $\mathbf{B}_\perp$ produces no net force to move the loop; it only tends to expand the loop (or, if the current is reversed, to contract it).
- Although the net force on the loop due to $\mathbf{B}_\parallel$ is zero, there is a torque that tends to rotate the loop about the x -axis so as to align the magnetic field produced by I with the external field $\mathbf{B}_\parallel$.

$$\begin{aligned} d\mathbf{T} &= \mathbf{a}_x(dF)2b \sin \phi \\ &= \mathbf{a}_x(I d\ell B_\parallel \sin \phi)2b \sin \phi \\ &= \mathbf{a}_x 2Ib^2 B_\parallel \sin^2 \phi d\phi \end{aligned}$$

$$\begin{aligned} \mathbf{T} &= \int d\mathbf{T} = \mathbf{a}_x 2Ib^2 B_\parallel \int_0^\pi \sin^2 \phi d\phi \\ &= \mathbf{a}_x I(\pi b^2) B_\parallel \end{aligned}$$

$$\mathbf{m} = \mathbf{a}_n I(\pi b^2) = \mathbf{a}_n IS$$

$$\mathbf{T} = \mathbf{m} \times \mathbf{B}_\parallel = \mathbf{m} \times (\mathbf{B}_\parallel + \mathbf{B}_\perp)$$

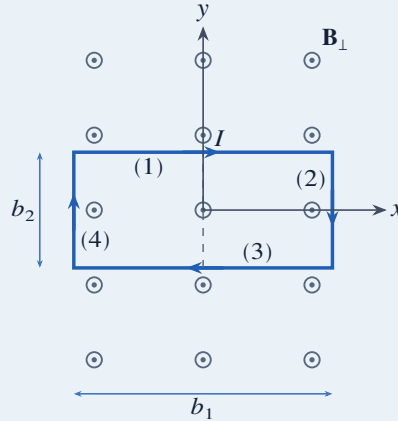
⚡ Important result

$$\mathbf{T} = \mathbf{m} \times \mathbf{B} \quad (\text{N} \cdot \text{m})$$

- It can be shown that this relation holds for a planar loop of any shape, as long as it lies in a uniform magnetic field

Example 5-17

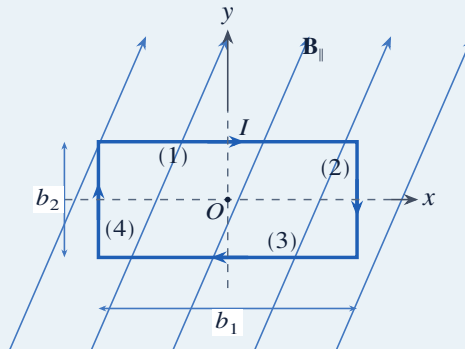
A rectangular loop in the xy -plane with sides b_1 and b_2 carries a current I and lies in the uniform magnetic field $\mathbf{B} = \mathbf{a}_x B_x + \mathbf{a}_y B_y + \mathbf{a}_z B_z$. Determine the force and the torque on the loop.



Solution.

$$\mathbf{B}_\perp = \mathbf{a}_z B_z, \quad \mathbf{B}_\parallel = \mathbf{a}_x B_x + \mathbf{a}_y B_y$$

- The normal component exerts the forces $Ib_1 B_z$ on the sides (1) and (3) and $Ib_2 B_z$ on the sides (2) and (4), toward the centre. → No torque is produced.
- The forces on the loop due to the component parallel to the plane

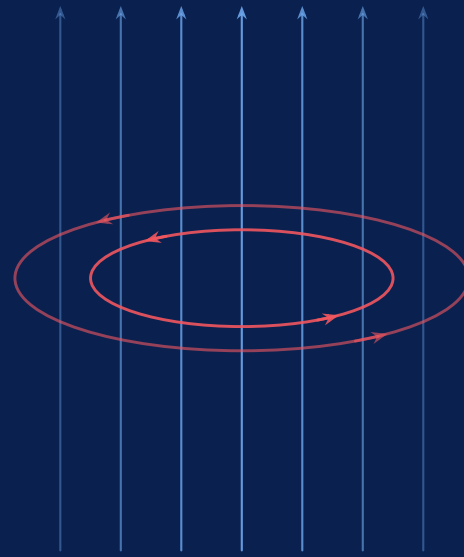


$$\begin{aligned} \mathbf{F}_1 &= Ib_1 \mathbf{a}_x \times (\mathbf{a}_x B_x + \mathbf{a}_y B_y) \\ &= \mathbf{a}_z Ib_1 B_y = -\mathbf{F}_3 \end{aligned}$$

$$\begin{aligned} \mathbf{F}_2 &= Ib_2 (-\mathbf{a}_y) \times (\mathbf{a}_x B_x + \mathbf{a}_y B_y) \\ &= \mathbf{a}_z Ib_2 B_x = -\mathbf{F}_4 \end{aligned}$$

$$\left. \begin{aligned} \mathbf{T}_{13} &= \mathbf{a}_x Ib_1 b_2 B_y \\ \mathbf{T}_{24} &= -\mathbf{a}_y Ib_1 b_2 B_x \end{aligned} \right\} \begin{aligned} \mathbf{T} &= \mathbf{T}_{13} + \mathbf{T}_{24} = Ib_1 b_2 (\mathbf{a}_x B_y - \mathbf{a}_y B_x) \quad (\text{N} \cdot \text{m}) \\ \mathbf{T} &= \mathbf{m} \times (\mathbf{a}_x B_x + \mathbf{a}_y B_y) = \mathbf{m} \times \mathbf{B} \end{aligned}$$

6



CHAPTER

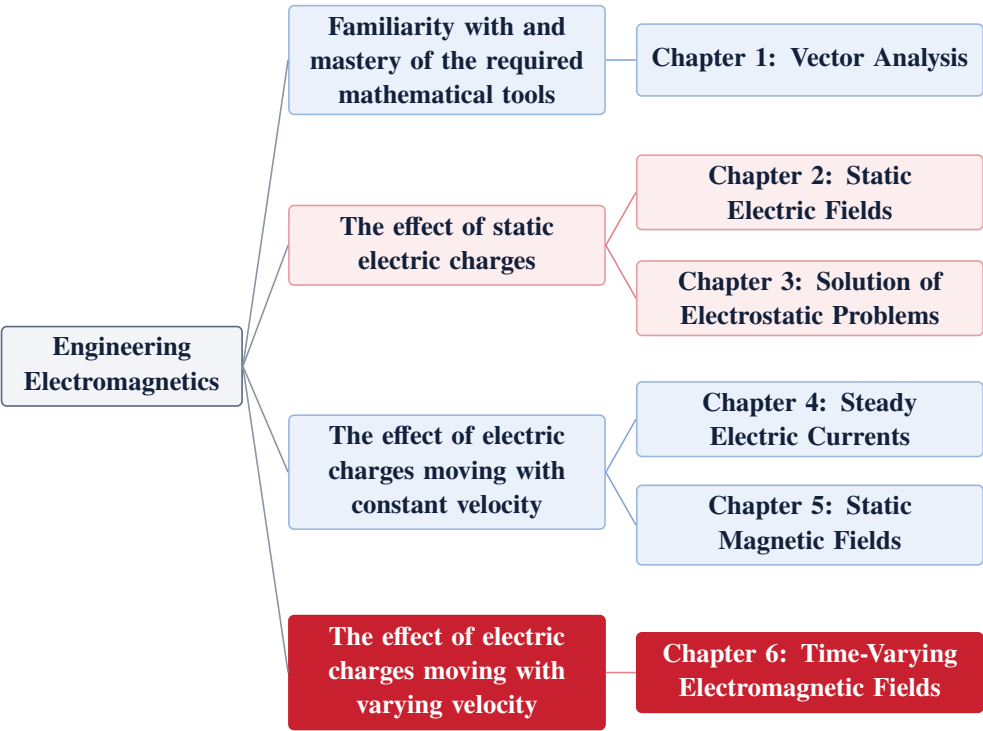
Time-Varying Electromagnetic Fields

میدان‌های الکترومغناطیسی متغیر با زمان

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6.1 Course Outline



6.2 Introduction

- The fundamental relations of electrostatics and magnetostatics

Fundamental Relations	Electrostatic Model	Magnetostatic Model
Governing equations	$\nabla \times \mathbf{E} = 0, \nabla \cdot \mathbf{D} = \rho$	$\nabla \cdot \mathbf{B} = 0, \nabla \times \mathbf{H} = \mathbf{J}$
Constitutive relations (linear and isotropic media)	$\mathbf{D} = \epsilon \mathbf{E}$	$\mathbf{H} = \frac{1}{\mu} \mathbf{B}$

- In the static (time-invariant) case, the electric field vectors and the magnetic field vectors form separate, independent pairs.
- We shall see that in the time-varying case a time-varying magnetic field produces an electric field, and vice versa

6.3 Faraday's Law of Electromagnetic Induction

- In 1820 **Hans Christian Ørsted** showed that a wire carrying an electric current deflects a compass needle.
 - The motion of the compass needle was caused by the magnetic force of the magnetic field produced by the electric current.
- After this discovery **Michael Faraday** reasoned that a magnetic field should, in the same way, have an electric effect and produce a current in a wire.
- After about ten years of experiments he made the discovery known as Faraday's law of electromagnetic induction.

≡ Definition | Faraday's law of electromagnetic induction

If the magnetic flux through a closed circuit changes with time, an electromotive force is induced in the circuit. This electromotive force drives a current.

- The polarity of the electromotive force, and hence the direction of the induced current, can be found from Lenz's law.

≡ Definition | Lenz's law

The current induced in a circuit always flows in the direction that opposes the change of the magnetic flux producing it.

- The mathematical statement of Faraday's law of electromagnetic induction

⚡ Important result

$$v_{emf} = -N \frac{d\Phi}{dt} = -N \frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s}$$

- The definition of the electromotive force induced in a circuit with the closed contour C

$$v_{emf} \triangleq \oint_C \mathbf{E} \cdot d\boldsymbol{\ell}$$

- For a stationary circuit

$$v_{emf} = \oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad \Rightarrow \quad \int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

- The last step uses Stokes's theorem.

⚡ Important result

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

- In the time-varying case the electric field is not conservative $\rightarrow \nabla \times \mathbf{E} \neq 0$
- In the time-varying case the scalar potential cannot be defined as $V = -\int \mathbf{E} \cdot d\ell$, because the potential difference between two points would depend on the path of integration and would not have a fixed value.
- At low frequencies $\rightarrow \nabla \times \mathbf{E} \simeq 0 \rightarrow$ the scalar potential can be defined (lumped-circuit theory)

🔗 Example 6-1

A circular loop of N turns of conducting wire lies in the xy -plane with its centre at the origin, in a magnetic field given by

$$\mathbf{B} = \mathbf{a}_z B_0 \cos\left(\frac{\pi r}{2b}\right) \sin(\omega t)$$

Here b is the radius of the loop and ω the angular frequency. Find the induced electromotive force.

► Solution.

$$\begin{aligned} \Phi &= \int_S \mathbf{B} \cdot d\mathbf{s} = \int_0^{2\pi} \int_0^b \left[B_0 \cos\left(\frac{\pi r}{2b}\right) \sin(\omega t) \mathbf{a}_z \right] \cdot (r dr d\phi \mathbf{a}_z) \\ &= \frac{8b^2}{\pi} \left(\frac{\pi}{2} - 1 \right) B_0 \sin(\omega t) \\ v &= -N \frac{d\Phi}{dt} = -\frac{8Nb^2}{\pi} \left(\frac{\pi}{2} - 1 \right) B_0 \omega \cos(\omega t) \end{aligned}$$

6.4 Displacement Current

- Ampère's circuital law for the static magnetic field

$$\nabla \times \mathbf{H} = \mathbf{J}$$

- James Clerk Maxwell doubted this equation, because

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J}$$

- which is not consistent with the equation of continuity

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$$

- To remove this inconsistency Maxwell modified the relation as follows

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} \quad \xrightarrow{\nabla \cdot \mathbf{D} = \rho} \quad \nabla \cdot (\nabla \times \mathbf{H}) = \nabla \cdot \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right)$$

⚡ Important result

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

- It can easily be shown that $\partial \mathbf{D} / \partial t$ has the dimension of a current density. It is called the **displacement current density**.
- The modified form of Ampère’s circuital law states that a time-varying electric field produces a magnetic field even when there is no current.

6.5 Maxwell’s Equations

- The equations governing time-varying electric and magnetic fields, or **Maxwell’s equations**

Differential Form	Integral Form	Significance
$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt}$	Faraday’s law
$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}$	Ampère’s circuital law
$\nabla \cdot \mathbf{D} = \rho$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$	Gauss’s law
$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$	No isolated magnetic charge